

STA630 Mega Subjective For Final Term All in 1 File August 2023

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Lec (19-42)

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UMAR

1- How would you differentiate between History effect and Mechanical loss in experimental Research?

Answer. History effect is specific event in the external environment occurring between the 1st and 2nd measurement that is beyond the control of the experimenter and that affects the validity of an experiment.

Mechanical loss is problem may be experienced due to equipment failure.

2 Write Note on three varieties of Non reactive observations?

Answer. (1) - Physical traces

(i) - Erosion $\Rightarrow$ wear and tear suggest ^{use.} greater

(ii) Accretion $\Rightarrow$ Accumulation of

physical evidence suggests behavior.

(2) - Archives

(i) Running Records $\Rightarrow$ Regularly produced public records may reveal lot of information.

(ii) Other Records $\Rightarrow$ irregular or private records

- (3) - observations.
- (i) - External appearance: How people appear may indicate social factors -
 - (ii) Count behaviors: Counting how many people do something can be informative
 - (iii) Time duration. How long people take to do things may indicate their intention.

3- Term Inductive to describe Qualitative Data Analysis and interpretation?

Answer. In inductive term answers should address methodological characteristics such as data collection and coding, developing categories and patterns, and interpreting relationships across patterns.

4- Discuss Solomon Four Group Design in experimental research with the help of example -

Answer. Solomon Four Group Design to gain more confidence in internal validity in experimental

designs, it is advisable to set up two experimental groups and two control groups. one experimental group and one control group can be given the both pretest and the posttest.

Group	Pretest	Treatment	Posttest.
Experimental.	01	X	02
Control.	03	-	04
Experimental.	-	X	05
Control	-	-	06

5- Ethical Dilemma of Field Research

Answer. The direct personal involvement of a field researcher in the social lives of other people raises many ethical dilemmas. Dilemmas arises when researcher is alone in the field and has little time to make a moral decision.

6- How Researchers locate evidence in Historical

Comparative research ?

Answer. Historical Comparative Research draw on four types historical evidence or data.

- (1) Primary sources.
- (2) Secondary sources.
- (3) - Running Records -
- (4) - Recollections.

Tradition historians can rely heavily on primary sources. H-C researchers often use secondary sources.

7- Comparative Historical Research Follow Scientific Approach ?

Answer. Historical Comparative Research follows scientific approach because of Reasons.

- (1) Can be survey of events in history
- (2) Historical-comparative researchers mostly do longitudinal analysis.
- (3) Historical-comparative researchers make cross-cultural comparisons of social reforms or economic reform.

8- Importance of Measurement and Coding in non-Research Answer. Example -

Careful Measurement is crucial in content analysis because a researcher takes different and murky symbolic communication and turns it into precise, objective, quantitative data. Coding to make replication possible. because coding system is set of instructions.

9- Advantages of Cluster Sampling?

Answer. ① Feasibility

② Economy.

③ Reduced variability.

10- Do you explain inferences from non-reactive data by using secondary survey data?

Answer. A researcher's ability to infer causality or to test a theory on the basis of non-reactive data is limited. It is difficult to use unobtrusive

measures to establish temporal order and eliminate alternative explanations.

11- Research process, once you collect the data how would perform the scoring index activity?
Answer.

- ① Remember the measurement of of Job Satisfaction.
- ② No of Statements on each element.
- ③ Likert Scale (5) response category.
Strong Agree, Agree, disagree, Neutral
- ④ There are positive or Negative statements
Positive Statements score ^{Strong} Agree = 5
Agree = 4, Neutral = 3, disagree = 2 -
disagree = 1 -

12- Would you explain spontaneity and confession by using secondary data?

Answer. Spontaneity actions or feelings can be recorded when they occurred rather than at a time specified by the

researcher. If respondent was keeping a diary, he or she may have been recording spontaneous feelings about a subject.

Confession A - A person may be more likely to confess in a document particularly one to be read only after his/her death than in an interview or mailed questionnaire.

13- No and duration in FDG?

Answer. The Number of focus group session to be conducted depends upon project needs, resources and new information is still coming from the sessions.

For duration, a focus group session typically lasts up to an hour and a half.

14 Suggest the Appropriate data collection tool that can be used by Research organization;

Answer. Secondary data are usually historical, already assembled and do not require access to respondents or subjects. Secondary data may be collected by large ~~be~~ bureaucratic organization like the bureau of statistics.

15 What is bracketing?

Answer. Researchers attempt to read primary sources with the eyes and assumptions of contemporary who lived in past.

16- Accretion and count behaviors?

Answer. Accretion: Accumulation of physical evidence suggests behavior.

Count behaviors. Counting how many people do something can be informative.

17- Independent variable?

Answer. It is the variable you have control over, what you can choose and manipulate.

18 Being Researcher, how would you criticize the use of secondary data in Research?
Answer:

Many documents used in Research were not originally intended for research purposes. Bias are -

- (i) - Selective ~~in~~ Survival
- (ii) - incompleteness.
- (iii) - Lack of Availability of documents
- (iv) - Limited to verbal behaviors.
- (v) - Sampling bias.
- (vi) - Lack of Standardized format.

19- Different measures of central tendency are used?

Answer: Three measures of central tendency, or measure of the center of frequency distribution: (i) Mean, Mode (ii) Median. which are often called averages. (less precise, less clear, same thing). Mode is most commonly occurring number. Median is Middle point. Mean also called Arithmetic Average.

20 Functions of the Recorder?

Answer. ① Data, time and place

② Names and characteristics of participants.

③ General principle of group dynamics.

④ opinion of participants.

⑤ Vocabularily used.

21 Historical Comparative depends upon some factors. List down.

① Answer. Historical Comparative depends upon tracking the development of social forms over time as well as its broad historical processes.

② Comparing those forms and its developmental processes across cultures.

22 Using APA style describe Quotation i.e. what is their format in writing a literature view?

Answer. American psychologist Association style was developed by social and behavioral scientists.

The writing of APA requires the writer to pay attention to mechanical details in the writing such as

margins, typeface, page headers and line spacing.

Writing literature article reviews is not an easy task for any level student. This is because student is required to read the literature material which are usually wide and provide their deeper meaning in the article reviews.

23 Parts of executive Summary?

Answer. Summary contains four elements.

- ① The objective of the report are stated including most important background and specific purpose of the project.
- ② The major results are presented.

- ③ Conclusion that are based on results.
- ④ The recommendation or suggestions for actions which are based on conclusion.

24. Parts of Research Report?

Answer. ① Prefatory parts-

- (i) Title page (ii) Title page (iii) Letter of transmittal (iv) Letter of authorization.

- (v) Table of contents (vi) Executive Summary.

② Main body -

- (i) intro (ii) Methodology (iii) Results (iv) Conclusions.
- (v) Reference.

3- Appended parts.

- (i) Data collection -
- (ii) Detailed calculations
- (iii) General Tables
- (iv) other Support material -
- (v) Bibliography.

25. Bivariate table ?

Answer. Bivariate contingency table is widely used. This table is based on cross tabulation.

26 How can we use Secondary Analysis ?

Answer. Secondary Analysis is increasingly used by researchers. It is inexpensive.

It can permits comparison.

It facilitate Replication.

27 Reference Format ?

Answer. Author $\Rightarrow$ Abraham Maslow -

Name Journal $\Rightarrow$ international journal of psychology.

Title $\Rightarrow$ Need Hierarchy theory.

Page No = 20-23.

Year of publish $\Rightarrow$ 1946.

28- What is bibliography?

Answer.

- (i) Bibliography is the list of works that are relevant to the topic of research.
- (ii) A reference list is subset of bibliography which includes details of all the citations used in the literature survey.

29 Explanatory Research?

Answer. When we encounter issues that is already known. we might begin to wonder why things are the way they are. The desire to know "why" to explain is the purpose of explanatory research. It builds upon explanatory or descriptive research.

30- Basic Research

(i) intrinsically satisfying

(ii) Freedom

(iii) Hi Standard Applied. of Sponsors.

(iv) Logic and rigorous

(v) Research design.

(vi) Basic knowledge.

Success = results published
impact on other scientists

Applied Research.

R → Part of job.

Contained on demands

R → Quick and dirty

may not meet hi standard.

Practical payoff

Success = used by sponsors.

32 Ratio scale properties -

Answer. ① Uniquely classifiers.

② Preserves order.

③ Equal intervals.

- Natural zero -

Weight and distance.

33- Problem definition -

Answer. Problem definition

is clear, precise, and succinct

statement of the questions

or issues that is to be investigated

with the goal of finding an

answer or solution.

34 Report writing?

Answer

① Communicating the Research finding

② Report is technical.

③ Historical document.

④ Academic purpose

35 Advantages of E-mail surveys?

Answer. ① Speed of distribution

② Lower distribution and processing cost.

③ More flexibility.

④ Comfortable level of response is high

36- Data collection techniques?

ANSWER. Every researcher collects data using one or more techniques. Techniques may be grouped into two categories.

(i) - Quantitative - collecting data in form of numbers -

(ii) - Qualitative - collecting data in form of words or pictures.

37. Probability Non Probability

There are several ways to taking a sample

In probability sampling every element in the population has a known non-zero probability of selection. Its types are	In non-probability sampling the probability of chosen is unknown. Its types are
--	---

- | | |
|-----------------------------|-------------------------|
| ① Simple Random Sample. | ① Convenience Sampling. |
| ② Systematic Random Sample. | ② Purposive Sampling. |
| ③ Stratified Random Sample. | ③ Quota Sampling. |
| ④ | ④ Snowball Sampling. |
| | ⑤ Sequential Sampling. |

38 Unit of Analysis?

ANSWER. Unit of Analysis refers to the level of aggregation of the data collected during the subsequent data analysis stage.

39- Pilot testing ?

Answer. Pilot testing is to identify potential problems with the methods logistics and the Questionnaire.

Pilot testing also called pre-testing means small scale trial run.

Pilot testing involves the use of small number of respondents to test the appropriateness of the questions and their comprehension.

40- Ethnography and Ethno-methodology?

Answer. Two modern extensions of field research are :

Ethnography or Ethno methodology - build on social constructionist perspective.

Ethnography comes from cultural anthropology. Ethno means people or ethnography refers to describing something. Ethnography means describing the culture. It is understanding of culture of people from their own perspective.

Ethno-methodology implies how people create reality and how they interpret it. It examines ordinary social interaction.

41. Survey interview.

Field interview

- | | |
|--|---|
| ① Clear start and clear end. | ① Start and end are not clear. |
| ② Same standard Questions are asked of all respondents in the same sequence. | ② Questions are 2. specific people. 01 Situations. |
| ③ interviewers appears neutral at all times. | ③ interviewer shows interest in responses |
| ④ It is always with one respondent alone. | ④ It is like a friendly conversational exchange but with more interviewer Questions |

42. Telephone interviewing?

Answer. Telephone interview is mainstay of commercial survey research. Data collected by telephone may be comparable to that collected in personal interviews.

Benefits are: ① High Speed. ② Save cost. ③ Callbacks.

43. Self Administered ?

Answer. Self - Administered Questionnaire has become ubiquitous in modern living. Self Administered mail Questionnaire are

delivered not only through postal services, but also via fax and courier service.

Advantages are:-

- ① Geographical flexibility.
- ② Sample Accessibility.
- ③ Save time.
- Save cost.

44 Role of pilot testing?

Answer. Role of pilot testing:

Pilot testing is rehearsal of your research study. Allowing you to ^{test your} research ~~your~~ with small number of test participants.

Pilot study is small scale study. Establish the validity of research based upon the expert knowledge of other researchers.

45- Experimental Research?

Answer. Builds on the principal of positive approaches more directly than do the other researches.

Its purpose is to control the research situation so that causal relationship among variables may be

evaluated.

46. Pre testing or post test?

Answer. Pre testing is the stage in survey research when survey questions and questionnaires are tested on members of target population/study population, to evaluate the reliability and validity of the survey instruments prior to their final distribution.

A post test design is an experiment where measurements are taken both before and after a treatment.

47. Human interactive Media. | Electronic interactive Media.

① Personal forms of communication. | Allows researcher to reach the large audience.

② Human being direct a message to and interact with another individual. | Controlled by user themselves. Actively involved in 2-way communication.

48- Population

- ① Aggregation of specified study elements.
- ② Pool of All Available elements is called population.

Target population.

Researcher wants to focus on is called target population.
Also called survey population.

49 Historical data researchers evidence ?

- Answer
- ① Primary sources.
 - ② Secondary sources.
 - ③ Running records.
 - ④ Recollections.

50 Purpose of FGD? formal

Answer The purpose of focus group is to obtain in-depth information on concepts

STA630 – Research Methods

Important Questions+Glossary

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Questions

1) Is there any single definition of Research?

There is no single definition of research. Several authors have described the word “research” in their own words. According to Uma Sekaran, “research is a process of finding solutions to a problem after a thorough study and analysis of the situational factors.” In words of Neuman “research is a way of going about finding answers to questions.”

2) What is meant by Research?

Research is an organized and systematic way of finding answers to questions by gathering the needed information.

3) What is the purpose and importance of Research?

Research is important because it 1-Liberates you from the false claims of others. 2-Allows you to learn more about your particular area of interest. As an educated person, knowing about research will give you the ability to: 1-Accurately and systematically collect data that will answer questions 2-Analyze and attack problems; and 3-Communicate and justify both your approach to a problem and your conclusions.

4) Define the Scientific method of Research?

The Scientific method of Research is a procedure to produce knowledge and to discover the principles & laws in this universe.

5) What is the positivism approach of Research?

Positivism is characterized mainly by an insistence that science can only deal with observable entities known directly to experience. The positivist aims to construct general laws, or theories, which express relationships between phenomena. Observation and experiment will then show whether the phenomena do or do not fit the theory; explanation of phenomena consists in showing that they are instances of the general laws or regularities. The process of positivism approach is like that Observation & Repetition & Re-observation By repeating the observation researchers want to be positive and definite.

6) What are the special features or steps of Scientific Method?

1-Empirical: To observe the phenomenon 2- Verifiable: To confirm or verify the observation 3- Cumulative: Knowledge grows 4- Deterministic: To explain why things happen 5- Ethical

and Ideological neutrality 6- Statistical Generalisability: To analyze the information by using statistical techniques 7- Rationalism: To apply logical rules Any one who is doing the research by these steps is said to do the scientific research.

7) What does mean by Exploratory Research?

It is the initial research that is conducted to define the nature of the problem and to explore new topics. It develops techniques and a sense of direction for future research.

8) Differentiate b/w Basic and Applied Research?

Basic Research: The basic research is pure and fundamental research which expands basic knowledge about the problem. The basic research lays down the foundation for the applied research. For example, basic science investigations for answers to questions such as: How did the universe begin? What are protons, neutrons, and electrons composed of? Applied Research: Applied research solves specific problems through the implementation of different policies. Applied research is designed to solve practical problems of the modern world, rather than to acquire knowledge. For example, applied researchers may investigate ways to improve agricultural crop production, to treat or cure a specific disease and to improve the energy efficiency of homes, offices, or modes of transportation.

9) Differentiate b/w Cross Sectional and Longitudinal Research?

The cross sectional research observes the phenomena at one point in time e-g snapshot study. While longitudinal research examine the phenomena at more than one time e-g time series study, panel study, cohort study.

10) What are the quantitative techniques for conducting a research project?

The quantitative techniques for conducting a research project are as follows 1- Experimental Techniques 2-Conducting Surveys 3-Content analysis 4-Use of statistical techniques

11) What are the qualitative techniques for conducting a research project?

The qualitative techniques for conducting a research project are as follows 1-Field Research 2-Case Study 3-Focus Group Discussions

12) What are the aims that researchers set when they conduct research?

The aims of the researchers are 1-To expand the Knowledge 2-To discover the truth 3-To make prediction

13) What does mean by Theory?

Theory is a systematic and general attempt to explain something.

14) Define Variable?

Anything that can take different values or varies over the period of time is called variable. Variation can be in quantity, intensity, type and amount.

15) What are the different types of Variables?

There are different types of variables. 1-Continuous and Discrete Variables 2-Dependent and Independent Variables 3-Moderating Variable 4-Intervening Variable 5-Extraneous Variables

16) Define the Hypothesis?

Hypothesis is a testable proposition. It is the statement about the variables which has been tested whether it is true or not.

17) What are different types of hypothesis?

There are two types of hypothesis. 1-Descriptive Hypothesis: It is also called univariate hypothesis. It is the proposition that states the existence, size, form or distribution of some variable. 2-Relational Hypothesis: It describes the r/s b/w two or more variables. 3-Null Hypothesis: It describes that the r/s b/w the variables is zero (no r/s). It is denoted by H_0 . 4-Alternative Hypothesis: It describes that there exist some r/s b/w the variables. It is alternative to H_0 and is denoted by H_1 . 5-Research Hypothesis: This is the hypothesis formulated by the researchers. This tells the r/s as well as the nature of the r/s b/w the variables.

18) Differentiate b/w Directional and non-directional Hypothesis?

In directional hypothesis, the direction of r/s is specified e-g positive, negative, less than, more than etc. while in non-directional hypothesis, the direction of r/s is not specified e-g money supply and prices are related but it is not mentioned that related in which direction.

19) What is meant by Descriptive Research?

The research that is conducted to describe the characteristics of the problem under study is named as descriptive research. This type of research provides basic background information.

20) Define the explanatory Research?

Explanatory research is also known as causal research because it identifies the cause and effect r/s among different factors. This research determines the accuracy of the theory.

Glossary

1) Action Research

That research role where the researcher and their role is involved in the research study and is interventionist. Here research is joined with action in order to plan, implement and monitor change.

2) Alternative hypothesis

Alternative hypothesis is a statement that expresses a relationship between two variables or indicating differences among groups. It is opposite to null hypothesis. It is usually denoted by H_1 .

3) Applied Research

Applied research solves specific problems and it is conducted when decision must be made for particular real-life problem. For example, a particular product may not be selling well and manager wants to know reasons in order to take actions.

4) Basic Research

Basic research is conducted to generate a body of knowledge which may not be immediately utilized. It is also known as Fundamental research or Pure research.

5) Bibliography

Bibliography is the list of all works that are relevant to the topic of interested research, arranged in alphabetical order of the last name of authors.

6) Broad Problem Area

Broad problem area refers to the entire situation where one sees a possible need of doing research and solving a problem. It is a general area of study from where research process starts.

7) Case study

Case study is in-depth, contextual analysis of a situation or number of specific situations i.e. cases. Case study is a problem solving technique that explores problem in as rich detail as possible. Example can be case study of highly successful organization.

8) Categorical variable

Categorical variable is a variable that has distinct value and which cannot be divided into fractions. For example, sex; male and female. Categorical variables are also known as discontinuous or classifactory variables.

9) Causal study

The study in which researcher wants to explain the cause of one or more problems is called causal study. For example, does smoking cause cancer? The given research question is causal study.

10) Causality

Relationship where a change in one variable produces a change in another variable. One variable affects, influences, or determines some other variable.

11) Chi-square test statistic

Measure of the goodness of fit between the numbers observed in the sample and the numbers we should have seen in the sample, given the null hypothesis is true.

12) Close ended question

Closed ended questions are the questions in which respondent has to choose one of the given options in a question that specifically reflects his/her answer. Close ended questions help respondents to make quick decisions to choose among several alternatives

13) Cluster sampling

Cluster sampling is form of probability sampling which is done when groups of elements have heterogeneity among members within.

14) Coding

Coding is assigning of numbers or symbols to the data before entering it into the computer. The assigned numbers and symbols make statistical analysis easy.

15) Communication research

An umbrella term for market research, social research, media research and audience research combined.

16) Computer assisted personal interviewing (CAPI)

Computer assisted personal interviewing (CAPI) is another type of Computer assisted interviews in which responses can be directly entered into the computer using microcomputers. It reduces the error and cost

17) Computer assisted telephone interviewing (CATI)

Computer assisted telephone interviewing (CATI) is a technique to collect data in which responses gathered from telephone interviews are directly entered into the computer by using advance software. Voice recording system assists CATI programs to record interviewees' responses.

18) Concept

An idea aimed at satisfying consumer wants and needs.

19) Confounds or confounding variables

Extraneous causal factors (variables) that can possibly affect the dependent variable and, therefore, must be controlled.

20) Content Validity

Establishes the representative sampling of a whole set of items that measures a concept, and reflects how well the dimensions and elements thereof are delineated.

21) Data analysis

Data analysis is the process to analyze data after collecting it. During the analysis stage, several interrelated procedures are performed to summarize and rearrange the data.

22) Data interpretation

Data interpretation refers to making inferences pertinent to the meaning and implications of the research investigation and drawing conclusions.

23) Data presentation

Data presentation plays an important role to interpret data and deduce results by using tables and graphs.

24) Data transformation

Data transformation is the process of changing data from their original form to a format that is more suitable to perform a data analysis that will achieve the research objectives.

25) Deductive reasoning

A logical process of developing specific predictions (hypotheses) from general principles. This type of reasoning moves from the general to the particular.

26) Dependent variable

In experimental research, the dependent variable is the variable presumed within the research hypothesis to depend on (be caused by) another variable (the independent variable); it is sometimes referred to as the outcome variable.

27) Descriptive Research

Research designed to provide a summary of some aspects of the environment when the hypotheses are tentative and speculative in nature.

28) Descriptive statistics

Statistical methods used to describe or summarize data collected from a specific sample (e.g. mean, median, mode, range, standard deviation).

29) Desk research

Research done by summarizing published sources – a form of secondary research.

30) Directional hypothesis (or one-tailed hypothesis)

A hypothesis that makes a specific prediction about the nature and direction of the relationship between the independent and dependent variables.

31) Double barreled question 18Aug2023

Double barreled question consists of two or more questions joined together. It makes the respondent's answer ambiguous. For example, if asked, "Are you satisfied and happy with your supervisor?"

32) Educational Research

It is the formal, systematic application of the scientific method to the study of educational problems.

33) Empirical methods

Research methods and data gathering techniques supported by measurable evidence, not opinion or speculation.

34) Ethics

The rules of conduct recognized in respect of a particular class of human actions or a particular group, culture, etc.

35) Ethnographic research

The collection of extensive narrative data on many variables over an extended period of time in a naturalistic setting in order to gain insights not possible using other types of research. For ethnographic research, observations are done at particular points of time. Data would include observations, recording and interpreting what is seen.

36) Ex post facto designs

Ex post facto designs are designs in which there is no manipulation of independent variable in lab or field setting rather researchers only report what has happened or what is happening.

37) Experimental design

A study in which there is random assignment of subjects to different groups so that there are no major differences between the control and comparison groups. An experimental study controls all the conditions: •The environment •Sample assignment (random assignment in two groups) •Treatment/Intervention (Practice/Program)

38) Experimental group

In experimental research the group of subjects who receive the experimental treatment or intervention under investigation.

39) Experimental research

A research methodology used to establish cause-and-effect relationships between the independent and dependent variables by means of manipulation of variables, control and randomization. A true experiment involves the random allocation of participants to experimental and control groups, manipulation of the independent variable, and the introduction of a control group (for comparison purposes). Participants are assessed before and after the manipulation of the independent variable in order to assess its effect on the dependent variable (the outcome).

40) Extraneous variable

A variable that interferes with the relationship between the independent and dependent variables and which therefore needs to be controlled for in some way.

41) Filter question

A question that is asked to determine which branching question, if any, will be asked.

42) Focus Group discussion

A generic type of qualitative research in which a small group of people provide information by discussing a topic. Some variants are the consensus group, focus group, nominal group, co-discovery conference.

43) Frequency distribution

The number of respondents who choose each alternative answer as well as the percentage and cumulative percentage of respondents who answer.

44) Generalizability

Generalizability refers to make research findings universally applicable. It describes the extent to which research findings can be applied to settings other than that in which they were originally tested.

45) Goodness of measure

Goodness of measure describes that tool developed to measure a particular concept should indeed measure it accurately and actually. The three major criteria for evaluating the tool are; reliability, validity and sensitivity.

46) Historical research

The systematic collection and evaluation of data related to past occurrences in order to describe causes, effects, trends of those events which may help to explain present events and anticipate future events. Data is often archival-including newspaper clippings, photographs, etc.-and may include interviews.

47) Hypothesis

A statement that predicts the relationship between variables (specifically the relationship between the independent and dependent variables). A hypothesis may be directional or non-directional.

48) Hypothesis Test

A formal test, using probability and sampling distributions to decide which of two conflicting hypotheses should be accepted. Sometimes referred to as a significance test (where the degree of error is emphasized).

49) Independent variable

The variable (or antecedent) that is assumed to cause or influence the dependent variable(s) or outcome. The independent variable is manipulated in experimental research to observe its effect on the dependent variable(s). It is sometimes referred to as the treatment variable.

50) Inductive reasoning

A logical process of reasoning used to develop more general rules from specific observations; this type of reasoning moves from the specific to the more generalized.

51) Inductivism/Inductive Reasoning

The process by which the truth of a proposition is made more probable by the accumulation of confirming evidence, a common pattern in sociological and scientific research.

52) Industrial research

Market research in which the respondents are organizations, not consumers. Much the same as business to business research.

53) Inferential statistics

Statistics that allow a researcher to make inferences about whether relationships observed in a sample are likely to occur in the wider population from which that sample was drawn. Inferential statistics use logic and mathematical processes in order to test hypotheses relating to a specific population based on data gathered from a sample of the population of interest.

54) Interval data

Measurements that allow us to tell how far apart two or more objects are with respect to attributes and consequently to compare the difference between the numbers assigned. Because the interval data lack a natural or absolute origin, the absolute magnitude of the numbers cannot be compared.

55) Judgemental sampling

Studies in which respondents are selected because it is expected that they are representative of the population of interest and/or meet the specific needs of the research study.

56) Likert scale

Scaling technique where a large number of items that are statements of belief or intention are generated. Each item is judged according to whether it reflects a favourable or unfavourable attitude toward the object in question. Respondents are then asked to rate the attitude towards the object on each scale item in terms of a five-point category labelled scale.

57) Loaded questions

Questions that suggest what the answer should be or indicate the researcher's position on the issue under study.

58) Mail surveys

Data-collection method that involves sending out a fairly structured questionnaire to a sample of respondents.

59) Mall-intercept personal survey

Survey method using a central-location test facility at a shopping mall; respondents are intercepted while they are shopping.

60) Maturation

Threat to internal validity; refers to changes in biology or psychology of the respondent that occur over time and can affect the dependent variable irrespective of the treatment conditions.

61) Mortality

Threat to internal validity; refers to the differential loss (refusal to continue in the experiment) of respondents from the treatment condition groups.

62) Nominal data

Measurement in which the numbers assigned allow us to place an object in one and only one of a set of mutually exclusive and collectively exhaustive classes with no implied ordering.

- 63)** Non-directional hypothesis (or two-tailed hypothesis)
A hypothesis that does not specify the nature and direction of the relationship between the independent and dependent variables.
- 64)** Non-probability samples
Form of sampling where there is no way of determining exactly what the chance is of selecting any particular element or sampling unit into the sample.
- 65)** Non-response error
Error that occurs because not all of the respondents included in the sample respond; in other words with non-response, the mean true value (on the variable of interest) of the sample respondents who do respond may be different from the entire sample's true mean value (on the variable of interest).
- 66)** Non-sampling error
Degree to which the mean observed value (on the variable of interest) for the respondent of a particular sample agrees with the mean true value of the particular sample of respondents (on the variable of interest).
- 67)** Null hypothesis
A statement that there is no relationship between the independent and dependent variables and that any relationship observed is due to chance or fluctuations in sampling.
- 68)** Observational methods
Observation of behaviour, directly or indirectly, by human or mechanical methods.
- 69)** Ordinal data
Measurement in which the response alternatives define an ordered sequence so that the choice listed first is less (greater) than the second, the second less (greater) than the third, and so forth. The numbers assigned do not reflect the magnitude of an attribute possessed by an object.
- 70)** Participative Research/Participant Observation
Conscious and systematic sharing, in so far as circumstances will permit in the life activities, and on occasion, in the interests and affects of a group of persons.
- 71)** Positivism
This paradigm assumes that human behavior is determined by external stimuli and that it is possible to use the principles and methods traditionally employed by the natural scientist to observe and measure social phenomena.
- 72)** Positivism/Positivist Approach
A doctrine in the philosophy of science, positivism is characterized mainly by an insistence that science can only deal with observable entities known directly to experience. The positivist aims to construct general laws, or theories, which express relationships between phenomena. Observation and experiment will then show whether the phenomena do or do not fit the theory; explanation of phenomena consists in showing that they are instances of the general laws or regularities.
- 73)** Primary Data
Data collected to address a specific research objective (as opposed to secondary data).
- 74)** Probability sampling designs
Samples drawn in such a way that each member of the population has a known, non-zero chance of being selected.

- 75)** Qualitative data
Information gathered in narrative (nonnumeric) form (e.g. a transcript of an unstructured interview).
- 76)** Qualitative Research
Qualitative research generally examines people's words or actions in narrative or descriptive ways more closely representing the experiences of the people involved, it focuses on understandings, meanings and it takes seriously lay accounts and concepts.
- 77)** Quantitative data
Information gathered in numeric form.
- 78)** Quantitative Research
Quantitative research generally converts observations into discrete units that can be compared to other units by statistical analysis, it focuses on explanation, prediction and proof.
- 79)** Quota sampling
Design that involves selecting specific numbers of respondents who possess certain characteristics known, or presumed, to affect the subject of the research study.
- 80)** Random sampling error
Error caused when the selected sample is an imperfect representation of the overall population; therefore, the true mean value for the particular sample of respondents (on the variable of interest) differs from the true mean value for the overall population (on the variable of interest).
- 81)** Ratio data
Measurements that have the same properties as interval scales, but which also have a natural or absolute origin.
- 82)** Reflexive Research
Reflexive research involves exploring the ability of people to collectively or individually transform social structures. The concepts of 'habitus' and 'structured improvisation' capture the intersection of structure and agency, & society and the individual.
- 83)** Reliability
The degree to which the test consistently measures what is suppose to measure.
- 84)** Replicability
A study is considered replicable if it can be repeated by another researcher. You can determine if a study is replicable by asking the following questions: •Is the description of the design detailed so that duplication of the study is possible? •Are the findings clearly described and reported? •Are the findings presented clearly and objectively? •Can the findings be enhanced with additional research?
- 85)** Research method
Specific procedures used to gather and analyze research data.
- 86)** Research methodology
Different approaches to systematic inquiry developed within a particular paradigm with associated epistemological assumptions (e.g. experimental research, grounded theory, ethno methodology).
- 87)** Research problem
A formal statement of question or hypothesis that implies empirical investigation.

88) Research question

A clear statement in the form of a question of the specific issue that a researcher wishes to answer in order to address a research problem. A research problem is an issue that lends itself to systematic investigation through research.

89) Response error

Error that occurs because respondents (who do respond) may give inaccurate answers, or a respondent's answers may be misrecorded.

90) Response rates

The total number of respondents sent questionnaires who complete and return them, expressed as a percentage.

91) Sample

The group of subjects from which data are collected. In other words, the group of people who participate in the study and whose data is used by the researcher.

92) Sampling frame

An explicit list of individuals or households that are eligible for inclusion in the sample.

93) Sampling interval

Computed by taking n/N together with r , the first chosen element to be included in the sample, determines which elements will be included in the sample.

94) Secondary data

Data that have been collected for another project and have already been published. Sources can be in-house or external.

95) Selection bias

Threat to internal validity; refers to the improper assignment of respondents to treatment conditions.

96) Snowball design

Sample formed by having each respondent, after being interviewed, identify others who belong to the target population of interest.

97) Social research

Uses the same techniques as market research, but focuses less on business and more on public issues. See also audience research.

98) Statistical significant

A research term used to indicate whether the results from a study are very probably not due to chance. Take, for example, a study that reports it found that student test scores improved after a reading intervention program was initiated at a school, and that these findings are "statistically significant." That means that the researcher is reasonably certain that student test scores did actually improve over time. If research findings are not statistically significant, any increase reported may be due to chance, rather than a result of the intervention.

99) Stratified sampling

Design that involves partitioning the entire population of elements into sub-population, called strata, and then selecting elements separately from each sub-population.

100) Survey

A method of gathering information from a number of individuals (the respondents, who collectively form a sample) in order to learn something about a larger target population from which the sample was drawn.

101) Survey Research

This is the systematic gathering of information about individuals and collectivities, using interview, questionnaire or response methods to elicit information directly, and interpreting the resulting data by means of statistical analysis. It provides an alternative to the experimental method or participant observation and is widely used in both qualitative and quantitative research.

102) Syndicated research

Research originated by a research company, with data sold to anybody who is interested – unlike an ad hoc survey, which is a one-off survey for an individual client.

103) Systematic methods

Data collection methods that are consistent and scientific–i.e. that follows the scientific method, used for collecting observations.

104) Systematic sampling

Design whereby the target sample is generated by picking an arbitrary starting point (in a list) and then picking every nth element in succession from a list.

105) Target population

Set of people, products, firms, markets, etc., that contain the information that is of interest to the researcher.

106) Telephone surveys

Survey that involves phoning a sample of respondents drawn from an eligible population and asking them a series of questions.

107) Theory

In its most general sense a theory describes or explains something. Often it is the answer to 'what', 'when', 'how' or 'why' questions.

108) Validity

Refers to the best approximation to truth or falsity of a proposition, including propositions concerning cause-and-effect relationships.

109) Variable

An attribute or characteristics of a person or an object that takes on different values (i.e. that varies) within the population under investigation (e.g. age, weight, pulse rate).

110) Variance

A measure of dispersion or variability (spread), calculated by squaring the value of the standard deviation.

STAT630 FINAL TERM SOLVED CURRENT PAPERS (SPRING 2020)

1. What is meant by probing?

Probing means the verbal prompts made by field worker when the respondent must be motivated to communicate his or her answer or to enlarge on, clarify or explain an answer. Probing may be needed for two types of situations.

- i. First, it is necessary when the respondent must be motivated to enlarge on, clarify, or explain his or her answer. The interviewer must encourage the respondent to clarify or expand on answers by providing a stimulus that will not suggest the interviewer's own ideas. The ability to probe with neutral stimuli is the mark of an experienced interviewer.
- ii. Second, probing may be necessary in situations in which the respondent begins to ramble or lose track of the question. In such cases the respondent must be led to focus on specific content of the interview and to avoid irrelevant and unnecessary information. Probing is also needed when the interviewer recognizes an irrelevant or inaccurate answer.

2. Criteria for selecting sampling technique.

Sampling Criteria is one must remember that two costs are involved in a sampling analysis viz., the cost of collecting the data and the cost of an incorrect inference resulting from the data. Researcher must keep in view the two causes of incorrect inferences viz., systematic bias and sampling error.

3. Probability and non probability. 15Aug2023

Probability:

In probability sampling every element in the population has a known nonzero probability of selection. The simple random is the best known probability sample, in which each member of the population has an equal probability of being selected. Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalizability.

Non Probability:

In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment.

It should be noted that there are no appropriate statistical techniques for measuring random sampling error from a non-probability sample. Thus projecting the data beyond the sample is statistically inappropriate. When time or other factors, rather than generalizability, become critical, non-probability sampling is generally used.

4. **5 suggestions for better interview**

- Know the culture of the people in advance.
- Appearance – wear acceptable dress.
- Pleasantness and flexibility.
- Carry the letter of authority.
- Establish credibility and rapport.
- Motivating individuals to respond.
- Familiarity with the questionnaire.
- Following the question wording/ question order
- Recording responses exactly.
- Probing for responses.
- Closing the interview.
- No false promises. Also don't burn your bridges.
- Edit the questionnaire in the first available opportunity.

5. **Frequency distribution table of 2 variables**

Several useful techniques for displaying data are in use. The easiest way to describe the numerical data of one variable is with a frequency distribution. It can be used with nominal-, ordinal-, interval-, or ratio-level data and takes many forms.

Note: Solution depends upon the given data

6. **Types of Quasi experimental designs.**

One-shot Case Study Design: 15Aug2023

Also called the one-group posttest-only design, the one-shot case study design has only one group, a treatment, and a posttest. Because it is only one group, there is no random assignment. For example, a researcher shows a group of students a horror film, and then measures their attitude with a questionnaire. A weakness of this design is that it is difficult to say for sure that the treatment caused the dependent variable. If subjects were the same before and after the treatment, the researcher would not know it.

One Group Pretest-posttest Design:

This design has one group, a pretest, a treatment, and a posttest. It lacks a control group and random assignment. Continuing with the previous example, the researcher gives a group of students an attitude questionnaire to complete, shows a horror film, then has them complete the same questionnaire second time. This is an improvement over the one-shot case

study because the researcher measures the dependent variable both before and after the treatment. But it lacks the control group for comparison. The researcher cannot know whether something other than the treatment occurred between the pretest and the posttest to cause the outcome.

Two Groups Posttest-only Design:

It has two groups, a random assignment of subjects, a posttest, and a treatment. It has all parts of the classical design except a pretest. Continuing with our previous example, the researcher forms two groups through randomization process. He shows group a horror film to one group i.e. the experimental group. The other group is not shown any film. Both groups then complete the questionnaire. The random assignment reduces the chance that the groups differed before the treatment, but without a pretest, a researcher cannot be as certain that the groups began the same on the dependent variable.

7. Limitations of Simple Random sampling.

Random sampling does not guarantee that every random sample perfectly represents the population. Instead, it means that most random samples will be close to the population most of the time, and that one can calculate the probability of a particular sample being inaccurate.

8. Survey interview

1. It has clear beginning and end.
2. The same standard questions are in which asked of all respondents in the same sequence.
3. The interviewer appears neutral all the time.
4. The interviewer asks questions, and the respondent answers questions
5. It is almost always with one respondent alone.
6. It has a professional tone and businesslike focus, diversions are ignored,
7. Closed-ended questions are common, with rare probes\
8. The interviewer alone controls the pace and direction of interview
9. The social context in which the interview occurs is ignored and assumed to make little difference
10. The interviewer attempts to mold the communication pattern into a standard framework.

9. Field research adopt different social role why?

Field researchers study people in a location or setting. It has been used to study entire communities. Field research has a distinct set of methodologies. Field researchers directly observe and interact with community members in natural settings to get inside their perspectives. They embrace an activist or social constructionist perspective on social life. They do not see people as a neutral medium through which social forces

operate, nor do they see social meanings as something “out there” to observe. Instead they believe that people create and define the social world through their interactions. Human experiences are filtered through a subjective sense of reality, which affects how people see and act on events. Thus they replace the positivist emphasis on “objective facts” with a focus on the everyday, face-to-face social processes of negotiation, discussion, and bargaining to construct social meaning.

10. Content analysis superior over survey experiment explain 5 points

1. **Can use large sample size:** Larger the sample closer the results to the population. In experimentation as well as in survey research there could be limitations due to the availability of the subjects or of the resources but in document analysis the researcher could increase the sample and can have more confidence in generalization. Let us assume that a researcher is studying the matrimonial advertisements in the newspapers over a long period of time, there should be no problem in drawing a sample as large as several thousand or more.
 2. **Spontaneity:** The spontaneous actions or feelings can be recorded when they occurred rather than at a time specified by the researcher. If the respondent was keeping a diary, he or she may have been recording spontaneous feelings about a subject whenever he or she was inspired to do so. The contents of such personal recording could be analyzed later on. Q33 Aug2023
 3. **Confessions:** A person may be more likely to confess in a document, particularly one to be read only after his or her death, than in an interview or mailed questionnaire study. Thus a study of documents such as diaries, posthumously published autobiographies, and suicide notes may be the only way to obtain such information.
 4. **Relatively low cost:** Although the cost of documentary analysis can vary widely depending on the type of document analyzed, how widely documents are dispersed, and how far one must travel to gain access to them, documentary analysis can be inexpensive compared to large-scale surveys. Many a time's documents are gathered together in a centralized location such as library where the researcher can study them for only the cost of travel to the repository.
 5. **High quality:** Although documents vary tremendously in quality, many documents, such as newspaper columns, are written by skilled commentators and may be more valuable than, for example, poorly written responses to mailed questionnaires.
11. Ex post factor experimental better than one shot experiment design, explain.
12. Population v/s targeted population define

Population:

A population is the theoretically specified aggregation of study elements. It is translating the abstract concept into workable concept. For example, let us look at the study of “college students.”

Targeted population:

Out of the conceptual variations what exactly the researcher wants to focus on. This may also be called a target population. Target population is the complete group of specific population elements relevant to the research project. Target population may also be called survey population i.e. that aggregation of elements from which the survey sample is actually selected.

13. Ways to reduce Interview bias**1. Standardize the Process**

Before you even start interviewing people, create a standard list of questions you plan to ask. Of course, there may be aspects of each candidate's background you want to learn more about or get specific details on, but the more you can level the playing field, the more you'll give everyone an equal chance of impressing you.

Similarly, attempt to have the interview in the same place for all your candidates. Don't let some come in person to the interview and others Skype in. Keep the process as similar as possible each and every time.

Good to note: The time and order of interviews matters as well—but more often than not, there's less you can do about it.

2. Take Good Notes

Human memory is notoriously unreliable. So, rather than relying on your recall abilities and opening yourself up to unintentional biases, try to take brief notes as candidates respond to questions. Ideally, write down as much of the interviewee's exact response as possible without your own interpretations. Then immediately post-interview, jot down your thoughts on the interviewee before you get too scattered and you're forced to trust your own unpredictable memory.

3. Use a Rubric

Ability and fit are both hard to quantify, but you're better off at least trying than avoiding it all together. Ideally, prior to the interview stage of the hiring

process, create a rubric for what you're seeking in the new hire. Include qualifications like specific skills and experiences, soft skills like communication and teamwork, and cultural fit with the company.

After interviewing all the candidates, select a numerical range and rate each person. Rubrics helps you avoid giving too much credit for one particular experience or qualification—it keeps things balanced.

4. Justify Your Decision

You might think a rubric is only marginally different than going with a gut feeling—and you would be right. Rubrics are only useful if you're able to justify your scores. Going through the process of reasoning out why you believe something is a huge help in trying to avoid all the subconscious business going on when interviewing and evaluating a job candidate. Get real evidence—such as the notes you took of the interviewee's responses—to back up your beliefs in order to avoid falling into the trap of cognitive biases.

5. Get Input From Others

Other people can have a huge influence on your decision. While you're going through your rubric and justifying your choices, it's best to do this alone to avoid the outside influence. But, once you're done with that, it's worth seeing what everyone else thought, too. Ideally, you want to receive feedback from others to add to the data you've already collected, not to *impact* the data you've collected

It's hard to admit that we can be biased even when we're trying not to be. But studies have shown that it takes more than intention to overcome them, it takes action to right them. For example, when symphonies introduced “blind auditions” by using opaque screens to hide the musician auditioning, the rate of women accepted into symphonies increased dramatically. In the end, the process might be more cumbersome, but it'll also be fairer—plus you'll be more likely to get the best candidate for the job.

14. Reactive, non-reactive research,

Experiments and surveys research are both **reactive**; that is, the people being studied are aware of the fact that they are being studied.

In non-reactive research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

15. content analysis,

18Aug2023

Content analysis is a technique for gathering and analyzing the content of a text. The content refers to words, meanings, pictures, symbols, ideas, themes, or any message that can be communicated. The text is anything written, visual, or spoken that serves as a medium of communication.

Possible artifacts for study could be books, newspaper or magazine articles, advertisements, poems, letters, laws, constitutions, dramas, speeches, official documents, films or videotapes, musical lyrics, photographs, articles of clothing, or works of arts. All these works may be called as documents.

The documents can be:

- Personal – letters, diary, autobiography.
- Non-personal – interoffice memos, official documents, proceedings of a meeting.
- Mass media – newspapers, magazines, fiction, films, songs, poems, works of arts.

Content analysis goes back nearly a century and is used in many fields – literature, history, journalism, political science, education, psychology, sociology, and so on. It is also called a study of communication, which means who says what, to whom, why, how, and with what effect.

In content analysis, the researcher uses objective and systematic counting and recording procedures to produce a quantitative description of the symbolic content in a text. It may also be called “textual coding.” There are qualitative versions of content analysis. The emphasis here is quantitative data about a text’s content.



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Content Analysis is Non-Reactive: It is non-reactive because the placing of words, messages, or symbols in a text to communicate to the reader or receiver occurs without influence from the researcher who analyzes its contents. There is no interaction between the researcher and the creator of the text under analysis.

Content analysis lets a researcher reveal the contents (i.e. messages, meanings, symbols, etc.) in a source of communication (i.e. a book, article, movie, etc.). It lets him/her probe into and discover content in a different way from ordinary way of reading a book or watching a television program.

With content analysis, a researcher can compare content across many texts and analyze it with quantitative techniques (table, charts). In addition, he or she can reveal aspects of the text’s content that are difficult to see. For example, you might watch television commercials and feel that women are

mostly portrayed working in the house, cooking food, using detergents, looking after children. Content analysis can document – in objective, quantitative terms – whether or not your vague feelings based on unsystematic observation are true. It yields repeatable, precise results about the text.

Content analysis involves random sampling, precise measurement, and operational definitions for abstract constructs. Coding turns aspects of content that represent variables into numbers. After a content analysis researcher gathers the data, he or she enters them into computers and analyzes them with statistics in the same way that an experiment or survey researcher would.

16. Loaded question,

Loaded questions suggest a socially desirable answer or are emotionally charged. “Should the city government repair all the broken streets?” Most of the people are going to agree with this question simply because this is highly socially desirable. A question which may be challenging the traditionally set patterns of behavior may be considered as emotionally charged i.e. it is loaded with such material which may hit the emotions of the people. Look at some behaviors associated with masculinity in Pakistani society.

17. Historical comparative research limitations, 5 marks

18. researchers ny teachers k unsolved questions ko mid-value satisfaction dy di, apna stand point btaen, 5 marks ka

19. Cause and effect relationship, 3 marks

A cause-effect relationship is a relationship in which one event causes another to happen. ... Whenever the cause occurs, the effect must also occur. There must not be another factor that can explain the relationship between the cause and effect.

20. Guideline of interviewing

- **Have integrity and be honest.** This is the cornerstone of all professional inquiry, regardless of its purpose.
- **Have patience and tact.** Interviewers ask for information from people they do not know. Thus all the rules of human relations that apply to inquiry situations – patience, tact, courtesy – apply “in spades” to interviewing.
- **Have attention to accuracy and detail.** Among the greatest interviewing “sins” are inaccuracy and superficiality, for the professional analyst can misunderstand, and in turn mislead, a client. Do not record the answer

unless you fully understand it yourself. Probe for clarification and to get detailed answers.

- **Exhibit a real interest in** the inquiry at hand, but keep your opinions to yourself. Impartiality is imperative.
- **Be a good listener**. Some interviewers talk too much, wasting time when respondents could be supplying more pertinent facts or opinions on the topic.
- **Keep the inquiry and respondents' responses confidential**. Do not discuss the studies you are doing with relatives, friends, or associates. Never quote one respondent's opinion to another.
- **Respect others' rights**. Survey research depends on the goodwill of others to provide information. There should be no coercion. Impress on prospective respondents that their cooperation is important and valuable.

21. Differentiate bw ethnography and ethno-methodology

Ethnography comes from cultural anthropology. Ethno means people or a folk distinct by their culture and graphy refers to describing something. Thus ethnography means describing a culture and understanding another way of life from the native point of view. It is just an understanding the culture of people from their own perspective.

Ethno-methodology implies how people create reality and how they interpret it. Ethnomethodologists examine ordinary social interaction in great detail to identify the rules for constructing social reality and common sense, how these rules are applied, and how new rules are created. They try to figure out how certain meanings are attached to a reality.

22. What are the limitations of tri-variate tables? Explain any two

Trivariate tables have three limitations.

1. First, they are difficult to interpret if a control variable has more than four categories.
2. Second, control variables can be at any level of measurement, but interval or ratio control variables must be grouped (i.e. converted to an ordinal level), and how cases are grouped can affect the interpretation of effects.
3. Finally, the total number of cases is a limiting factor because the cases are divided among cells in partials. The number of cells in the partials equals the number of cells in the bivariate relationship multiplied by the number of categories in the control variables.

For example if the control variable has three categories, and a bivariate table has 12 cells, the partials have $3 \times 12 = 36$ cells. An average of five cases per cell is recommended, so the researcher will need $5 \times 36 = 180$ cases at minimum.

23. What do you mean by double barreled questions? Give example

A double barreled question consists of two or more questions joined together. It makes the respondent's answer ambiguous. For example, if asked, "Does this company have pension and health insurance benefits?" a respondent at the company with health insurance benefits only might answer either yes or no. The response has an ambiguous meaning and the researcher cannot be certain of the respondent's intentions.

24. Enlist the steps in field research. Q22 Aug2023

1. Prepare yourself, read the literature and defocus.
2. Select a site and gain access.
3. Enter the field and establish social relations with members.
4. Adopt a social role, learn the ropes, and get along with members.
5. Watch, listen, and collect quality data.
6. Begin to analyze data, generate and evaluate working hypothesis.
7. Focus on specific aspects of the setting and use theoretical sampling.
8. Conduct field interviews with member informants.
9. Disengage and physically leave the setting.
10. Complete the analysis and write the report.

25. Identify type of sampling in which researcher gets information from Salma and Salma also refer other respondents?

26. Write dilemmas of field research

The direct personal involvement of a field researcher in the social lives of other people raises many ethical dilemmas. The dilemmas arise when the researcher is alone in the field and has little time to make a moral decision. Although he or she may be aware of general ethical issues before entering the field, they arise unexpectedly in the course of observing and interacting in the field. Let us look at some of these dilemmas:

- i. **Deception:** Deception arises in several ways in field research: The research may be covert; or may assume a false role, name, or identity; or may mislead members in some way. The most hotly debated of the ethical issues arising from deception is that of covert versus overt field research. Some support it and see it as necessary for entering into and aiming a full knowledge of many areas of social life. Others oppose it and argue that it undermines a trust between researchers and society. Although its moral status is questionable, there are some field sites or activities that can only be studied covertly. One may have to look into the cost and benefit equation; where the researcher is the best judge.

Covert research is never preferable and never easier than overt research because of the difficulties of maintaining a front and the constant fear of getting caught.

- ii. **Confidentiality:** A researcher learns intimate knowledge that is given in confidence. He or she has a moral obligation to uphold the confidentiality of data. This includes keeping information confidential from others in the field and disguising members' names in field notes. **Involvement with deviants:** Researchers who conduct research on deviants who engage in illegal behavior face additional dilemmas. They know of and are sometimes involved in illegal activity. They might be getting 'guilty knowledge.' Such knowledge is of interest not only to law enforcement officials but also to other deviants. The researcher faces a dilemma of building trust and rapport with the deviants, yet not becoming so involved as to violate his or her basic personal moral standards. Usually, the researcher makes an explicit arrangement with the deviant members.
- iii. **The powerful:** Field researchers tend to study those without power in society (e.g., street people, the poor, children, and lower level workers). Powerful elites can block access and have effective gatekeepers. Researchers are criticized for ignoring the powerful, and they are also criticized by the powerful for being biased toward the less powerful.
- iv. **Publishing field reports:** The intimate knowledge that a researcher obtains and reports creates a dilemma between the right of privacy and the right to know. A researcher does not publicize member secrets, violate privacy, or harm reputations. Yet if he or she cannot publish anything that might offend or harm someone, some of what the researcher learned will remain hidden, and it may be difficult for others to believe the report if critical details are omitted. Some researchers suggest asking members of the group under study to look at a report to verify its accuracy and to approve of their portrayal in print. For marginal groups (addicts), this may not be possible, but the researchers must always respect member privacy. On the other hand, censorship or self-censorship can be a danger. A compromise position is that truthful but unflattering material may be published only if it is essential to the researchers' larger arguments.

27. What are the consequences if the interview is not properly terminate?

Fieldworkers should not close the interview before all the information has been secured. The interviewer whose departure is hasty will not be able to

record those spontaneous comments respondents sometimes offer after all formal questions have been asked. Avoiding hasty departures is also a matter of courtesy.

28. In a study researcher observe one group and measure the variables before and after changes. What type of quasi experimental design it is and how?

It is One Group Pretest-posttest Design. This design has one group, a pretest, a treatment, and a posttest.

29. Explain the specification pattern with the help of example?

It occurs when one partial replicate the initial bivariate relationship but other partials do not. For example, we find a strong (negative) bivariate relationship between age of the respondents and attitude towards women empowerment. We control for gender and discover the relationship holds only for males (i.e. the strong negative relationship was in the partial for males, but not for females). This is specification because a researcher can specify the category of the control variable in which the initial relationship persists.

30. The researchers want to study the crisis team management cell. Scenario base question. Identify unit of analysis and unit of observation also provide explanation.

31. What are filtered question?

Filter question is that question which screens out respondents not qualified to answer a second question.

Asking a question that doesn't apply to the respondent or that the respondent is not qualified to answer may be irritating or may cause a biased response. Including filter question minimizes the chance of asking questions that are inapplicable

For example the researcher wants to know about the bringing up of one's children. "How much time do you spend playing games with your oldest child?" What if the respondent is unmarried? Even if the respondent is married but does not have the child. In both these situations the question is inapplicable to him/her. Before this question the person may put a filter question whether or not the respondent is married.

32. Different between field editing and in home editing

Field Editing

In large projects, field supervisors are often responsible for conducting preliminary field edits. The purpose of field editing the same day as the interview is to catch technical omissions (such as a blank page), check legibility

of the handwriting, and clarify responses that are logically or conceptually inconsistent. If a daily field editing is conducted, a supervisor who edits completed questionnaires will frequently be able to question the interviewers, who may be able to recall the interview well enough to correct any problems. The number of “no answers,” or incomplete answers can be reduced with a rapid follow-up simulated by a field edit. The daily edit also allows fieldworkers to re-contact the respondent to fill in omissions before the situation has changed. The field edit may also indicate the need for further training of interviewers.

In-House Editing

Although almost simultaneous editing in the field is highly desirable, in many situations (particularly with mail questionnaires), early reviewing of the data is not possible. In-house editing rigorously investigates the results of data collection. The in-house editor’s task is to ensure that inconsistent or contradictory responses are adjusted and that answers will not be a problem for coders and keyboard punchers.

33. Which type of research is used to compare Indian and Pakistani culture?

Historical comparative research: Historical-comparative research is a collection of techniques and approaches. It is a distinct type of research that puts historical time and /or cross-cultural variation at the center of research – that is, which treats what is studied as part of the flow of history and situated in cultural context.

34. Data transformation with examples

Data transformation is the process of changing data from their original form to a format that is more suitable to perform a data analysis that will achieve the research objectives.

Researchers often modify the values of a scalar data or create new variables.

For example many researchers believe that response bias will be less if interviewers ask consumers for their year of birth rather than their age, even though the objective of the data analysis is to investigate respondents’ age in years. This does not present a problem for the research analyst, because a simple data transformation is possible. The raw data coded at birth year can be easily transformed to age by subtracting the birth year from the current year.

35. Pre test and post test

Pretests and Posttests: Frequently a researcher measures the dependent variable more than once during an experiment. The pretest is the measurement of the dependent variable prior to the introduction of the

treatment. The posttest is the measurement of the dependent variable after the treatment has been introduced into the experimental situation

36. Differentiate between parameters and statistics

Parameter

A parameter is the summary description of a given variable in a population. The mean income of all families in a city and the age distribution of the city's population are parameters. An important part portion of survey research involves the estimation of population parameters on the basis of sample observation.

Statistic

A statistic is the summary description of a given variable in a survey sample. Thus the mean income computed from the survey sample and the age distribution of that sample are statistics. Sample statistics are used to make estimates of the population parameters.

37. Sampling and its types

Sampling

Sampling is the process of using a small number of items or parts of a larger population to make conclusions about the whole population. It enables the researchers to estimate unknown characteristics of the population.

38. Types of probability and non probability/ Types of sampling

Types of Non probability

- Convenience sampling
- Purposive Sampling
- Quota Sampling
- Snowball Sampling
- Sequential Sampling
- Theoretical Sampling

Types of probability Q28 Aug2023

- Simple Random Sample
- Systematic Random Sample
- Stratified Random Sample
- Cluster Sampling
- Double Sampling

39. Parts of interview

Stages of an Interview

1. Making Initial Contact and Securing the Interview

The interview proceeds through stages, beginning with introduction and entry. Interviewers are trained to make appropriate opening remarks that will convince the person that his or her cooperation is important.

By indicating that telephone call is a long distance, interviewers attempt to capitalize on the fact that most people feel a long distance call is something special, unusual, or important.

Giving one's personal name personalizes the call.

Personal interviewers may carry a letter of identification that will indicate that the study is bona fide research project and not a salesman's call. The name of the research agency is used to assure the respondent that the caller is trustworthy

2. Asking the Questions

The purpose of the interview is, of course, to have the interviewer ask questions and record the respondent's answers. Training in the art of stating questions can be extremely beneficial, because interviewer bias can be a source of considerable error in survey research.

- There are five major principles for asking questions:
 - Ask the questions exactly as they are worded in the questionnaire.
 - Read each question very slowly.
 - Ask the question in the order in which they are presented in the questionnaire.
 - Repeat questions that are misunderstood or misinterpreted.

Although interviewers are generally trained in these procedures, when working in the field many interviewers do not follow them exactly. Do not take shortcuts when the task becomes monotonous. Interviewers may shorten questions or rephrase unconsciously when they rely on their memory of the question rather than reading the question as it is worded. If the respondents do not understand a question, they will usually ask for some clarification.

The recommended procedure is to repeat the question, or if the respondent does not understand a word, the interviewer should respond with "just whatever it means to you."

Often the respondents volunteer information relevant to a question that is supposed to be asked at a later point in the research. In this situation the response should be recorded under the question that deals specifically with that subject. Then rather than skip the question that was answered out of sequence, the interviewers should be trained to say something like "We have briefly discussed this, but let me ask you" By asking every question, the interviewer can be sure that complete answers are recorded.

3. Probing

Probing means the verbal prompts made by field worker when the respondent must be motivated to communicate his or her answer or to enlarge on, clarify or explain an answer.

Probing may be needed for two types of situations. First, it is necessary when the respondent must be motivated to enlarge on, clarify, or explain his or her answer. The interviewer must encourage the respondent to clarify or expand on answers by providing a stimulus that will not suggest the interviewer's own ideas. The ability to probe with neutral stimuli is the mark of an experienced interviewer. Second, probing may be necessary in situations in which the respondent begins to ramble or lose track of the question. In such cases the respondent must be led to focus on specific content of the interview and to avoid irrelevant and unnecessary information. Probing is also needed when the interviewer recognizes an irrelevant or inaccurate answer.

The interviewer has several possible probing tactics to choose from, depending on the situation:

- Repetition of the question. The respondent who remains completely silent may not have understood the question or may not have decided how to answer it. Mere repetition may encourage the respondent to answer in such cases. For example, if the question is "What is there that you do not like about your supervisor?" and the respondent does not answer, the interviewer may probe: "just to check, is there anything you do not like about your supervisor?"
- An expectant pause. If the interviewer believes the respondent has more to say, the "silent probe," accompanied by an expectant look may motivate the respondent to gather his/her thoughts and give a complete response.
- Repetition of the respondent's reply. As the interviewer records the response, he or she may repeat the respondent's reply verbatim. This may stimulate the respondent to expand on the answer.
- Neutral questions or comments. Asking neutral question may indicate the type of information that the interviewer is seeking. For example, if the interviewer believes that the respondent's motives should be clarified, he or she might ask, "Why do you feel that way?" If the interviewer feels that there is a need to clarify a word or phrase, then he/she might ask, "What do you mean by _____?"

4. Recording the Responses

The rules for recording responses to closed ended questions vary with the specific question.

The general rule, however, is to place a check in the box that correctly reflects the respondent's answer.

The general instructions for recording answers to open-ended response questions is to record the answer verbatim, a task that is difficult for most people. Some of the suggestions are:

- ☐ **Record responses during the interview.**

- ☐ **Use the respondent's own words.**
- ☐ **Do not summarize or paraphrase the respondent's answer.**
- ☐ **Include everything that pertains to the question objectives.**
- ☐ **Include all your probes.**

5. Terminating the Interview

Fieldworkers should not close the interview before all the information has been secured. The interviewer whose departure is hasty will not be able to record those spontaneous comments respondents sometimes offer after all formal questions have been asked. Avoiding hasty departures is also a matter of courtesy.

Fieldworkers should also answer to the best of their ability any questions the respondent has concerning the nature and purpose of the study. Always leave by observing the local cultural customs. "Don't burn your bridges." Because the fieldworker may be required to re-interview the respondent at some future time, he or she should leave the respondent with positive feeling about having cooperated in a worthwhile undertaking. It is extremely important to thank the respondent for his or her cooperation. The interviewer then goes to a quiet and private place to edit the questionnaire and record other details such as the date, time, and place of interview; a thumbnail sketch of the respondent and interview situation, the respondent's attitude; and any unusual circumstances. The interviewer also records personal feelings and anything that was suspected.

Experimental research

Experimental research builds on the principles of positivist approach more directly than do the other research techniques. Researchers in the natural sciences (e.g. chemistry and physics), related applied fields (e.g. engineering, agriculture, and medicines) and the social sciences conduct experiments.

The purpose of experimental research is to allow the researcher to control the research situation so that causal relationships among variables may be evaluated. The experimenter, therefore, manipulates a single variable in an investigation and holds constant all other, extraneous variables.

40. Exclusive and exhaustive

Exhaustive – that is, coding categories should be provided for all subjects or objects or responses. With a categorical variable such as sex, making categories exhaustive is not a problem. However, when the response represents a small number of subjects or when the responses might be categorized in a class not typically found, there may be a problem.

Mutually exclusive and independent:

This means that there should be no overlap between the categories, to ensure that a subject or response can be placed in only one category. This frequently requires that an “other” code category be included, so that the categories are all inclusive and mutually exclusive. For example, managerial span of control might be coded 1, 2, 3, 4, and “5 or more.” The “5 or more” category ensures everyone a place in a category.

41. Participants-non participants observations

In **participant observation** because of the much familiarity with the events, sometimes the **observer** does not realize the significance of same events and neglects them. But in **non- participant observation** the researcher does not even miss a minute thing

42. Double blinding

The double-blind experiment is designed to control experimenter expectancy. In it, people who have direct contact with subjects do not know the details of the hypothesis or the treatment. It is double blind because both the subjects and those in contact with them are blind to details of the experiment. For example a researcher wants to see if new drug is effective. Using capsules of three colors – green, yellow, and pink – the researcher puts the new drug in the yellow capsule, puts an old drug in the pink one, and take the green capsule a placebo – a false treatment that appears to be real (e.g., a sugar capsule without any physical effects). Assistants who give the capsules and record the effects do not know which color contains the new drug. Only another person who does not deal with subjects directly knows which colored capsule contains the drug and examines the results.

43. Code book

A book that identifies each variable in a study and its position in the data matrix is called as code book. The book is used to identify a variable’s description, code name, and field. Here is a sample:

Q/V No.	Field/ col. No.	Code values
• --	1-5 Study number •	- 6 City
•		1 = Lahore
•		2 = Rawalpindi
•		3 = Karachi
•	7 -9	Interview No.
• Sex	10	1 = Male
•		2 = Female
• Age	11-12	Actual

44. In which situation researcher can select historical comparative research (3marks)

Historical-comparative research is suited for examining the combination of societal factors that produce a specific outcome (e.g., civil war). It is also appropriate for comparing entire social system to see what is common across societies and what is unique, and to study long term change. An H-C researcher may apply a theory to specific cases to illustrate its usefulness. And he or she compares the same social processes and concepts in different cultural or historical contexts

45. Three parts of appendix (3marks)**Appended Parts**

Appendix: The appendix presents the “too ...” material. Any material that is too technical or too detailed to go to the body should appear in appendix. This includes materials of interest only to some readers, or subsidiary materials not directly related to the objectives. Some examples of appendix material are data collection forms (instruments), detailed calculations, discussions of highly technical questions, detailed or comprehensive tables of results, and a bibliography (if appropriate).

46. Difference between purposive and sequential sampling (3marks) 16Aug2023

Sequential sampling is similar to purposive sampling with one difference. In purposive sampling, the researcher tries to find as many relevant cases as possible, until time, financial resources, or his or her energy is exhausted. The principle is to get every possible case. In sequential sampling, a researcher continues to gather cases until the amount of new information or diversity is filled. The principle is to gather cases until a saturation point is reached.

47. Disadvantages of content analysis (5marks) Q24 Aug2023**Disadvantages**

1. Bias: Many documents used in research were not originally intended for research purposes. The various goals and purposes for which documents are written can bias them in various ways.

For example, personal documents such as confessional articles or autobiographies are often written by famous people or people who had some unusual experience such as having been a witness to a specific event. While often providing a unique and valuable research data, these documents usually are written for the purpose of making money. Thus they tend to exaggerate and even fabricate to make good story. They also tend to

include those events that make the author look good and exclude those that cast him or her in a negative light.

2. Selective survival: Since documents are usually written on paper, they do not withstand the elements well unless care is taken to preserve them. Thus while documents written by famous people are likely to be preserved, day-to-day documents such as letters and diaries written by common people tend either to be destroyed or to be placed in storage and thus become inaccessible. It is relatively rare for common documents that are not about some events of immediate interest to the researcher (e.g., suicide) and not about famous occurrence or by some famous person to be gathered together in a public repository that is accessible to researchers.

3. Incompleteness: Many documents provide incomplete account to the researcher who has had no prior experience with or knowledge of the events or behaviour discussed. A problem with many personal documents such as letters and diaries is that they were not written for research purposes but were designed to be private or even secret. Both these kinds of documents often assume specific knowledge that researcher unfamiliar with certain events will not possess. Diaries are probably the worst in this respect, since they are usually written to be read only by the author and can consist more of “soul searching” and confession than of description. Letters tend to be little more complete, since they are addressed to a second person. Since many letters assume a great amount of prior information on the part of the reader.

4. Lack of availability of documents: In addition to the bias, incompleteness, and selective survival of documents, there are many areas of study for which no documents are available. In many cases information simply was never recorded. In other cases it was recorded, but the documents remain secret or classified, or have been destroyed.

5. Sampling bias: One of the problems of bias occurs because persons of lower educational or income levels are less likely to be represented in the sampling frames. The problem of sampling bias by educational level is more acute for document study than for survey research.

It is a safe generalization that a poorly educated people are much less likely than well educated people to write documents.

6. Limited to verbal behavior: By definition, documents provide information only about respondent’s verbal behavior, and provide no direct information on the respondent’s nonverbal behavior, either that of the document’s author or other characters in the document.

7. Lack of standardized format: Documents differ quite widely in regard to their standardization of format. Some documents such as newspapers appear frequently in a standard format. Large dailies always contain such

standard components as editorial page, business page, sports page, and weather report. Standardization facilitates comparison across time for the same newspapers and comparison across different newspapers at one point in time. However, many other documents, particularly personal documents have no standard format. Comparison is difficult or impossible, since valuable information contained in the document at one point in time may be entirely lacking in an earlier or later documents.

8. Coding difficulties: For a number of reasons, including differences in purpose for which the documents were written, differences in content or subject matter, lack of standardization, and differences in length and format, coding is one of the most difficult tasks facing the content analyst. Documents are generally written arrangements, rather than numbers are quite difficult to quantify. Thus analysis of documents is similar to analysis of open-ended survey questions.

9. Data must be adjusted for comparability over time: Although one of the advantages of document study is that comparisons may be made over a long period of time, since external events cause changes so drastic that even if a common unit of measure is used for the entire period, the value of this unit may have changed so much over time that comparisons are misleading unless corrections are made. Look at the change in measuring distance, temperature, currency, and even literacy in Pakistan.

48. How to eliminate selection bias during experiment

Selection bias is the threat that subjects will not form equivalent groups. It is a problem in design without random assignment, hence differential selection of the subjects for the comparison groups. It occurs when subjects in one experimental group have a characteristic that affects the dependent variable. For example, in an experiment on physical aggressiveness, the experimental group unintentionally contains subjects who are sportsmen, whereas the control group is made up of musicians, chess players, and painters.

49. About facilitator in focus group discussion

Functions of the Facilitator:

The facilitator should not act as an expert on the topic. His or her role is to stimulate and support discussion. He should perform the following functions:

1. Introduce the session:

He or she should introduce himself/herself as facilitator and introduce the recorder. Introduce the participants by name or ask them to introduce themselves (or develop some new interesting way of introduction). Put the participants at ease and explain the purpose of the FGD, the kind of

information needed, and how the information will be used (e.g., for planning of a health program, an education program, et.).

2. Encourage discussion:

The facilitator should be enthusiastic, lively, and humorous and show his/her interest in the group's ideas. Formulate questions and encourage as many participants as possible to express their views. Remember there is no "right" or "wrong" answers. Facilitator should react neutrally to both verbal and nonverbal responses.

**3. Encourage involvement: **

Avoid a question and answer session. Some useful techniques include: asking for clarification (can you tell me more?); reorienting the discussion when it goes off the track (saying: wait, how does this relate to the issue? Using one participant's remarks to direct a question to another); bringing in reluctant participants (Using person's name, requesting his/her opinion, making more frequent eye contact to encourage his participation); dealing with dominant participants (avoiding eye contact or turning slightly away to discourage the person from speaking, or thanking the person and changing the subject).

4. Avoid being placed in the role of expert:

When the facilitator is asked for his/her opinion by a respondent, remember that he or she is not there to educate or inform. Direct the question back to the group by saying: "What do you think?" "What would you do?" Set aside time, if necessary, after the session to give participants the information they have asked. Do not try to give comments on everything that is being said. Do not feel you have to say something during every pause in the discussion. Wait a little and see what happens.

5. Control the timing of the meeting but unobtrusively:

Listen carefully and move the discussion from topic to topic. Subtly control the time allocated to various topics so as to maintain interest. If the participants spontaneously jump from one topic to the other, let the discussion continue for a while because useful additional information may surface and then summarize the points brought up and reorient the discussion.

6. Take time at the end of the meeting to summarize, check for agreement and thank the participants:

Summarize the main issues brought up, check whether all agree and ask for additional comments. Thank the participants and let them know that their ideas had been a valuable contribution and will be used for planning the proposed research/intervention/or whatever the purpose of FGD was. Listen to the additional comments made after the meeting. Sometime some valuable information surfaces, which otherwise may remain hidden.

Important topics

Interview

Validity

Central tendency

Sampling measures

Two questions was based on scenario

Element

Population

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1. Historical comparative research limitation
2. Content analysis
3. Simple random sampling and stratified random sampling

Simple Random Sample 15Aug2023

The simple random sample is both the easiest random sample to understand and the one on which other types are modeled. In simple random sampling, a research develops an accurate sampling frame, selects elements from sampling frame according to mathematically random procedure, then locates the exact element that was selected for inclusion in the sample.

Stratified Random Sample

When the population is heterogeneous, the use of simple random sample may not produce representative sample. Some of the bigger strata may get over representation while some of the small ones may entirely be eliminated. Look at the variables that are likely to affect the results, and stratify the population in such a way that each stratum becomes homogeneous group within itself. Then draw the required sample by using the table of random numbers. Hence in stratified random sampling a sub-sample is drawn utilizing simple random sampling within each stratum. (Randomization is not done for quota sampling).

4. Why we prefer similar population?
5. Why editing is required by field worked during data collection process?
6. Experimental research is a qualitative research which build on interpretivists principles agree or not agree?
7. Researchers who conduct research on deviant, engaged in illegal behavior face an additional dilemma mentioned the dilemma for the field researcher?
8. How we make questions for field work research

Data collection in a sponsored study is rarely carried out by the person who designs the research project. For a student, depending upon the sample size, data collection is usually done by the student himself/herself. However, the data collection stage is crucial, because the research project is no better than the data collected in the field. Therefore, it is important that the research administrator selects capable people who may be entrusted to collect the data. There are Field Interviewing Services, who specialize in data gathering. These agencies perform door-to-door surveys, central location telephone interviewing, and other forms of fieldwork for fee.

9. Double blind experiment

The double-blind experiment is designed to control experimenter expectancy. In it, people who have direct contact with subjects do not know the details of the hypothesis or the treatment. It is double blind because both the subjects and those in contact with them are blind to details of the experiment. For example a researcher wants to see if new drug is effective.

10. Maturity effect and test effect sampling techniques

Cause and effect relationship can also be contaminated by the effects of the passage of time – another uncontrollable variable. Such contamination is called maturation effect. The maturation effects are a function of the processes – biological and psychological – operating within the subjects as a result of the passage of time. Examples of maturation processes could include growing older, getting tired, feeling hungry, and getting bored. In other words there could be maturation effect on the dependent variable purely because of the passage of time

11. Parts of the table

- Give each table a number.
- Give each table a title, which names variables and provides background information
- Label the row and columns variables and give name to each of the variable categories.
- Include the totals of the columns and rows. These are called marginal. They equal the univariate frequency distribution for the variable.
- Each number or place that corresponds to the intersection of a category for each variable is a cell of a table.
- The numbers with the labeled variable categories and the totals are called the body of the table.
- If there is missing information, report the number of missing cases near the table to account for all original cases.
- Researchers convert raw count tables into percentages to see bivariate relationship. There are three ways to percentage a table: by row, by column, and for the total. The first two are often used and show relationship.

12. Probability or Non-probability

Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalizability. When time or other factors, rather than generalizability become critical then non-probability sampling is generally used.

In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment.

13. Pretest posttest

Frequently a researcher measures the dependent variable more than once during an experiment. The pretest is the measurement of the dependent variable prior to the introduction of the treatment. The posttest is the measurement of the dependent variable after the treatment has been introduced into the experimental situation.

14. Why we use Sample population researchers.

15. Random sampling

Researchers often use random sampling in content analysis. First, they define the population and the sampling element. For example, the population might be all words, all sentences, all paragraphs, or all articles in certain type of documents over a period of specified period. Likewise, it could include each conversation, situation, scene, or episode of a certain type of television program over a specified time period.

16. Experiment research

- a. Experimental research builds on the principles of positivist approach more

directly than do the other research techniques.

- b. The purpose of experimental research is to allow the researcher to *control* the research situation so that *causal* relationships among variables may be evaluated.
- c. The goal of the experimental design is the confidence that it gives the researcher that his experimental treatment is the cause of the effect he measures.
- d. Experimentation is a research design that allows evaluation of causal relationship among variables.

17. How construct frequency distribution in two variable.

18. Difference in Internal or external validity

Internal validity:

Internal validity is high in the laboratory experiment, reason being the control over all the confounding factors. External validity (generalizability) is not sure because of the effect of variety of factors. Field experiments have more external validity but less internal validity because it is closer to the real situations.

External Validity:

Even if the researcher eliminates all concerns for internal validity, external validity remains a potential problem. External validity is the ability to generalize experimental findings to real life situations. Without external validity, the findings are of little use for both basic and applied research i.e. we shall not be able to develop any theories that could be applicable to similar other situations.

19. Ethical dilemmas

The direct personal involvement of a field researcher in the social lives of other people raises many ethical dilemmas. The dilemmas arise when the researcher is alone in the field and has little time to make a moral decision. Although he or she may be aware of general ethical issues before entering the field, they arise unexpectedly in the course of observing and interacting in the field. Let us look at some of these dilemmas:

Deception:

Deception arises in several ways in field research. The research may be covert; or may assume a false role, name, or identity; or may mislead members in some way.

Confidentiality:

A researcher learns intimate knowledge that is given in confidence. He or she has a moral obligation to uphold the confidentiality of data. This includes keeping information confidential from others in the field and disguising members' names in field notes.

Involvement with deviants:

Researchers who conduct research on deviants who engage in illegal behavior face additional dilemmas. They know of and are sometimes involved in illegal activity. They might be getting „guilty knowledge.“ such knowledge is of interest not only to law enforcement officials but also to other deviants.

The powerful:

Field researchers tend to study those without power in society (e.g., street people, the poor, children, and lower level workers). Powerful elites can block access and have effective

gatekeepers. Researchers are criticized for ignoring the powerful, and they are also criticized by the powerful for being biased toward the less powerful.

Publishing field reports:

The intimate knowledge that a researcher obtains and reports creates a dilemma between the right of privacy and the right to know. A researcher does not publicize member secrets, violate privacy, or harm reputations.

20. Difference b/w simple and systematic random sample

Simple Random Sample

The simple random sample is both the easiest random sample to understand and the one on which other types are modeled. In simple random sampling, a research develops an accurate sampling frame, selects elements from sampling frame according to mathematically random procedure, then locates the exact element that was selected for inclusion in the sample.

Systematic Random Sample Q19 Aug2023

Systematic random sampling is simple random sampling with a short cut for random selection. Again, the first step is to number each element in the sampling frame. Instead of using a list of random numbers, researcher calculates a sampling interval, and the interval becomes his or her own quasi random selection method. The sampling interval (i.e. 1 in K where K is some number) tells the researcher how to select elements from a sampling frame by skipping elements in the frame before one for the sample.

21. Two basic rules for coding construction.

There are two basic rules for code construction. First, the coding categories should be exhaustive – that is, coding categories should be provided for all subjects or objects or responses. With a categorical variable such as sex, making categories exhaustive is not a problem. However, when the response represents a small number of subjects or when the responses might be categorized in a class not typically found, there may be a problem. Second, the coding categories should also be mutually exclusive and independent. This means that there should be no overlap between the categories, to ensure that a subject or response can be placed in only one category. This frequently requires that an “other” code category be included, so that the categories are all inclusive and mutually exclusive

22. Importance of instrumentation in internal validity 15Aug2023

Instrumentation effects are yet another source of threat to internal validity. These might arise because of a change in the measuring instrument between pretest and posttest, and not because of the instrument’s differential impact at the end. For example, in a weightless experiment, the springs on the scale weaken during the experiment, giving lower readings in the posttest. A change in the wording of questions (may be done to avoid testing effects), change in interviewers, or change in other procedures to measure the dependent variable can cause instrumentation effect. Performance of the subjects measured by the units of output in the pretest, but when measuring the output in posttest the researcher measures it by the number of units rejected, and the amount of resources expended to produce the units.

23. What type of research to be used for economic study of Pakistan?

24. Explain the role of pilot testing in development of research instruments.

It is important to pilot test the instrument to ensure that the questions are understood by the respondents and there are no problems with the wording or measurement. Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their

comprehension. Usually, the draft questionnaire is tried out on a group that is selected on a convenience and that is similar in makeup to the one that ultimately will be sampled. Making a mistake with 25 or so subjects can avert the disaster of administering an invalid questionnaire to several hundred individuals. Hence the main purpose of pilot testing is to identify potential problems with the methods, logistics, and the questionnaire.

25. A researcher takes data of 15 persons first. And then again take interview of 5 persons among these 15.. What type of sampling is used?

26. Pilot testing long 16Aug2023

Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire.

Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension. Usually, the draft questionnaire is tried out on a group that is selected on a convenience and that is similar in makeup to the one that ultimately will be sampled.

What aspects to be evaluated during pilot testing?

i. Reactions of Respondents:

The reactions of the respondents can be looked at from different angles. The researcher may be familiar with the local culture; still getting the first-hand experience is always useful. Going to the field, contacting the people, and their reactions to the different aspects of research may be a learning experience.

- Acceptability of the questions asked.
- Willingness of the respondents to co-operate.
- ii. Discovering errors in the instrument:
 - Do the tools provide you the information? Reliability. Suitability for analysis
 - Time taken/needed to interview/conduct the observation.
 - If there is any need to revise the format of the tool
- iii. Sampling procedure can be checked:
 - The extent to which instructions given are followed
 - How much time is needed to locate the respondents?
- iv. Staffing and activities of research team can be checked:
 - How successful the training has been?
 - What is the work output of each member?
 - How well the research team works together?
 - Is the logistical support adequate?
- v. Procedure for data processing and analysis can be evaluated:
 - Make dummy tables. S

27. FGD long

In research, focus group discussions (FGD) are a more formal way of getting groups of people to discuss selected issues. A focus group discussion is a group discussion of 6-12 persons guided by a facilitator, during which group members talk freely and spontaneously about a certain topic. There may be some disagreement about the exact number of participants in the discussion, as one comes across variations in numbers (6 to10, 6 to12, 6 to15, 8 to10, 5 to 7) in different books on research methods.

Uses of Focus Group Discussions

- The primary advantage of focus groups is its ability to quickly and inexpensively grasp the core issues of the topic.
- As part of exploratory research, focus group discussions help the researcher to focus on the issue and develop relevant research hypotheses.
- Focus group discussion is an excellent design to get information from non-literates.
- Focus groups discussions are a good means to discover attitudes and opinions that might not be revealed through surveys.
- Focus group discussions are well accepted in the folk communities, as this form of communication already exists whereby the local communities try to sort out controversial issues.
- Focus group discussions generate new ideas, questions about the issues under consideration. It may be called serendipity (surprise ideas). It is more often the case in a group than in an individual interview that some idea will drop out of the blue.
- Focus group discussions can supplement the quantitative information on community knowledge, attitude, and practice (KAP), which may have already been collected through survey research.
- Focus group discussions are highly flexible with respect to topic, number of participants, time schedule, location, and logistics of discussion.
- Focus group discussions provide a direct link between the researcher and the population under study.
- For some researchers, focus group discussions may be a fun. They enjoy discussing the issues directly with the relevant population.

28. **Field research** Q23 Aug2023

Observation studies can be participant or non-participant. In participant observation the researcher directly observes and participates in small scale social settings in the present time. Such a study is also referred to as **field research**, ethnography, or anthropological study. Here the researchers:

- Study people in their natural settings, or in situ.
- Study people by directly interacting with them.
- Gain an understanding of the social world and make theoretical statements about members' perspective.

Field researchers study people in a location or setting. It has been used to study entire communities. Field research has a distinct set of methodologies. Field researchers directly observe and interact with community members in natural settings to get inside their perspectives

29. **Two situations**

Situation 1: when you are going to a company and ask them to talk about the features of the mobile you used r like...

Situation 2: when you are visiting the automobile survey and assessing its methods...

Tell about both situations which type of research type is used in it?

30. **Non-reactive research**

Experiments and surveys research are both reactive; that is, the people being studied are aware of the fact that they are being studied. In non-reactive research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

31. Specification Pattern

Is the next easiest pattern. It occurs when one partial replicate the initial bivariate relationship but other partials do not.

The interpretation pattern

Describes the situation in which the control variable intervenes between the original independent variable and the dependent variables.

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Mostly paper conceptual

CONDUCTS THE FIELDWORK

Data collection in a sponsored study is rarely carried out by the person who designs the research project. For a student, depending upon the sample size, data collection is usually done by the student himself/herself. However, the data collection stage is crucial, because the research project is no better than the data collected in the field. Therefore, it is important that the research administrator selects capable people who may be entrusted to collect the data. There are **Field Interviewing Services**, who specialize in data gathering. These agencies perform door-to-door surveys, central location telephone interviewing, and other forms of fieldwork for fee.

NON-REACTIVE RESEARCH

Experiments and surveys research are both reactive; that is, the people being studied are aware of the fact that they are being studied. In non-reactive research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

PROBABILITY AND NON-PROBABILITY SAMPLING

There are several alternative ways of taking a sample. The major alternative sampling plans may be grouped into probability techniques and non-probability techniques. In probability sampling every element in the population has a *known nonzero probability* of selection. The simple random is the best known probability sample, in which each member of the population has an equal probability of being selected. Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalizability. When time or other factors, rather than generalizability become critical then non-probability sampling is generally used. In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment. It should be noted that there are no appropriate statistical techniques for measuring random sampling error from a non-probability sample. Thus projecting the data beyond the sample is statistically inappropriate.

Types of non-probability sampling

In non-probability sampling designs, the elements in the population do not have any probabilities attached to their being chosen as sample subjects. This means that the findings from the study of the sample cannot be confidently generalized to the population. Sometimes non-probability could be the only way to collect the data.

1. Convenience Sampling

2. Purposive Sampling

3. Quota Sampling
4. Snowball Sampling
5. Sequential Sampling
6. Theoretical Sampling.

The replication pattern is the easiest to understand. It is when the partials replicate or reproduce the same relationship that existed in the bivariate table before considering the control variable. It means that the control variable has no effect.

The specification pattern

Is the next easiest pattern. It occurs when one partial replicate the initial bivariate relationship but other partials do not.

The interpretation pattern

Describes the situation in which the control variable intervenes between the original independent variable and the dependent variables.

EXPERIMENTAL RESEARCH

- Experimental research builds on the principles of positivist approach more directly than do the other research techniques.
- The purpose of experimental research is to allow the researcher to *control* the research situation so that *causal* relationships among variables may be evaluated.
- The goal of the experimental design is the confidence that it gives the researcher that his experimental treatment is the cause of the effect he measures.
- Experimentation is a research design that allows evaluation of causal relationship among variables.

TYPES OF PROBABILITY SAMPLING and TYPES

Probability samples that rely on random processes require more work than nonrandom ones. A researcher must identify specific sampling elements (e.g. persons) to include in the sample. For example, if conducting a telephone survey, the researcher needs to try to reach the specific sampled person, by calling back several times, to get an accurate sample. Random samples are most likely to yield a sample that truly represents the population. In addition, random sampling lets a researcher statistically calculate the relationship between the sample and the population – that is the size of *sampling error*. A non-statistical definition of the sampling error is the deviation between sample result and a population parameter due to random process.

1. Simple Random Sample:
2. Systematic Random Sample:
3. Stratified Random Sample:
4. Multistage Sampling:
5. Cluster Sampling:
6. Double Sampling:

Cluster Sampling

- □ Purpose: sample economically and retaining probability sampling.

- ☐ Heterogeneity within clusters but homogeneity between clusters.
- ☐ Random selection of clusters.
- ☐ Random selection of elements within the selected clusters.
- ☐ Cluster sampling addresses two problems
- ☐ Researchers lack good sampling frame for dispersed population
- ☐ Reaching each sampled element is costly.

Coding

- Coding involves assigning numbers or other symbols –responses can be grouped into classes or categories
- Helps in efficient analysis of data.
- Keep the data in raw form so far it is possible. Computer can help in grouping.
- ABU SULTAN's SHORT-NOTES – RESEARCH METHODS (STA630) – LECTURE#23 to 37-----Page 18 of 34
- Codes allow data processing in a computer.
- Organize data into field, record and file.
- A field: collection of characters (single No., letter, symbol) representing single type of data.
- A record: collection of related fields.
- A file: collection of related records

Validity in Experiments

Experiments are judged by two measures; internal validity and external validity. The first, internal validity indicates whether the independent variable was the sole cause of the change in the dependent variable. It implies the researcher's ability to eliminate alternative explanations of the dependent variable. Variables, other than the treatment, that affect the dependent variable are threats to internal validity. They threaten the researcher's ability to say that the treatment was the true causal factor producing change in the dependent variable. The second measure, external validity, indicates the extent to which the results of the experiment are applicable in the real world. Internal validity is high in the laboratory experiment, reason being the control over all the confounding factors. External validity (generalizability) is not sure because of the effect of variety of factors. Field experiments have more external validity but less internal validity because it is closer to the real situations. Factors Affecting Internal Validity: There are eight major types of extraneous variables that may jeopardize internal validity:

1. History effect
2. Maturation effect
3. Testing effect
4. Instrumentation effect
5. Selection bias effect
6. Selection bias effect
7. Statistical regression
8. Mortality
9. Mechanical loss

INDEX CONSTRUCTION

An index is combination of items into a single numerical score. Various components or subgroups of a construct are each measured and then combined into one measure. There are many types of indexes. For example, if you take an exam with 25 questions, the total number of questions correct is a kind of index. It is a composite measure in which each question measures a small piece of knowledge, and all the questions scored correct or incorrect are totaled to produce a single measure. One way to demonstrate that indexes are not a very complicated is to use one. Answer yes or no to the seven questions that follow on the characteristics of an occupation.

PILOT TESTING OF THE Questionnaire

Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire. Conventional wisdom suggests that pre-testing not only is an established practice for discovering errors but also is useful for extra training the research team.

Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension. The main purpose of pilot testing is to identify potential problems with the methods, logistics, and the questionnaire.

WHAT ASPECTS TO BE EVALUATED DURING PILOT TESTING

1. Reactions of Respondents:
2. Discovering errors in the instrument:
3. Sampling procedure can be checked:
4. Staffing and activities of research team can be checked:
5. Procedure for data processing and analysis can be evaluated

Systematic Random Sample

- ☐ Short cut for simple random sampling.
- ☐ Number each element in the sampling frame.
- ☐ Instead of using the table of random numbers, researcher develops a system for the selection of elements.
- ☐ Calculate the sampling interval. Basis of random selection

Probing

Probing means the verbal prompts made by field worker when the respondent must be motivated to communicate his or her answer or to enlarge on, clarify or explain an answer. Probing may be needed for two types of situations. First, it is necessary when the respondent must be motivated to enlarge on, clarify, or explain his or her answer.

Snowball Sampling

Snowball sampling (also called network, chain referral, or reputational sampling) is a method for identifying and sampling (or selecting) cases in the network. It is based on an analogy to a snowball, which begins small but becomes larger as it is rolled on wet snow and picks up additional snow. It begins with one or a few people or cases and spreads out on the basis of links to the initial cases. This design has been found quite useful where respondents are difficult to

identify and are best located through referral networks. In the initial stage of snowball sampling, individuals are discovered and may or may not be selected through probability methods.

Reliability and Validity

Validity problems occur when the researcher's theoretical definition does not match that of the government agency or organization that collected the information. Official policies and procedures specify definitions for official statistics.

Reliability

Stability reliability problems develop when official definition or the method of collecting information changes over time. Official definitions of work injury, disability, unemployment, literacy, poverty, and the like change periodically. Even if the researcher learns of such changes, consistent measurement over time is impossible. Equivalence reliability can also be a problem.

Secondary Data

Secondary data refer to information gathered by someone other than the researcher conducting the present study. Secondary data are usually historical, already assembled and do not require access to respondents or subjects.

Factors Affecting Internal Validity

There are eight major types of extraneous variables that may jeopardize internal validity:

1. History effect
2. Maturation effect
3. Testing effect
4. Instrumentation effect
5. Selection bias effect
6. Selection bias effect
7. Statistical regression
8. Mortality
9. Mechanical loss

Types of Designs: Researchers combine parts of experiment (e.g. pretests, control groups, etc.) together into an experimental design. For example some designs lack pretests, some do not have control groups, and others have many experimental groups. Certain widely used standard designs have names. Classical Experimental Design: All designs are variations of the classical experimental design, which has random assignment of subjects, a pretest and a posttest, an experimental group, and a control group.

1. Quasi Experimental Designs
2. True Experimental Designs
3. Solomon's Four Group Design

Quasi-Experimental Designs

One-shot Case Study Design

Also called the one-group posttest-only design.

The one-shot case study design has only one group, a treatment, and a posttest. Because it is only one group, there is no random assignment.

One Group Pretest-posttest Design

This design has one group, a pretest, a treatment, and a posttest. It lacks a control group and random assignment.

Two Groups Posttest-only Design:

It has two groups, a random assignment of subjects, a posttest, and a treatment. It has all parts of the classical design except a pretest.

True Experimental Designs

Experimental designs, which have at least two groups, a random assignment of subjects to experimental and control groups, only experimental group is exposed to treatment, both groups record information before and after the treatment, are known as ex-post facto experimental designs.

Random Assignment

Random assignment is a method for assignment cases (e.g. individuals, organizations) to groups for the purpose of making comparisons. It is a way to divide or sort a collection of cases into two or more groups in order to increase one's confidence that the groups do not differ in a systematic way.

Parts of Experiments

We can divide the experiments into seven parts and for each part there is a term. Not all experiments have all these parts, and some have all seven parts plus others. The following seven usually make up a true experiment. 1. Treatment or independent variable. 2. Dependent variable. 3. Pretest. 4. Posttest. 5. Experimental group. 6. Control group. 7. Assignment of subjects.

BIVARIATE TABLES

The bivariate contingency table is widely used. The table is based on cross-tabulation (cross classification); that is the cases are organized in the table on the basis of two variables at the same time. A contingency table is formed by cross-tabulating the two or more variables.

PRINCIPLES OF INTERVIEWING —5

Have integrity and be honest. This is the cornerstone of all professional inquiry, regardless of its purpose.

1 Have patience and tact. Interviewers ask for information from people they do not know. Thus all the rules of human relations that apply to inquiry situations – patience, tact, courtesy – apply “in spades” to interviewing.

2 Have attention to accuracy and detail. Among the greatest interviewing “sins” are inaccuracy and superficiality, for the professional analyst can misunderstand, and in turn mislead, a client. Do not record the answer unless you fully understand it yourself. Probe for clarification and detailed full answers.

3 Exhibit a real interest in the inquiry at hand, but keep your opinions to yourself. Impartiality is imperative.

4 Be a good listener. Some interviewers talk too much, wasting time when respondents could be supplying more pertinent facts or opinions on the topic.

5 Keep the inquiry and respondents' responses confidential. Do not discuss the studies you are doing with relatives, friends, or associates. Never quote one respondent's opinion to another.

6 Respect others' rights. Survey research depends on the goodwill of others to provide information

Steps in Field Research

Naturalism and direct involvement mean that field research is more flexible or less structured than quantitative research. This makes it essential for a researcher to be well organized and prepared for the field. It also means that the steps of project are not entirely predetermined but serve as an approximate guide or road map. Here is just the listing of these steps.

1. Prepare yourself, read the literature and defocus.
2. Select a site and gain access.
3. Enter the field and establish social relations with members.
4. Adopt a social role, learn the ropes, and get along with members.
5. Watch, listen, and collect quality data.
6. Begin to analyze data, generate and evaluate working hypothesis.
7. Focus on specific aspects of the setting and use theoretical sampling.
8. Conduct field interviews with member informants.
9. Disengage and physically leave the setting.
10. Complete the analysis and write the report.

Quantitative approach

Positivist researchers reject the idea that there is a distinct H-C method. They measure variables, test hypotheses, analyze quantitative data, and replicate research to discover generalizable laws that hold across time and societies. They see no fundamental distinction between quantitative social research and historical-comparative research. They apply quantitative research techniques, with some minor adjustments, to study the past or other cultures

Qualitative approach

There are no easily listed steps to follow in the analysis of historical data. Max Weber used the German term *verstehen* –“understanding” – in reference to an essential quality of research in behavioral sciences. He meant that the researcher must be able to take on, mentally, the circumstances, views, and feelings of those being studied to interpret their actions appropriately. The historical-comparative researcher must find patterns among the voluminous details describing the subject matter of study.

Similarities to Field Research

Stratified Random Sample

When the population is heterogeneous, the use of simple random sample may not produce representative sample. Some of the bigger strata may get over representation while some of the small ones may entirely be eliminated. Look at the variables that are likely to affect the results, and stratify the population in such a way that each stratum becomes homogeneous group within itself. Then draw the required sample by using the table of random numbers. Hence in stratified random sampling a sub-sample is drawn utilizing simple random sampling within each stratum. (Randomization is not done for quota sampling)

TOOLS FOR DATA COLLECTION----5

Asslam O Alikum

STA630 Research Methods current paper subjective solved by Hira Al

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Best regard's

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If u like me than raise your hand with me

If not than raise ur standard

That's about me ... !

Paper # 01

Q1. Explanatory Research.

When we encounter an issue that is already known and have a description of it, we might begin to wonder *why* things are the way they are. The desire to know “why,” to explain, is the purpose of *explanatory research*. It builds on exploratory and descriptive research and goes on to identify the reasons for something that occurs. Explanatory research looks for causes and reasons. For example, a descriptive research may discover that 10 percent of the parents abuse their children, whereas the explanatory researcher is more interested in learning *why* parents abuse their children.

Goals of Explanatory Research

- Explain things not just reporting. Why?
- Which explanation is better.
- Determine the accuracy of theory.
- Advance knowledge about underlying process
- Build and elaborate a theory → complete.
- Extend a theory or principle into new areas.
- Provide evidence to support or refute an explanation or prediction.

Q2: Basic and applied research compared

Basic:

- ○ Intrinsically satisfying
- ○ Freedom.
- ○ Hi standards applied.

- Logic and rigorous research design
- Basic knowledge.
- Success → results published impact on other scientists.

Applied:

- R → part of job
- Constrained to demands of sponsors
- R → quick and dirty → may not meet hi standards.
- Apply to areas of interest to sponsors.
- Practical payoffs.
- Success → results are used by sponsors.

Q3) What purpose of research design? (5 M)

- A master plan specifying the methods and procedures for collecting and analyzing the data.
- A strategy or blueprint that plans the action for carrying through the research data.
- Study can be exploratory, descriptive, and explanatory.
- Present study can't be exploratory. We have already covered long distance.
- Can be either descriptive or explanatory.
- Who, what, where, when, or how much?
- Why? Explain the relationship.
- Decide about the purpose.
- Within explanatory →
- Correlational: identification of factors associated with the problem.
- Causal: establish definite cause-and-effect relationship.
- Help in deciding the mode of observation

Q4) Rational scale?

Ratio scales incorporate all three powers of the previous scales plus the provision for absolute zero or origin. Ratio data represent the actual amounts

of variable. Measures of physical dimensions such as weight, height, distance, and area are the examples. The absolute zero represents a point on the scale where there is an absence of the given attribute. If we hear that a person has zero amount of money, we understand the zero value of the amount.

Ratio Scale Properties

- Uniquely classifies
- Preserves order
- Equal intervals
 - Natural zero
 - Weight and distance

Q5: Problem Definition.

Problem definition or problem statement is a clear, precise, and succinct statement of the question or issue that is to be investigated with the goal of finding an answer or solution.

For example the problem could pertain to (1) existing business problems where the manager is looking for a solution, (2) situation that may not pose any current problems but which the manager feels have scope for improvement, (3) areas where some conceptual clarity is needed for better theory building, or (4) situations in which a researcher is trying to answer a research question empirically because of interest in the topic.

- Translate the broad issue into a research question.
- Management dilemma into management question – symptoms of the problem. Productivity decline. Demand for daycare facilities

Q6. Report Writing:

- Communicating the research findings.
- Report is technical. Managers are selective readers. Interested only in recommendations.
- Historical document.
- Academic purpose – dissertations. Specific format. Writing of research papers

Q7: Advantages of E-Mail Surveys:

- Speed of distribution.
- Lower distribution and processing cost.
- More flexibility.
- Less handling of paper questionnaires.
- Many respondents more comfortable in impersonal communication.

Q8: Disadvantages of E-Mail surveys:

- Eavesdropping of e-mails by organization
- Maintaining anonymity is difficult. Sender's address is there.
- Not all e-mails have the same capacity: software limitations

Q9: Debriefing:

It involves several activities following the collection of data:

Explanation of any deception.

Description of the hypothesis, goal, or purpose of the study.

Post study sharing of the results.

Post study follow-up medical or **psychological attention.**

Q10: Ethics in research context?

Ethics are norms or standards of behavior that guide moral choices about our behavior and our relationships with others. The goal of ethics in research is to ensure that no one is harmed or suffers adverse consequences from research activities.

Ethics means to avoid plagiarism

- ▶ Ethics in the research is applied at all stages
- ▶ Goals of the research should be informed to the respondents

STA630 Research Methods current paper subjective solved by Hira Al

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Best regard's

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If u like me than raise your hand with me

If not than raise ur standard

That's about me ... !

STA630

Morning 10:00 AM

Date 01 March 2023

👉 🧠 MCQs are all conceptual.

👉 🧠 Similarities of Historical Comparative and Field Research.

Answer:

First, both H-C research and field research recognize that the researcher's point of view is an avoidable part of research. Both involve interpretation, which introduces the interpreter's location in time, place, and world-view. H-C research does not try to produce a single, unequivocal set of objective facts. Rather, it is a confrontation of old with new or different world-views. It recognizes that the researcher's reading of historical or comparative evidence is influenced by an awareness of the past and by living in the present. Our present day conscio

usness of history is fundamentally different from the manner in which the past appeared to any foregoing people.

Second, both field and H-C research examine a great diversity of data. In both, the researcher becomes immersed in data to gain an emphatic understanding of events and people. Both capture subjective feelings and note how every day, ordinary activities signify important social meaning. The researcher inquires, selects, and focuses on specific aspects of social life from the vast array of events, actions, symbols, and words. An H-C researcher organizes data and focuses attention on the basis of evolving concepts. He or she examines rituals and symbols and dramatize culture and investigates the motives, reasons, and justifications for behaviors.

Third, both field and H-C researchers often use grounded theory. Theory usually emerges during the process of data collection. Both examine data without beginning with fixed hypotheses. Instead, they develop and modify concepts and theory through a dialogue with the data, then apply theory to reorganize evidence. [Historically grounded theory means that concepts emerge from the analytic problem of history: ordering the past into structures, conjectures and events. History and theory can

thus be simultaneously constructed.]

Fourthly, both field and H-C research involve a type of translation. The researcher's meaning system usually differs from that of people he or she studies, but he or she tries to penetrate and understand their point of view. Once the life, language, and perspective of the people being studied have been mastered, the researcher "translates" it for others who read his or her report.

Fifth, both field and H-C researchers focus on action, process, and sequence and see time process as essential. Both say that people construct a sense of social reality through actions that occur over time. Both see social reality simultaneously as something created and changed by people and as imposing a restriction on human choice.

👉 🧩 Supervisor said that it is not appropriate to use frequency distribution, bar chart and pie chart. Give your opinion why?

👉 🧩 A scenario regarding Snowball Sampling is provided, Elaborate the scenario is related with Snowball Sampling.

👉 🧩 A scenario is given, Now elaborate advantages of Personal Interviews with respect to the given scenario.

Answer:

1. The Opportunity for Feedback
2. Probing Complex Questions
3. Length of Interview
4. High Completion Rate
5. Props and Visual Aids
6. High Participation Rate
7. Observation of the Non-Verbal Behavior
8. Non-Literates can participate in Study
9. Interviewer can Prescreen Respondent
10. CAPI – Computer Assisted Personal Interviewing

👉 🧩 Why Cover letter is important.

The cover letter that accompanies the questionnaire or is printed on the first page of the questionnaire is an important means of inducing a reader to complete and return the questionnaire. In the letter tell why the study is important, who is sponsoring the study, how was the respondent selected, assuring the anonymity of the respondent could help in establishing rapport and motivating the respondent to respond.

A personalized letter addressed to a specific individual shows the respondent that he or she is important. Including an individually typed letter on letterhead versus printed form is an important element in increasing the response rate in mail surveys.

👉 🧩 What consideration should made before developing Questionnaires.

Answer:

A good questionnaire should be valid, reliable, clear, interesting and succinct.

Valid. A valid questionnaire should ask what it intends to ask, i.e. the questions should be phrased in such a way that the respondent understands the objective of the question. ...

Reliable. ...

Interesting. ...

Succinct.

👉 🧩 Why a sample is considered as the part of research. Give explanations.

Answer:

In statistics, a sample is an analytic subset of a larger population. The use of samples allows researchers to

conduct their studies with more manageable data and in a timely manner. Randomly drawn samples do not have much

bias if they are large enough, but achieving such a sample may be expensive and time-consuming.

👉 🧩 Explain One Shot Case Study Design.

Answer:

What is a one shot case study design?

a research design in which a single group is observed on a single occasion after experiencing some event,

treatment, or intervention. Because there is no control group against which to make comparisons, it is a weak design; any changes noted are merely presumed to have been caused by the event.

👉🍀 Prepare your self accordingly.

Wish you all best of Luck 😊

Sta630 02:30PM

Few MCQs from file. Mostly new MCQs.

Questions:

Pilot testing 16Aug2023

Answer:

PILOT TESTING OF THE QUESTIONNAIRE: Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire. Conventional

wisdom suggests that pre-testing not only is an established practice for discovering errors but also is useful for

extra training the research team.

Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and

their comprehension. The main purpose of pilot testing is to identify potential problems with the methods,

logistics, and the questionnair

Internal and external validity of experiment

Answer:

The first, internal validity indicates whether the independent variable was the sole cause of the change in

the dependent variable. It implies the researcher's ability to eliminate alternative explanations of the

dependent variable. Variables, other than the treatment, that affect the dependent variable are threats to

internal validity. They threaten the researcher's ability to say that the treatment was the true causal factor producing change in the dependent variable.

The second measure, external validity, indicates the extent to which the results of the experiment are applicable in the real world.

Internal validity is high in the laboratory experiment, reason being the control over all the confounding factors. External validity (generalisability) is not sure because of the effect of variety of factors. Field experiments have more external validity but less internal validity because it is closer to the real situation

Interview research problems

Answer:

Because an interview is a social interaction, the interviewer's appearance or behavior may influence the respondent's answers. This is a problem as it can bias the results of the study and make them invalid.

Characteristics of historical research

Answer:

Historical research involves the careful study and analysis of data about past events.

The purpose is to gain a clearer understanding of the impact of the past on the present and future events related to the life process.

It is a critical investigation of events, their development, experiences the past.

Involves the review of written materials but may include oral documentation as well.

It covers categories such as historical legal, documentary, bibliographical, biographical institutional and organizational.

Typically relies on available data which are in form of diaries, letters, newspapers, reports, and so on.

Sta630 today 12:00

Conceptual mcqs thy zyda

variable sy related thy

Or quiz files main sy b ay thy 2 3

Questions:

1. Cover letter main kia kia chzn ati hain.

Answer:

Key Elements of a Cover Letter

Information about you.

Date.

Contact Person's Name, Title, Employer, and Address.

Salutation.

Opening Paragraph.

Middle Paragraph.

Second Middle Paragraph.

Contact Information and Closing.

2. variable extrnal or internal

repeat

3. field experimnt or lab experiment

Experimental studies can be contrived and non-contrived. Research can be done in the natural environment where work proceeds normally (i.e. in non-contrived setting) or in artificial, contrived setting. Correlational studies are invariably conducted in non-contrived settings, whereas most rigorous causal studies are done in contrived lab settings. Correlational studies done in organizations are called field studies. Studies conducted to establish cause-and-effect relationship using the same natural environment are called field experiments.

Experiments done to establish cause and effect relationship beyond the possibility of the least doubt require the creation of an artificial, contrived environment in which all the extraneous factor are strictly controlled. Similar subjects are chosen carefully to respond to certain manipulated stimuli. These studies are referred to as lab experiments.

4.5 principles of interview

Handouts Pg # 86

5.quasi experiment sy related

Handouts Pg # 118

6.ethical dilemma have to face an interviewer

7.SSP sy related question tha

question. Because the respondent had already clearly identified his answer, the interviewer may have avoided asking the subsequent question. The interviewer may have wanted to avoid hearing "I have already answered that earlier" and to maintain rapport with the respondent and therefore skipped the question. To make the response appear in the same order as on other questionnaires, the editor may remove the out-of-order answer to the section related to the skipped question.

lec#29

Coding

Coding involves assigning numbers or other symbols to answers so the responses can be grouped into limited number of classes or categories. The classifying of data into limited categories sacrifices some data detail but is necessary for efficient analysis. Nevertheless, it is recommended that try to keep the data in raw form so far it is possible. When the data have been entered into the computer you can always ask the computer to group and regroup the categories. (In case the data have been entered in the computer in grouped form, it will not be possible to disaggregate it.)

Although codes are generally considered to be numerical symbols, they are more broadly defined as the rules for interpreting, classifying, and recording data. Codes allow data to be processed in a computer. Researchers organize data into fields, records, and files. A **field** is a collection of characters (a character is a single number, letter of the alphabet, or special symbol such as the question mark) that represent a single type of data. A **record** is collection of related fields. A **file** is a collection of related records. File, records, and fields are stored on magnetic tapes, floppy disks, or hard drives.

Researchers use a coding procedure and codebook. A **coding procedure** is a set of rules stating that certain numbers are assigned to variable attributes. For example, a researchers codes males as 1 and females as 2. Each category of variable and missing information needs a code. A **codebook** is a document (i.e. one or more pages) describing the coding procedure and the location of data for variables in a format that computers can use.

When you code data, it is very important to create a well-organized, detailed codebook and make multiple copies of it. If you do not write down the details of the coding procedure, or if you misplace the codebook, you have lost the key to the data and may have to recode the data again.

Researchers begin thinking about a coding procedure and a codebook before they collect data. For example (a survey researcher pre-codes a questionnaire before collecting the data. **Pre-coding** means placing the code categories (e.g. 1 for male, 2 for female) on the questionnaire. Sometimes to reduce dependence on codebooks, researchers also place the location in the computer format on the questionnaire.)

If the researcher does not pre-code, his or her first step after collecting and editing of data is to create a codebook. He or she also gives each case an identification number to keep track of the cases. Next, the researcher transfers the information from each questionnaire into a format that computers can read.

Code Construction

When the question has a fixed-alternative (closed ended) format, the number of categories requiring codes is determined during the questionnaire design stage. The codes 8 and 9 are conventionally given to "don't know" (DK) and "no answer" (NA) respectively. However, many computer program fields recognize a blank field or a certain character symbol, such as a period (.), as indicating a missing value (no answer).

There are two basic rules for code construction. First, the coding categories should be **exhaustive** - that is, coding categories should be provided for all subjects or objects or responses. With a categorical variable such as sex, making categories exhaustive is not a problem. However, when the response represents a small number of subjects or when the responses might be categorized in a class not typically found, there may be a problem.

How $\cup$ modify the question before going to interviewer. *of which type of error is present.*

if

DATA TRANSFORMATION

& importance

data conversion

Data transformation is the process of changing data from their original form to a format that is more suitable to perform a data analysis that will achieve the research objectives. Researchers often modify the values of a scalar data or create new variables. For example many researchers believe that response bias will be less if interviewers ask consumers for their year of birth rather than their age, even though the objective of the data analysis is to investigate respondents' age in years. This does not present a problem for the research analyst, because a simple data transformation is possible. The raw data coded at birth year can be easily transformed to age by subtracting the birth year from the current year.

Age $\rightarrow$ Year of birth

Collapsing or combining categories of a variable is a common data transformation that reduces the number of categories. For example five categories of Likert scale response categories to a question may be combined like the "strongly agree" and the "agree" response categories are combined. The "strongly disagree" and the "disagree" response categories are combined into a single category. The result is the collapsing of the five-category scale down to three. *Undecided*

Creating new variables by re-specifying the data numeric or logical transformations is another important data transformation. For example, Likert summated scale reflect the combination of scores (raw data) from various attitudinal statements. The summative score for an attitude scale with three statements is calculated as follows

$$\text{Summative Score} = \text{Variable 1} + \text{Variable 2} + \text{Variable 3}$$

This calculation can be accomplished by using simple arithmetic or by programming a computer with a data transformation equation that creates the new variable "summative score."

The researchers have created numerous different scales and indexes to measure social phenomenon. For example scales and indexes have been developed to measure the degree of formalization in bureaucratic organization, the prestige of occupations, the adjustment of people in marriage, the intensity of group interaction, the level of social activity in a community, and the level of socio-economic development of a nation.

Keep it in mind that every social phenomenon can be measured. Some constructs can be measured directly and produce precise numerical values (e.g. family income). Other constructs require the use of surrogates or proxies that indirectly measure a variable (e.g. job satisfaction). Second, a lot can be learned from measures used by other researchers. We are fortunate to have the work of thousands of researchers to draw on. It is not always necessary to start from a scratch. We can use a past scale or index, or we can modify it for our own purposes. The process of creating measures for a construct evolves over time. Measurement is an ongoing process with constant change; new concepts are developed, theoretical definitions are refined, and scales or indexes that measure old or new constructs are improved.

Indexes and Scales *measuring instrument*

Scales and indexes are often used interchangeably. One researcher's scale is another's index. Both produce ordinal- or interval- level measures of variable. To add to the confusion, scale and index techniques can be combined in one measure. Scales and indexes give a researcher more information about variables and make it possible to assess the quality of measurement. Scales and indexes increase reliability and validity, and they aid in data reduction; that is condense and simplify the information that is collected. *advantages*

measured with at least one indicator. Of course, it is better to measure the parts of a construct with multiple indicators.

Another example of an index is college quality index. Our theoretical definition says that a high quality college has six distinguished characteristics: (1) fewer students per faculty member, (2) a highly educated faculty, (3) more books in the library, (4) fewer students dropping out of college, (5) more students who go to advanced degrees and (6) faculty members who publish books or scholarly articles. We score 100 colleges on each item, and then add the score for each to create an index score of college quality that can be used to compare colleges.

Indexes can be combined with one another. For example, in order to strengthen the college quality index. We add a sub-index on teaching quality. The index contains eight elements: (1) average size of classes, (2) percentage of class time devoted to discussion, (3) number of different classes each faculty member teaches, (4) availability of faculty to students outside the classroom, (5) currency and amount of reading assigned, (6) degree to which assignments promote learning, (7) degree to which faculty get to know each student, and (8) student ratings of instruction. Similar sub-index measures can be created for other parts of the college quality index. They can be combined into a more global measure of college quality. This further elaborates the definition of a construct "quality of college."

Weighting a amount given to increase/decrease value of an item

An important issue in index construction is whether to weight items. Unless it is otherwise stated, assume that an index is un-weighted. Likewise, unless we have a good reason for assigning different weights, use equal weights. A weighted index gives each item equal weight. It involves adding up the items without modification, as if each were multiplied by 1 (or -1 for negative items that are negative).

lec#3C

Scoring and Score Index

In one of our previous discussions we had tried to measure job satisfaction. It was operationalized with the help of dimensions and elements. We had constructed number of statements on each element with 5 response categories using Likert scale i.e. strongly agree, agree, undecided, disagree, and strongly disagree. We could score each of these items from 1 to 5 depending upon the degree of agreement with the statement. The statements have been both positive as well as negative. For positive statements we can score straight away from 5 to 1 i.e. strongly agree to strongly disagree. For the negative statements we have to reverse the score i.e. 1 for "strongly agree," 2 for "agree," 3 for "undecided" to 4 for "disagree," and 5 for "strongly disagree." Reason being that negative multiplied by a negative becomes positive i.e. a negative statement and a person strongly disagreeing with it implies that he or she has a positive response so we give a score of 5 in this example. In our example, let us say there were 23 statements measuring for different elements and dimensions measuring job satisfaction. When on each statement the respondent could get a minimum score of 1 and a maximum score of 5, on 23 statements a respondent could get a minimum score of (23 X 1) and a maximum score of (23 X 5) 115. In this way the score index ranges from 23 to 115, the lower end of the score index showing minimum job satisfaction and upper end as the highest job satisfaction. In reality we may not find any on the extremes, rather the respondents could be spread along this continuum. We could use the raw scores of independent and dependent variable and apply appropriate statistics for testing the hypothesis. We could also divide the score index into different categories like high "job satisfaction" and "low satisfaction" for presentation in a table. We cross-classify job satisfaction with some other variable; apply appropriate statistics for testing the hypothesis.

underlying data. It is suggested that data analysis should begin with visual inspection.

The presented data has to be interpreted. The purpose of interpretation is to explain the meaning of data so that we can make inferences and formulate conclusions. Therefore interpretation refers to making inferences pertinent to the meaning and implications of the research investigation and drawing conclusions. In order for interpretation, the data have to be meaningfully analyzed. For purposes of analysis the researchers use statistics.

The word statistics has several meanings. It can mean a set of collected numbers (e.g. numbers telling how many people living in a city) as well as a branch of applied mathematics used to manipulate and summarize the features of numbers. Social researchers use both types of statistics. Here, we focus on the second type - ways to manipulate and summarize numbers that represent data from research project.

Descriptive statistics describe numerical data. They can be categorized by the number of variables involved: univariate, bivariate, or multivariate (for one, two, and three or more variables). Univariate statistics describe one variable.

Researchers often want to summarize the information about one variable into a single number. They use three measures of central tendency, or measures of the center of the frequency distribution: mean, median and mode, which are often called averages (a less precise and less clear way to say the same thing). The mode is simply the most common or frequently occurring number. The median is the middle point. The mean also called the arithmetic average, is the most widely used measure of central tendency. A particular central tendency is used depending upon the nature of the data.

2 variables Bivariate Tables

The bivariate contingency table is widely used. The table is based on cross-tabulation (cross-classification); that is the cases are organized in the table on the basis of two variables at the same time.

A contingency table is formed by cross-tabulating the two or more variables. It is contingent because the cases in each category of a variable get distributed into each category of a second variable. The table distributes cases into categories of multiple variables at the same time and shows how the cases, by the category of one variable, are "contingent upon" the categories of the other variables.

Constructing Percentage Tables

It is to construct a percentage table, but there are ways to make it look professional. Let us take two variables like the age of the respondents and their attitude towards "women empowerment." Assuming that age affects the attitude towards women empowerment let us hypothesize: the lower the age, the higher the favorable attitude towards "women empowerment." The age range of the respondents is 25 to 70, and the attitude index has three categories of "highly favorable," "medium favorable," and "low favorable." The age variable has so many categories that making a table with that number becomes unwieldy and meaningless. Therefore, we regroup (recode) the age categories into three i.e. under 40 years, 40 - 60 years, and 61 + years.

lec#31

Level of Attitude	Under 40		Age 40-50		Age 50+	
	F	%	F	%	F	%
High	350	70	200	50	20	1
Medium	150	30	150	38	220	44
Low	-	-	50	12	260	52
Total	500	100	400	100	500	100

copying lec#32

The replication pattern is the easiest to understand. It is when the partials replicate the same relationship that existed in the bivariate table before considering the control variable that the control variable has no effect.

The specification pattern is the next easiest pattern. It occurs when one partial bivariate relationship but other partials do not. For example, we find a strong relationship between age of the respondents and attitude towards women empowerment for gender and discover the relationship holds only for males (i.e. the strong negative in the partial for males, but not for females). This is specification because a research category of the control variable in which the initial relationship persists.

The interpretation pattern describes the situation in which the control variable into original independent variable and the dependent variables.

The suppressor variable pattern occurs when the bivariate tables suggest relationship appears in one or both of the partials. For example, the age of the respondents and attitudes towards women empowerment are independent in a bivariate table. Once "gender" is introduced, the relationship between the two variables appears in the partials. The control variable is suppressor variable because it suppressed the true relationship that appears in partials.

Multiple Regression Analysis

Multiple regression controls for many alternative explanations of variables simultaneously. It is possible to use more than one control variable using percentaged tables. Multiple regression is a technique whose calculation you may have learnt in the course on statistics.

Note

In the preceding discussion you have been exposed to the descriptive analysis of data. There are statistical tests which can be applied to test the hypothesis, which you will learn in your course on statistics.

are those individuals, organizational units, sales territories, or other entities may be the test units. Terminology is used on different component parts of the experiments.

Parts of Experiments:

We can divide the experiments into seven parts and for each part there is a term. Not all have all these parts, and some have all seven parts plus others. The following seven usually true experiment.

1. Treatment or independent variable.
2. Dependent variable.
3. Pretest.
4. Posttest.
5. Experimental group.
6. Control group.
7. Assignment of subjects.

} difference

Treatment or independent variable:

"X"

The experimenter has some degree of control over the independent variable. (The independent because its value can be manipulated by the experimenter to whatever he wishes to be.) In experimental design the variable that can be manipulated to be whatever wishes (Its value may be changed or altered independently of any other variable.)

In most experiments, a researcher creates a situation or enters into an ongoing situation. The treatment (or the stimulus or manipulation) is what the researcher modifies (from medicine, in which a physician administers a treatment to patients; the physician physical or psychological condition to change it. It is the independent variable or the independent variables.)

↓ it is X variable or combination

In experiments, for example, the researcher creates a condition or situation. Look at "fear or anxiety"; the levels are high-fear or low-fear situation. Instead of asking the subjects in surveys, whether they are fearful, experimenter puts the subjects into either in a high fear situation. They measure the independent variable by manipulating condition. Subjects feel a lot of fear and others feel little.

Researchers go to great lengths to create treatments. They want the treatment to have produce specific reactions, feelings, or behaviors.

It also possible the researchers look at the alternative manipulations of the independent variable. The independent variable is often categorized into qualitative aspects of management. The treatment that is given to the subjects.

measured by paper-and-pencil indicators, observations, interviews, or physiological responses (e.g., heartbeat, or sweating palms).

Selection of dependent variable is crucial decision in the design of an experiment.

Pretests and Posttests:

Frequently a researcher measures the dependent variable more than once during an experiment. ^{or before}
pretest is the measurement of the dependent variable prior to the introduction of the treatment.
posttest is the measurement of the dependent variable after the treatment has been introduced into the experimental situation.)

Experimental and Control Groups:

Experimental researchers often divide subjects into two or more groups for purposes of comparison. (A simple experiment has only two groups, only one of which receives the treatment. The experimental group is the group that receives the treatment or in which the treatment is present.)

(The group that does not receive the treatment is called the "control group." When the independent variable takes on many different values, more than one experimental group is used.)

* In the simplest type of experiment, only two values of the independent variable are manipulated. For example, consider measuring the influence of a change in work situation, such as playing music on an intercom during working hours, on employee productivity. In the experimental condition (the treatment administered to the experimental group), music is played during working hours. In the control condition (the treatment administered to the control group), the work situation remains the same, without change. By holding conditions constant in the control group, the researcher controls for potential sources of error in the experiment. Productivity, (the dependent variable) in the two groups is compared at the end of the experiment to determine whether playing the music (the independent variable) has any effect.

Several experimental treatment levels can also be used. The music/productivity experiment, with one experimental and one control group, may not tell the researcher everything he or she wishes to know about the music/productivity relationship. If the researcher wished to understand the functional form of the relationship between music and productivity at several treatment levels, additional experimental groups with music played for only 2 hours, only for 4 hours, and only for 6 hours might be used. This type of design would allow the experimenter to get a better idea about the impact of music on productivity.

Assignment of Subjects/Test Units:

Social researchers frequently want to compare. When making comparisons, the researcher wants to compare the cases that do not differ with regard to variables that offer alternative explanations. Therefore, the groups should be similar in characteristics in such a way that the change in the dependent variable is presumably the outcome of the manipulation of the independent variable, having no alternative explanations.

Random assignment (Randomization) is a method for assigning the cases (e.g., individuals or organizations) to groups for the purpose of making comparisons. It is a way to divide a random collection of cases into two or more groups in order to increase one's confidence that the groups do not differ in a systematic way. It is a mechanical method; the assignment is automatic.

what are different types of Quasi-experiments design. (5)

VU

Lesson 34

EXPERIMENTAL RESEARCH (Cont.)

Steps In Conducting an Experiment

Following the basic steps of the research process, experimenters decide on a topic, narrow it into a testable research problem or question, and then develop a hypothesis with variables. Once a researcher has the hypothesis, the steps of experimental research are clear. Broadly there are about 12 steps in conducting an experiment, which are as below:

1. Begin with a straightforward hypothesis that is appropriate for experimental research.
2. Decide on an experimental design that will test the hypothesis within practical limitations. The researcher decides the number of groups to use, how and when to create treatment conditions, the number of times to measure the dependent variable, and what the groups of subjects will experience from beginning till end.
3. Decide how to introduce the treatment or create a situation that induces the independent variable.
4. Develop a valid and reliable measure of the dependent variable.
5. Set up an experimental setting and conduct a pilot test of the treatment and dependent variable measures.
6. Locate appropriate subjects or cases.
7. Randomly assign subjects to groups (if random assignment is used in the chosen research design) and give careful instructions.
8. Gather data for the pretest measure of the dependent variable for all groups (if pretest is used in the chosen design).
9. Introduce the treatment to the experimental group only (or to the relevant groups if there are multiple experimental groups) and monitor all groups.
10. Gather data for posttest measure of the dependent variable.
11. Debrief the subjects by informing them of the true purpose and reasons for the experiment. Ask subjects what they thought was occurring. Debriefing is crucial when subjects have been deceived about some aspect of the treatment.
12. Examine data collected and make comparisons between different groups. Where appropriate, use statistics and graphs to determine whether or not the hypothesis is supported.

Types of Designs

(Researchers combine parts of experiment (e.g. pretests, control groups, etc.) together into an experimental design.) For example some designs lack pretests, some do not have control groups, and others have many experimental groups. Certain widely used standard designs have names.

Classical Experimental Design

All designs are variations of the classical experimental design, which has random assignment of subjects, a pretest and a posttest, an experimental group, and a control group.

Quasi-Experimental Designs

One-shot Case Study Design

Also called the one-group posttest-only design, the one-shot case study design has only one group, a treatment, and a posttest. Because it is only one group, there is no random assignment. For example, a researcher shows a group of students a horror film, then measures their attitude with a questionnaire. A weakness of this design is that it is difficult to say for sure that the treatment caused the dependent variable. If subjects were the same before and after the treatment, the researcher would not know it.

experimental group. The random assignment of subjects to treatment, but without a pretest, a researcher cannot be as certain that the groups began the study with the same level of the dependent variable.

True Experimental Designs

Experimental designs, which have at least two groups, a random assignment of subjects to experimental and control groups, only experimental group is exposed to treatment, both groups are given the same information before and after the treatment, are known as ex-post facto experimental designs.

Pretest and Posttest Experimental and Control Group Design:

Two groups, one control group and the other experimental group, are formed randomly. Both groups are exposed to pretest and posttest. The experimental group is exposed to treatment while the control group is not. Measuring the difference between the differences in the post- and pretest scores for the two groups would give the net effects of the treatment.

Experimental Group: Pretest (O1) X Posttest (O2)

Control Group: Pretest (O3) - Posttest (O4)

Randomization used for setting up the group.

$[(O2 - O1) - (O4 - O3)] = \text{Treatment effect (could be anywhere between 0 to -1 or +1)}$

Solomon's Four Group Design:

To gain more confidence in internal validity in experimental designs, it is advisable to use two experimental groups and two control groups. One experimental group and one control group are given both the pretest and the posttest. The other two groups will be given only the posttest. The effects of treatment can be calculated in several different ways as shown in figure 1:

lec#34

Diff b/w internal & external validity

is it possible to control "maturation" effect in e. design, why or why not. Lesson 35

EXPERIMENTAL RESEARCH (Cont.)

Validity
 internal internal

Validity in Experiments

Experiments are judged by two measures. The first, internal validity indicates whether the independent variable was the sole cause of the change in the dependent variable. It implies the researcher's ability to eliminate alternative explanations of the dependent variable. Variables, other than the treatment, that affect the dependent variable are threats to internal validity. They threaten the researcher's ability to say that the treatment was the true causal factor producing change in the dependent variable. The second measure, external validity, indicates the extent to which the results of an experiment are applicable in the real world.

Internal validity is high in the laboratory experiment, reason being the control over all the confounding factors. External validity (generalizability) is not sure because of the effect of variety of contexts. Field experiments have more external validity but less internal validity because it is closer to real situations.

Factors Affecting Internal Validity

When choosing or evaluating experimental research design, researchers must determine whether they are internal and external validity. There are eight major types of extraneous variables that may confound internal validity: History effect, maturation effect, testing effect, instrumentation effect, selection bias effect, statistical regression, mortality, and mechanical loss.

History Effect:

A specific event in the external environment occurring between the first and second measurement that is beyond the control of the experimenter and that affects the validity of an experiment. Example: Advertisement of a particular product (mineral water) and its sale is affected by an event in the society (contamination of drinking water). The researcher does not have control on such happenings which may have an impact on the X and Y relationship.

Maturation Effect:

Cause and effect relationship can also be contaminated by the effects of the passage of time - another uncontrollable variable. Such contamination is called maturation effect. The maturation effects are a function of the processes - biological and psychological - operating within subjects as a result of the passage of time. (Examples of maturation processes could include growing older, getting tired, feeling hungry, and getting bored.) In other words there could be a maturation effect on the dependent variable purely because of the passage of time. For example, let us say that an R & D director intends that an increase in the efficiency of workers would result within a few months' time if advanced technology is introduced in the work setting. If at the end of three months increased efficiency is indeed found, it will be difficult to claim that the advanced technology alone increased the efficiency of workers, because with the passage of time, employees would have gained experience, resulting in better performance and therefore improved efficiency. Thus, the internal validity also gets reduced owing to the effects of maturation in as much as it is difficult to pinpoint how much of the increase is attributable to the introduction of the enhanced technology.

Testing Effects:

Frequently, to test the effects of treatment, subjects are given what is called a pretest (say, a short questionnaire eliciting their feelings and attitudes). That is, a measure of the dependent variable is taken (pretest), then the treatment given, and after that a second test, called a posttest, administered. The difference between the posttest and the pretest scores is then attributed to the treatment. However, the very fact that the subjects were exposed to the pretest might influence

their responses on the posttest, which will adversely impact on internal validity. It is also called sensitization through previous testing.

4. **Instrumentation Effects:** ^{importance} Instrumentation effects are yet another source of threat to internal validity. These might arise because of a change in the measuring instrument between pretest and posttest, and not because of the instrument's differential impact at the end. For example, in a weight loss experiment, the springs on the scale weaken during the experiment, giving lower readings in the posttest.

A change in the wording of questions (may be done to avoid testing effects), change in interviewers, or change in other procedures to measure the dependent variable can cause instrumentation effect.

Performance of the subjects measured by the units of output in the pretest, but when measuring the output in posttest the researcher measures it by the number of units rejected, and the amount of resources expended to produce the units.

5. **Selection Bias Effect:** ^{effect} Selection bias is the threat that subjects will not form equivalent groups. It is a problem in design without random assignment, hence differential selection of the subjects for the comparison groups. ^{when we do wrong selection} It occurs when subjects in one experimental group have a characteristic that affects the dependent variable. For example, in an experiment on physical aggressiveness, the experimental group unintentionally contains subjects who are sportsmen, whereas the control group is made up of musicians, chess players, and painters.

6. **Statistical Regression:** Statistical regression is not easy to grasp intuitively. It is a problem of extreme values or a tendency for random error to move group results toward the average. If extremes are taken then they tend to regress toward the mean. Those who are on either end of the extreme would not truly reflect the cause and effect relationship.

One situation arises when subjects are unusual with regard to dependent variable. Because they begin as unusual or extreme, subjects are likely to respond further in the same direction. For example, a researcher wants to see whether violent films make people act violently. The researcher chooses a group of violent criminals from a high security prison, gives them a pretest, shows violent films, and then administers a posttest. To the researcher's surprise, the criminals are slightly less violent after the film, whereas a control group of non-prisoners who did see the film are slightly more violent than before. Because the violent criminals began at an extreme, it is unlikely that a treatment could make them more violent; by random chance alone, they appear less extreme when measured a second time.

* If participants chosen for experimental group have extreme scores on the dependent variable to begin with then the laws of probability say that those with very low scores on a variable have a greater probability to improve and scoring closer to mean on the posttest after treatment. This phenomenon of low scorers tending to score closer to the mean is known as "regressing toward the mean."

Likewise, those with high scores have a greater tendency to regress toward the mean - will score lower on the posttest than on pretest. Thus the extremes will not "truly" reflect the causal relationship - a threat to internal validity.

7. **Mortality:** ^{فانی ہونے} Mortality, or attrition, arises when some subjects do not continue throughout an experiment. Although the word mortality means death, it does not necessarily mean that subjects have died. If a subset of subjects leaves partway through an experiment, a researcher cannot be sure the results would have been different had the subjects stayed. Even with departure of few subjects, groups do not remain balanced.

Consider for example of a training experiment that investigates the effects of close supervision of salespersons (high pressure) versus low supervision (low supervision). The high pressure group may misleadingly appear to be superior.

8. **Mechanical Loss:** A problem may be experienced due to equipment failure. For example, in an experiment if the subjects are told that their behavior is being videotaped, and during the experiment the video equipment failed to work for some subjects, then the validity of the results could become doubtful.

9. **Experimenter Expectancy:** In addition to the usually listed eight factors affecting the internal validity sometimes experimenter expectancy may threaten the causal logic of the relationship between the variables. A researcher may threaten internal validity, not purposefully unethical behavior but by indirectly communicating experimenter expectancy to the subjects. Researchers may be highly committed to the hypothesis and indirectly communicate desired findings to subjects. For example, a researcher studying reactions towards disabled deeply believes that females are more sensitive toward the disabled than the males are. Through eye contact, tone of voice, pauses, and other nonverbal communication, the researcher unconsciously encourages female subjects to report positive feelings toward the disabled, the researcher's nonverbal behavior is the opposite for male subjects.

PPV
GMP The double-blind experiment is designed to control experimenter expectancy. In it, people who have direct contact with subjects do not know the details of the hypothesis or the treatment. It is double blind because both the subjects and those in contact with them are blind to details of the experiment. For example a researcher wants to see if new drug is effective. Using capsules of three colors - green, yellow, and pink -- the researcher puts the new drug in the yellow capsule, puts an old drug in the pink one, and take the green capsule a placebo - a false treatment that appears to be real (e.g., a sugar capsule without any physical effects). Assistants who give the capsules and record the effects do not know which color contains the new drug. Only another person who does not deal with subjects directly knows which colored capsule contains the drug and examines the results.

External Validity

Even if the researcher eliminates all concerns for internal validity, external validity remains a potential problem. External validity is the ability to generalize experimental findings to real life situations. Without external validity, the findings are of little use for both basic and applied research i.e. we shall not be able to develop any theories that could be applicable to similar other situations.

Reactivity: A Threat to External Validity

Subjects may react differently in an experiment than they would in real life; because they know they are in a study. The Hawthorn Effect, a specific kind of reactivity to the experimental situation is a good example in this respect. The experiment was conducted in the Hawthorn Electric Company where the performance of the participants was supposed to change due to the change in the environmental conditions i.e. improvement on the environmental conditions will have a positive effect on the performance. The researchers modified many aspects of the working conditions and measured productivity. Productivity rose after each modification. Productivity rose even if there was no real modification but it was announced that there is a modification. The behavior change was simply a reaction to the announcement of modification and some other factors like the participants were being watched and had a feeling of being 'very important persons.'

Here the workers did not respond to treatment (modification of working conditions) but to the additional attention they received (being in the experiment and being the focus of attention).

Demand characteristic (discussed earlier) is another type of reactivity. Here the participants change their behavior as a reaction to the demands of the experimenter who may have inadvertently told the subjects about the expected outcome of the treatment. They change their behavior as demanded by the experimenter.

Reactive Research
 * People being studied & aware of that effect
 * Experiments & Surveys are both reactive

Analysis report?

Lesson 36

NON-REACTIVE RESEARCH

Experiments and surveys research are both reactive that is, the people being studied are aware of the fact that they are being studied. In non-reactive research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

The Logic of Non-Reactive Research

with testing effect
 gmp ppy (3)

The critical thing about non-reactive or unobtrusive measures (i.e. the measures that are not obtrusive or intrusive) is that the people being studied are not aware of it but leave evidence of their social behavior or actions naturally. The researcher infers from the evidence to behavior or attitudes without disrupting those being studied. (Unnoticed observation is also a type of non-reactive measure.) For example, a researcher may be observing the behavior of drivers from a distance whether drivers stopped at red sign of the traffic lights. The observations can be made both at the day time and at night. It could also be noted whether the driver was a male or a female; whether the driver was alone or with passengers; whether other traffic was present; and whether the car came to a complete stop, a slow stop, or no stop.

Varieties of Non-Reactive Observations

Non-reactive measures are varied, and researchers have been creative in inventing indirect ways to measure behaviors. Because the measures have little in common except being non-reactive, they are best learned through examples like:

Physical Traces:

- Erosion: Wear and tear suggests a greater use. For example, a researcher examines children's toys at a children's play center that were purchased at the same time. Worn out toys suggest greater interest of children in them.
- Accretion: Accumulation of physical evidence suggests behavior. A researcher examines the soft drink cans or bottles in the garbage collection. That might indicate the brands and types of soft drinks that are very popular.

Archives:

- Running Records: Regularly produced public records may reveal lot of information. For example, a researcher may examine marriage records for brides' and grooms' recorded ages. The differences might indicate that males marrying younger females are greater than the other way around.
- Other Records: Irregular or private records can reveal a lot. For example, a researcher may look into the number of reams of paper purchased by a college principal's office for the last 10 years and compare it with students' enrollment.

Observations:

- External Appearance: How people appear may indicate social factors. For example, a researcher watches students to see whether they are more likely to wear their college's colors and symbols after the college team won or lost.
- Count Behaviors: Counting how many people do something can be informative. For example a researcher may count the number of men and women who come to a full stop and those who come to a rolling stop at a traffic stop sign. This suggests gender difference in driving behavior.
- Time Duration: How long people take to do things may indicate their intention. For example, a researcher may measure how long men and women pause in front of a particular painting to gauge their interest in the painting.

lec#36

Recording and Documentation

Creating non-reactive measures follows the logic of quantitative measurement, although qualitative researchers also use non-reactive observations. A researcher first conceptualizes a construct, and then links the construct to non-reactive empirical evidence, which is its measure. The operational definition of the variable includes how the researcher systematically notes and records observations.

Content Analysis

Content analysis is a technique for gathering and analyzing the content of a text. The content refers to words, meanings, pictures, symbols, ideas, themes, or any message that can be communicated. The text is anything written, visual, or spoken that serves as a medium of communication. Possible artifacts for study could be books, newspaper or magazine articles, advertisements, poems, letters, laws, constitutions, dramas, speeches, official documents, films or videotapes, musical lyrics, photographs, articles of clothing, or works of arts. All these works may be called as documents. The documents can be:

TYPES

- Personal – letters, diary, autobiography.
- Non-personal – interoffice memos, official documents, proceedings of a meeting.
- Mass media – newspapers, magazines, fiction, films, songs, poems, works of arts.

Content analysis goes back nearly a century and is used in many fields – literature, history, journalism, political science, education, psychology, sociology, and so on. It is also called a study of communication, which means who says what, to whom, why, how, and with what effect.

In content analysis, the researcher uses objective and systematic counting and recording procedures to produce a quantitative description of the symbolic content in a text. It may also be called "textual coding". There are qualitative versions of content analysis. The emphasis here is quantitative data about a text's content.

Content Analysis is Non-Reactive: It is non-reactive because the placing of words, messages, or symbols in a text to communicate to the reader or receiver occurs without influence from the researcher who analyzes its contents. There is no interaction between the researcher and the creator of the text under analysis.

Content analysis lets a researcher reveal the contents (i.e. messages, meanings, symbols, etc.) in a source of communication (i.e. a book, article, movie, etc.). It lets him/her probe into and discover content in a different way from ordinary way of reading a book or watching a television program.

With content analysis, a researcher can compare content across many texts and analyze it with quantitative techniques (table, charts). In addition, he or she can reveal aspects of the texts content that are difficult to see. For example, you might watch television commercials and feel that women are mostly portrayed working in the house, cooking food, using detergents, looking after children. Content analysis can document – in objective, quantitative terms – whether or not your vague feelings based on unsystematic observation are true. It yields repeatable, precise results about the text.

Content analysis involves random sampling, precise measurement, and operational definitions for abstract constructs. Coding turns aspects of content that represent variables into numbers. After a content analysis researcher gathers the data, he or she enters them into computers and analyzes them with statistics in the same way that an experiment or survey researcher would.

USE OF SECONDARY DATA

Existing statistics/documents

Prior to the discussion of secondary data, let us look at the advantages and disadvantages of content analysis that was covered in the last lecture. In a way content analysis is also a study of documents through which the writers try to communicate, though some of the documents (e.g., population census) may simply contain figures.

Advantages

Names

1. Access to inaccessible subjects: (One of the basic advantages of content analysis is research on subjects to which the researcher does not have physical access.) These could be old civilizations, say their marriage patterns. These could also be the documents from the speeches of the past leaders (Quaid-e-Azam) who are not alive, the suicide notes, old films, poems, etc.

2. Non-reactivity: Document study shares with certain types of observations (e.g., observation or non-participant observation through one-way mirror) the advantage of non-reactivity, particularly when the document was written for some other purpose. (This is not true for all documents. Even the creator of that document, and for that matter the characters in the document, is not always aware of the researcher, who may not be alive.)

3. Can do longitudinal analysis: Like observation and unlike experiments and surveys, document study is especially well suited to study over a long period of time. Many times the objective of research could be to determine a trend. One could pick up different periods in past and present and make comparisons and figure out the changes (in the status of women) that may have occurred. Take two martial periods in Pakistan, study the newspapers and look at the reported changes in the press.

4. Use Sampling: The researcher can use random sampling. One could decide on the population, develop sampling frame and draw simple random sample by following the appropriate procedure. For example, how women are portrayed in weekly English news magazines. One could pick up a sample of English news magazines, make a listing of articles that have appeared in the magazines (develop a sampling frame), and draw a simple random sample.

5. Can use large sample size: Larger the sample closer the results to the population. In experimentation as well as in survey research there could be limitations due to the availability of subjects or of the resources but in document analysis the researcher could increase the sample size and have more confidence in generalization. Let us assume that a researcher is studying the advertisements in the newspapers over a long period of time, there should be no problem in getting a sample as large as several thousand or more.

6. Spontaneity: The spontaneous actions or feelings can be recorded when they occur. If the respondent was keeping a diary, he or she may be recording spontaneous feelings about a subject whenever he or she was inspired to do so. The data from such personal recording could be analyzed later on.

7. Confessions: A person may be more likely to confess in a document, particularly one written only after his or her death, than in an interview or mailed questionnaire study. Thus, documents such as diaries, posthumously published autobiographies, and suicide notes provide one of the only ways to obtain such information.

time's documents are gathered together in a centralized location such as library where the researcher can study them for only the cost of travel to the repository.

High quality: Although documents vary tremendously in quality, many documents, such as newspaper columns, are written by skilled commentators and may be more valuable than, for example, poorly written responses to mailed questionnaires.

Disadvantages

1. Bias: Many documents used in research were not originally intended for research purposes. Various goals and purposes for which documents are written can bias them in various ways. For example, personal documents such as confessional articles or autobiographies are often written by famous people or people who had some unusual experience such as having been a witness to a specific event. While often providing a unique and valuable research data, these documents are written for the purpose of making money. Thus they tend to exaggerate and even fabricate to tell a good story. They also tend to include those events that make the author look good and exclude those that cast him or her in a negative light.

2. Selective survival: Since documents are usually written on paper, they do not withstand elements well unless care is taken to preserve them. Thus while documents written by famous people are likely to be preserved, day-to-day documents such as letters and diaries written by common people tend either to be destroyed or to be placed in storage and thus become inaccessible. It is relatively easy to find documents for common documents that are not about some events of immediate interest to the researcher (e.g., suicide) and not about famous occurrence or by some famous person to be gathered together in a public repository that is accessible to researchers.

3. Incompleteness: Many documents provide incomplete account to the researcher who has no prior experience with or knowledge of the events or behavior discussed. A problem with personal documents such as letters and diaries is that they were not written for research purposes. They were designed to be private or even secret. Both these kinds of documents often assume that the researcher is familiar with certain events and will not possess. Diaries are probably the worst in this respect, since they are usually written to be read only by the author and can contain a great deal of "soul searching" and confession rather than of description. Letters tend to be little more complete than they are addressed to a second person. Since many letters assume a great amount of prior information on the part of the reader.

4. Lack of availability of documents: In addition to the bias, incompleteness, and selective survival of documents, there are many areas of study for which no documents are available. In many cases information simply was never recorded. In other cases it was recorded, but the documents were secret or classified, or have been destroyed.

5. Sampling bias: One of the problems of bias occurs because persons of lower educational levels are less likely to be represented in the sampling frames. The problem of sampling bias by educational level is more acute for document study than for survey research. It is a generalization that a poorly educated person is much less likely than well educated person to write documents.

By definition, documents provide information on the past. Direct information on the present is the domain of other methods.

Basis of F.R.
major duties of F.R.
non participant

ethnography
Participant
Anthropological

Lesson 38

OBSERVATION STUDIES/FIELD RESEARCH

Observation studies are primarily part of qualitative research. Though qualitative and quantitative researches differ yet they complement each other. (Qualitative research produces soft data: impressions, words, sentences, photos, symbols. Usually it follows an interpretive approach, the goal of which is to develop an understanding of social life and discover how people construct meanings in natural settings. The research process follows a non-linear approach (spiral).)

(Quantitative research produces hard data: numbers. It follows a positivist approach to research in which the researcher speaks the language of variables and hypotheses. There is a much emphasis on precise measurement of variables and the testing of hypotheses. The researcher tries to establish causality. In most of the case there is a linear approach i.e. it follows sequential steps in doing research.)

Participant/Non-Participant Observation

Observation studies can be participant or non-participant. In participant observation the researcher directly observes and participates in small scale social settings in the present time. Such a study is also referred to as field research, ethnography, or anthropological study. Here the researchers:

- Study people in their natural settings, or in situ.
- Study people by directly interacting with them.
- Gain an understanding of the social world and make theoretical statements about members' perspective.

The people could be a group who interact with each other on regular basis in a field setting: a street corner, a tea shop, a club, a nomad group, a village, etc.

Non-participant studies are such where the research tries to observe the behavior of people without interacting with them. It could be observing the behavior of shoppers in a departmental store through a mirror or on a closed circuit TV. Somebody might be counting the number of vehicles crossing a particular traffic light.

Field researchers study people in a location or setting. It has been used to study entire communities. Field research has a distinct set of methodologies. Field researchers directly observe and interact with community members in natural settings to get inside their perspectives. They embrace an activist or social constructionist perspective on social life. They do not see people as a neutral medium through which social forces operate, nor do they see social meanings as something "out there" to observe. Instead they believe that people create and define the social world through their interactions. Human experiences are filtered through a subjective sense of reality, which affects how people see and act on events. Thus they replace the positivist emphasis on "objective facts" with a focus on the everyday, face-to-face social processes of negotiation, discussion, and bargaining to construct social meaning.

Ethnography and Ethno-methodology

Two modern extensions of field research, ethnography and ethno-methodology, build on the social constructionist perspective.

Ethnography comes from cultural anthropology. Ethno means people or a folk distinct by their culture and ethnography refers to describing something. Thus ethnography means describing a culture and understanding another way of life from the native point of view. It is just an understanding the way of people from their own perspective.

weeks. He or she also needs to decide how to tell members and how much to give. The best way to exist is to follow the local norms and continuing relations.

10. **Complete the analysis and write the report.** After disengaging from the field, the researcher writes the report. The researcher may share the written report with members to verify the accuracy and get their approval of its portrayal in print. (determining the validity of the findings) However, it may not be possible to share with marginal groups like addicts, and some deviant groups.

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Jaf

Ethical Dilemmas of Field research

The direct personal involvement of a field researcher in the social lives of other people creates ethical dilemmas. The dilemmas arise when the researcher is alone in the field and has to make a moral decision. Although he or she may be aware of general ethical issues before entering the field, they arise unexpectedly in the course of observing and interacting in the field. Some of these dilemmas:

~2005

Deception: Deception arises in several ways in field research. The researcher may assume a false role, name, or identity; or may mislead members in some way. One of the ethical issues arising from deception is that of covert versus overt field research. Covert researchers see it as necessary for entering into and gaining a full knowledge of many areas. Others oppose it and argue that it undermines a trust between researchers and social members. If the moral status is questionable, there are some field sites or activities that can only be studied covertly. One may have to look into the cost and benefit equation; where the researcher is the beneficiary.

Covert research is never preferable and never easier than overt research because of the need for maintaining a front and the constant fear of getting caught.

Confidentiality: A researcher learns intimate knowledge that is given in confidence. There is a moral obligation to uphold the confidentiality of data. This includes keeping information confidential from others in the field and disguising members' names in field notes.

Jaf

Involvement with deviants: Researchers who conduct research on deviants who engage in illegal behavior face additional dilemmas. They know of and are sometimes involved in illegal behavior. They might be getting 'guilty knowledge.' Such knowledge is of interest not only to law enforcement officials but also to other deviants. The researcher faces a dilemma of building trust with the deviants, yet not becoming so involved as to violate his or her basic personal principles. Usually, the researcher makes an explicit arrangement with the deviant members.

The powerful: Field researchers tend to study those without power in society (e.g., the poor, children, and lower level workers). Powerful elites can block access and

organized because the researcher will return to them over and over again. Once written, the notes are private and valuable. A researcher treats them with care and protects confidentiality.

Field researcher is supposed to collect quality data. What does the term high-quality data mean in the field research, and what does a researcher do to get it? For a quantitative researcher, high quality data are reliable and valid; they give precise, consistent measures of the "objective" truth for all researchers. An interpretive approach suggests a different kind of data quality. Instead of assuming one single, objective truth, field researchers hold that members subjectively interpret experiences within social context. What a member takes to be true results from social interaction and interpretation. Thus high quality field data capture such processes and provide an understanding of the member's viewpoint.

A field researcher does not eliminate subjective views to get quality data: rather, quality data include his or her subjective responses and experiences. Quality field data are detailed descriptions from the researcher's immersion and authentic experiences in the social world of members.

6. Begin to analyze data generate and evaluate working hypothesis. Right in the field try to look into the research questions and the kind of answers the researcher is getting. The analysis of the answers might help in the generation of hypotheses. Over time are such hypotheses being supported by further field research?

7. Focus on specific aspects of the setting and use theoretical sampling. Field researcher first gets a general picture, and then focuses on a few specific problems or issues. A researcher decides on specific research questions and develops hypotheses only after being in the field and experiencing it first-hand. At first, everything seems relevant; later, however, selective attention focuses on specific questions and themes.

Field research sampling differs from survey sampling, although sometime both use snowball sampling. A field researcher samples by taking a smaller, selective set of observations from all possible observations. It is called theoretical sampling because it is guided by the researcher's developing theory. Field researchers sample times, situations, types of events, locations, types of people, or context of interest.

For example, field researcher samples time by observing a setting at different times. He or she observes at all time of the day, on every day of the week, and in all seasons to get a full sense of how the field site stays the same or changes. Another example, when the field researcher samples locations because one location may give depth, but narrow perspective. Sitting or standing in different locations helps the researcher to get a sense of the whole site. Similarly the field researchers sample people by focusing their attention or interaction on different kinds of people (young, adult, old).

8. Conduct field interviews with member informants. Field researchers use unstructured, non-directive, in-depth interviews, which differs from formal survey research interviews in many ways. The field interview involves asking question, listening, expressing interest, and recording what was said.

(Field interview is a joint production of a research and a member. Members are active participants whose insights, feelings, and cooperation are essential parts of a discussion process that reveals subjective meaning. The interviewer's presence and form of involvement

Field research interviews go by many names: unstructured, depth, ethnographic, open ended, informal, and long. Generally, they involve one or more people being present, occur in the field, and are informal and nondirective.

(A comparison of the field research interview and a survey interview could be as below:

Survey Interview

Field Interview

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- | | |
|--|--|
| 1. It has <u>clear</u> beginning and end. | 1. The beginning and end are not clear. The interview can be picked up later. |
| 2. The same standard questions are asked of all respondents in the same sequence. | 2. The questions and the order in which they are asked are tailored to specific people and situations. |
| 3. The interviewer appears <u>neutral</u> at all times. | 3. The interviewer shows interest in responses, encourages elaboration. <i>تفصيل</i> |
| 4. The interviewer asks questions, and the respondent answers. | 4. It is like a <u>friendly</u> conversational exchange but with more interviewer questions. |
| 5. It is almost always with one respondent <u>alone</u> . | 5. It can occur in group setting or with others in area, but varies. |
| 6. It has a <u>professional</u> tone and businesslike focus, diversions are ignored. | 6. It is interspersed with jokes, aside, stories, diversions, and anecdotes, which are recorded. |
| 7. <u>Closed-ended</u> questions are common, with rare probes. | 7. <u>Open-ended</u> questions are common, and probes are frequent. |
| 8. The interviewer <u>alone</u> controls the pace and direction of interview. <i>لها</i> | 8. The interviewer and member jointly control the pace and direction of the interview. |
| 9. The <u>social context</u> in which the interview occurs is ignored and assumed to make little difference. | 9. The social context of the interview is noted and seen as important for interpreting the meaning of responses. |
| 10. The interviewer attempts to mold the communication pattern into a standard framework. | 10. The interviewer adjusts to the member's norms and language usages. <i>معا</i> |

b/w bibliography & reference.

Research Methods - STA630

Lesso

REFERENCING

There is a general mix up or referencing with bibliography; though the purposes are different. **bibliography** is the listing of the works that are relevant to the topic of research interest arranged in alphabetical order of the last names of authors. A **reference list** is a subset of the bibliography, it includes details of all the citations used in literature survey and elsewhere in the report, arranged again, in the alphabetical order of the last names of authors. These citations have the goals of credit to the author and enabling the reader to find the works cited.

Giving references in the report or thesis is a must, whereas the bibliography is additional information and is certainly optional. There should be no mixing up of the meanings.

There are different modes of referencing being followed by different disciplines. Find out what is followed in your discipline. For example, psychologists follow the publication manual of American Psychological Association (APA), and sociologists follow guidelines given in the manual of American Sociological Association. Similarly other subjects follow their professional association's manual. Each of these manuals specifies, with examples, how books, journals, newspapers, dissertations, and other materials are to be referenced in manuscripts. Whichever the style you pick up, follow it consistently. Since APA format is followed for referencing in many journals in management area, we shall present that here as a specimen. All the citations mentioned in the research report should find place in the References section at the end of the report.

Specimen Format for citing different Types of References

Book by a single author

Leshin, C. B. (1997). *Management on the World Wide Web*. Englewood Cliffs, NJ: Prentice-Hall.

Start with the last name, put a comma and then initials with full stop. It is followed by the year of publication in parentheses with a full stop. Then we have the title of the publication; all in small words (unless there is some name which has to be with capital letter as it is in this title) and in italic. Give full stop at the end. It is followed by place of publication with a colon at the end. After the colon give the name of the publisher. Second line of the reference should be indented by giving five spaces.

Give two spaces for separating the references.

Book by more than one author

Step in which situation a researcher can select → **HISTORICAL COMPARATIVE RESEARCH** ←

History has several meanings; one of which could refer to 'the events of the past.' Historiography is the method of doing historical research or of gathering and analyzing historical evidence.

Historical-comparative research is a collection of techniques and approaches. It is a distinct type of research that puts historical time and /or cross-cultural variation at the center of research – that is which treats what is studied as part of the flow of history and situated in cultural context.

Major questions

Historical comparative research is a powerful method for addressing big questions: How did major societal change take place? What fundamental features are common to most societies? Why do current social arrangements take a certain form in some societies but not in others? For example, historical-comparative researchers have addressed the questions of what caused societal revolutions in China, France and Russia; how major social institutions, medicine, have developed and changed over two centuries; how basic relationships, like feelings about the value of children, change; why public policy toward the treatment of elderly developed in one way instead of another way in an industrial country; why South Africa developed a system of greater racial separation as the United States moved toward a greater racial integration.

Historical-comparative research is suited for examining the combination of societal factors that produce a specific outcome (e.g., civil war). It is also appropriate for comparing entire social systems to see what is common across societies and what is unique, and to study long term change. A researcher may apply a theory to specific cases to illustrate its usefulness. And he or she compares the same social processes and concepts in different cultural or historical contexts.

Researchers also use H-C method to reinterpret data or challenge old explanations. By asking different questions, finding new evidence, or assembling evidence in a different way, the researcher raises questions about old explanations and finds support for new ones by interpreting data in its cultural-historical context.

Historical-comparative research can strengthen conceptualization and theory building. By looking at historical events or diverse cultural contexts, a researcher can generate new concepts and broaden her perspective. Concepts are less likely to be restricted to a single historical time or to a culture; they can be grounded in the experiences of people living in a specific cultural and historical context.

Historical-Comparative research focuses on:

- Tracing the development of social forms (patterns) overtime as well as its broad historical processes, and
- Comparing those forms and its developmental processes across cultures (countries/nations)

Step Historical-Comparative research follows scientific approach:

- Can be a survey of events in history – could be through the study of documents. Organizations generally document themselves, so if one is studying the development of an organization he/she should examine its official documents: charters, policy statements, speeches by the leaders, and so on. Often, official government documents provide the needed for analysis. To better appreciate the history of race relations in the United States, one could examine 200 years of laws and court cases involving race.

have supported the argument that the role of Protestantism in that regard.

Logic of Historical-Comparative Research

Confusion over terms reigns H-C research. Researchers call what they do historical-comparative, but mean different things. The key question is: Is there a comparative method and logic, or is there just social research that happens to be about the past or in several societies? Some researchers use positivist, quantitative approaches to historical or comparative issues, while others rely on qualitative approach.

Imp

Quantitative approach:

Positivist researchers reject the idea that there is a distinct H-C method. They hypothesize, analyze quantitative data, and replicate research to discover general patterns across time and societies. They see no fundamental distinction between quantitative and historical-comparative research. They apply quantitative research techniques and adjustments, to study the past or other cultures.

- The researcher can focus on the issue in one society few societies or many societies.
- The researcher can focus on the issue in one time in the past or examine many years/periods in the past.
- The researcher can focus on the issue in the present or a recent past.
- The researcher's analysis could be based primarily on quantitative data.
- Nevertheless, the debate continues.

H-C researchers sometimes use *time-series* data to monitor changing conditions. Data on population, crime rates, unemployment, infant mortality rates, and such data sometimes requires sophistication for purposes of comparability. If the concept vary, it becomes difficult to make comparisons. The definitions of nations but also these could vary within the same country over time (In literacy changed from what it was in first population census of 1951 and what it is now).

Imp

Qualitative approach:

There are no easily listed steps to follow in the analysis of historical data. The German term *verstehen* – “understanding” – in reference to an essential feature of the behavioral sciences. He meant that the researcher must be able to understand the circumstances, views, and feelings of those being studied to interpret their actions.

The historical-comparative researcher must find patterns among the voluminous data on a subject matter of study. Often this takes the form of what Weber called *ideal types*, composed of the essential characteristics of the phenomena. Thus, for example, Weber conducted lot of research on bureaucracy. Having observed numerous bureaucracies, he identified those qualities essential to bureaucracies in general: jurisdictional area, hierarchy, authority, written files, and so on. Weber did not merely list those characteristics.

- Focus group discussions are highly flexible with respect to schedule, location, and logistics of discussion.
- Focus group discussions provide a direct link between the research and work. In fact most of the focus group discussions are held on the part of it.
- For some researchers, focus group discussions may be a fun. They interact directly with the relevant population.

Limitations *IMP*

- Results of the focus group discussions cannot usually be used to generalize to the population from where the participants in FGD came. One implication is that their representativeness about other populations.
- It is often seen that participants usually agree with the responses (for different reasons). Without a sensitive and effective facilitator, a single participant may dominate the session. Researchers have to be aware of this.
- The moderator may influence focus group discussion and may lead to biased results.
- Focus group discussions may have limited value in exploring issues which they may not share in open discussion.
- It is possible that focus group discussions may paint a picture of the community rather than what is actually occurring or is believed to be occurring. It may be a good image of their community to strangers; hence the results may be contaminated.

CASE STUDY

Case study is a comprehensive description and analysis of a single situation or a group of situations i.e. cases. It is an intensive description and analysis of a qualitative approach to explore the case in as rich a detail as possible. It is a study of a highly successful organization, a project (Orangi Pilot Project), a teacher, and a patient. In a way it is more like a clinical approach to study a case. The researcher is looking at highly successful organization then he tries to find out the factors which have contributed to its success. The factors may be management, the work environment, etc.

Facilitator is → **FOCUS GROUP DISCUSSION**

A visitor to a locality stops by a house and inquires about the address of a resident he wants to see. May be he starts talking with a couple of persons asking for their help. In the meantime, some other passersby, or coming out of other houses join, showing their curiosity about the issue. They ask for some more information about the resident concerned, and then start discussing among them to come up with the exact identification of the resident. As an outcome of this discussion they would guide the visitor to reach the destination. This is quite a common feature in a folk society (village, neighborhood in a city) where we may start talking with a couple of persons and others come and join the conversation. This is an example of *informal* focus group discussion, which is built upon the social networks that operate in a natural setting. These social networks include both kinsfolk and other neighbors. In some cases the participants may be the local decision makers.

In research, focus group discussions (FGD) are a more formal way of getting groups of people to discuss selected issues. A focus group discussion is a group discussion of 5-12 persons guided by a facilitator, during which group members talk freely and spontaneously about a certain topic. There may be some disagreement about the exact number of participants in the discussion, as one comes across variations in numbers (6 to 10, 6 to 12, 6 to 15, 8 to 10, 5 to 7) in different books on research methods. The trend has been toward smaller groups due to some problems with the larger groups which like:

- In a bigger group each participant's speaking time is substantially restricted. Dominant/submissive relationships are almost inevitable.
- Frustration or dissatisfaction among group members is likely to result because of some members' inability to get a turn to speak. This produces lower quality and quantity of data.
- Participants are often forced into long speeches, often containing irrelevant information, when they get to speak only infrequently.
- The tendency for side conversations between participants increases.

In contrast, smaller group sessions are felt to provide greater depth response for each participant. The group is often more cohesive and interactive, particularly when participants are professionals, such as physicians or pharmacists.

The key factor concerning group size is generally the of group purpose. If the purpose of the group is to generate as many ideas as possible, a larger group may be most useful. If the purpose of the group is to maximize the depth of expression from each participant, a smaller group works better.

The Purpose of FGD

The purpose of an FGD is to obtain in-depth information on concepts, perceptions, and ideas of a group. An FGD aims to be more than a question-answer interaction (Focus group interview is different). Here the idea is that group members discuss the topic among themselves.

Formal Focus Groups

Formal groups are formally constituted, that is these are organized in advance by inviting the selected individuals to participate in the discussion on a specific issue. They are structured groups brought together in which the participants are expected to have similar background, age, sex, education, religion, or similar experiences. Similarity in background is likely to make them comfortable when they could express their viewpoint frankly and freely. If the big boss and his junior officer working in an organization together participate in an FGD, the junior officer may not be able to express his or

he or she organizes the evidence into a coherent whole. He or she sorts and resorts them into piles or files on the basis of categories and writes down the links or connections he or she sees within them.

Synthesis links specific evidence with an abstract model of a phenomenon. A researcher often looks for new evidence to verify or modify an explanatory model is developed. He or she evaluates how well the model fits the evidence and adjusts it accordingly.

Historical-comparative researchers also identify critical indicators or explanations. A *critical indicator* is unambiguous evidence that supports a specific theoretical relationship. Researchers seek these indicators to confirm a model. Indicators critically confirm a theoretical inference and provide a clear interpretation.

Writing a Report

Combine evidence, concepts, and synthesis into a research report. Writing a report is key in H-C research. Assembling evidence, and writing a report is always a crucial step; but more than in quantitative approaches, a good explanation makes or breaks H-C research. A researcher drafts a report and prepares extensive footnotes. She or he weaves together the evidence into a coherent, convincing picture to readers.

Data and Evidence in Historical context

Historical-comparative researchers draw on four types' historical evidence:

1. Primary sources;

Question No : 22 of 27

Marks: 1 (Budgeted Time 1 Min)

Which of the following is the most critical area while reviewing a journal article?

- ☐ Introduction
- ☒ Review of literature
- ☐ Method
- ☐ Discussion

Made By: Waqar Siddhu

Question No : 23 of 27

Marks: 3 (Budgeted Time 6 Min)

"Research is gathering information to find solution of a problem" What type of problem is a research problem?

Answer (Please [click here](#) to Add Answer)



Made By: Waqar Siddhu

Question No : 24 of 27

Marks: 3 (Budgeted Time 6 Min)

Goodwill of the sponsoring agency can increase the response rate but may bias the respondents. Elaborate.

Answer (Please [click here](#) to Add Answer)



Made By: Waqar Siddhu

Question No : 25 of 27

Marks: 3 (Budgeted Time 6 Min)

Take stress as an independent variable and job satisfaction as dependent variable and develop directional and non-directional hypotheses from the given variables.

Answer (Please [click here](#) to Add Answer)

Rich text editor toolbar with icons for bold, italic, underline, text color, background color, bulleted list, numbered list, link, unlink, undo, redo, and a 100% zoom level. The editor area is currently empty.

Made By: Waqar Siddhu

Question No : 26 of 27

Marks: 5 (Budgeted Time 10 Min)

"Because literature survey is a time-consuming exercise, a good in-depth interview will be sufficient to develop a theoretical framework". Discuss this statement.

Answer (Please [click here](#) to Add Answer)

Rich text editor toolbar with icons for bold, italic, underline, text color, background color, bulleted list, numbered list, link, unlink, undo, redo, and a 100% zoom level. The editor area is currently empty.

Made By: Waqar Siddhu

Question No : 27 of 27

Marks: 5 (Budgeted Time 10 Min)

Judge the following cover letter on the following characteristics:

1. Sponsoring agency of the research
2. Objectives of the research
3. Selection criteria of the respondents
4. The assurance of the confidentiality

Answer (Please [click here](#) to Add Answer)

Rich text editor toolbar with icons for bold, italic, underline, text color, background color, bulleted list, numbered list, link, unlink, undo, redo, and a 100% zoom level. The editor area is currently empty.

Made By: Waqar Siddhu

Question No : 27 of 27

Marks: 5 (Budgeted Time 10 Min)

Dear respondent

It is to request you that I am conducting a research on a very important topic and want your effective participation in it. A very famous company is sponsoring our research and the token amount for filling the questionnaire is attached with the survey.

We have drawn a random sample and you are one of the 1000 respondents. We assure u that your contribution means a lot.

Thanking you in advance

Answer (Please [click here](#) to Add Answer)

Rich text editor toolbar with icons for bold, italic, underline, text color, background color, bulleted list, numbered list, link, unlink, undo, redo, and a 100% zoom level. The editor area is currently empty.

Made By: Waqar Siddhu

Question No : 27 of 27

Marks: 5 (Budgeted Time 10 Min)

We have drawn a random sample and you are one of the 1000 respondents. We assure u that your contribution means a lot.

Thanking you in advance

For any queries related to survey contact on the numbers enclosed

X Y Z

Research scholar

Answer (Please [click here](#) to Add Answer)

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Made By: Waqar Siddhu

Question No : 22 of 27

Marks: 1 (Budgeted Time 1 Min)

Which of the following is the most critical area while reviewing a journal article?

Answer (Please select your correct option)

☐ Introduction

☐ Review of literature

☐ Method

☒ Discussion

Made By: Waqar Siddhu

Question No : 23 of 27

Marks: 3 (Budgeted Time 6 Min)

What might be the possible reasons for the low completion rate of the mailed questionnaire? And how it can be overcome?

Answer (Please click here to Add Answer)

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Made By: Waqar Siddhu

Question No : 24 of 27

Marks: 3 (Budgeted Time 6 Min)

"Strong customer base increases the probability of success of a brand." Classify this hypothesis among four type of hypothesis and justify your choice.

Answer (Please click here to Add Answer)

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Made By: Waqar Siddhu

Question No : 25 of 27

Marks: 3 (Budgeted Time 6 Min)

A researcher is conducting research on employee commitment and has taken high morale as one indicator of employee commitment. "High morale" itself is too broad and vague that it must be defined through some elements that make an employee morale high. Write just three elements that indicate high morale rate in an employee.

Answer ([Please click here to Add Answer](#))

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Made By: Waqar Siddhu

Question No : 26 of 27

Marks: 5 (Budgeted Time 10 Min)

How research question helps to formulate hypothesis. Explain with the help of an example.

Answer ([Please click here to Add Answer](#))

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Made By: Waqar Siddhu

Question No : 27 of 27

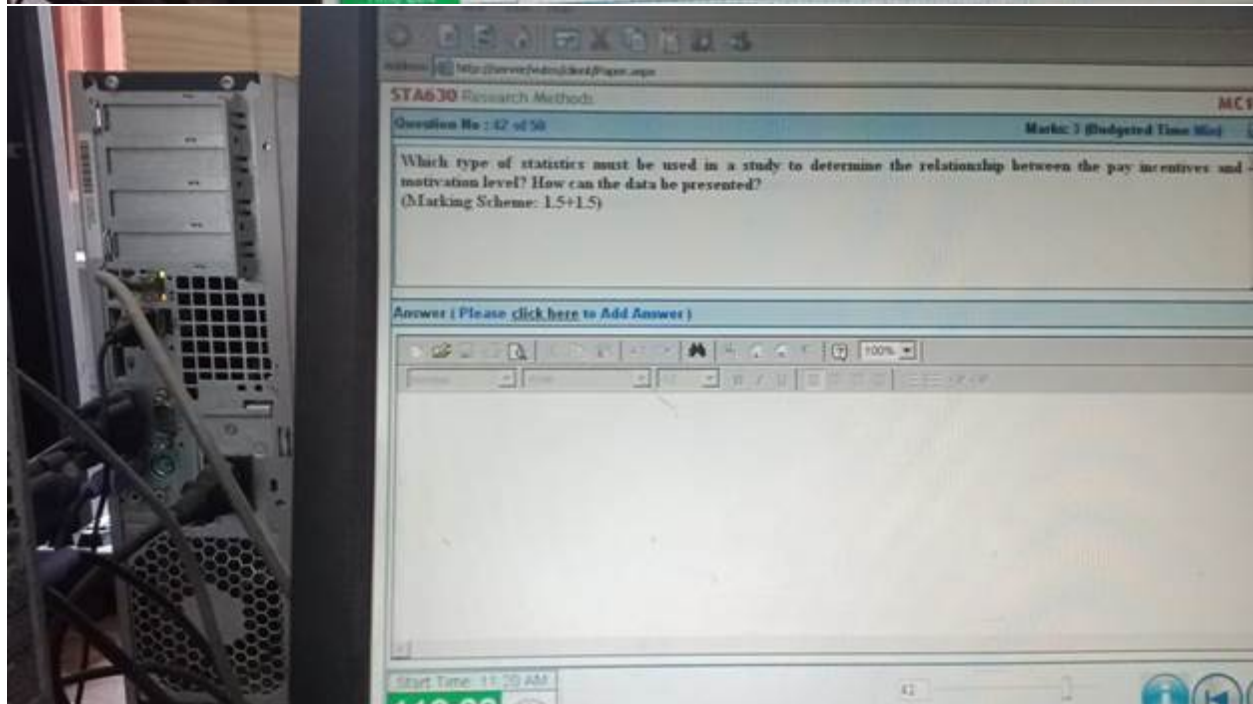
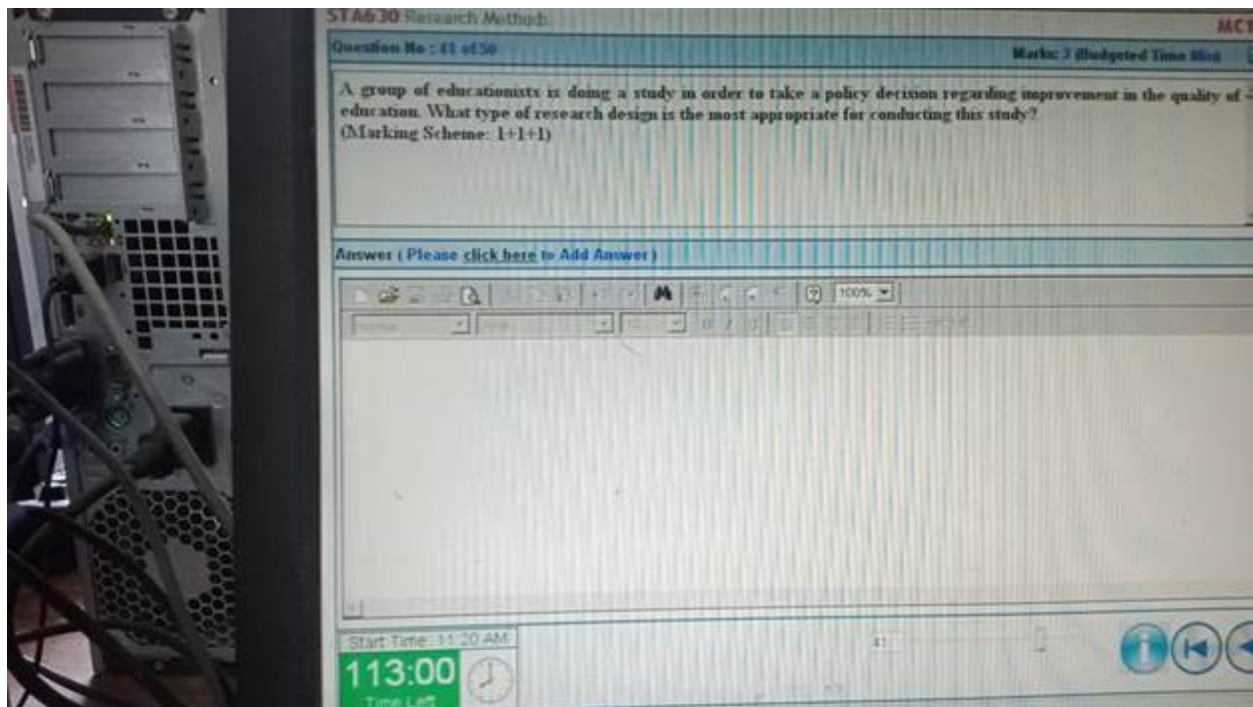
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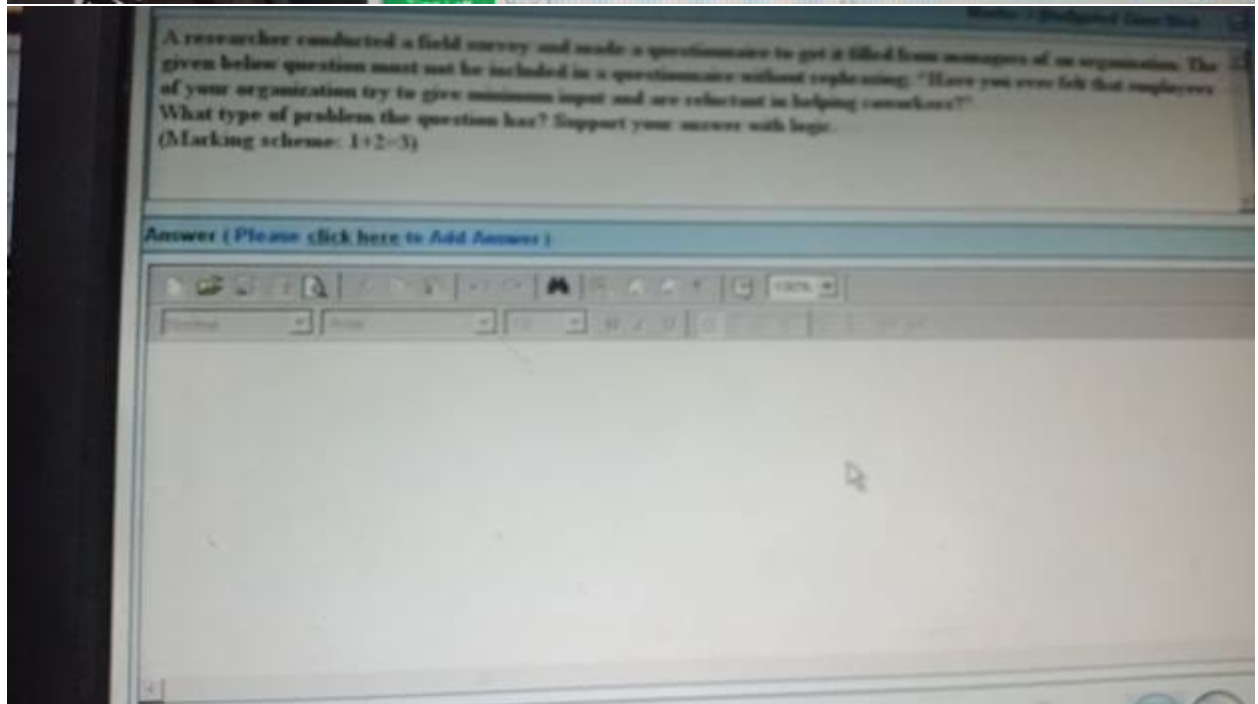
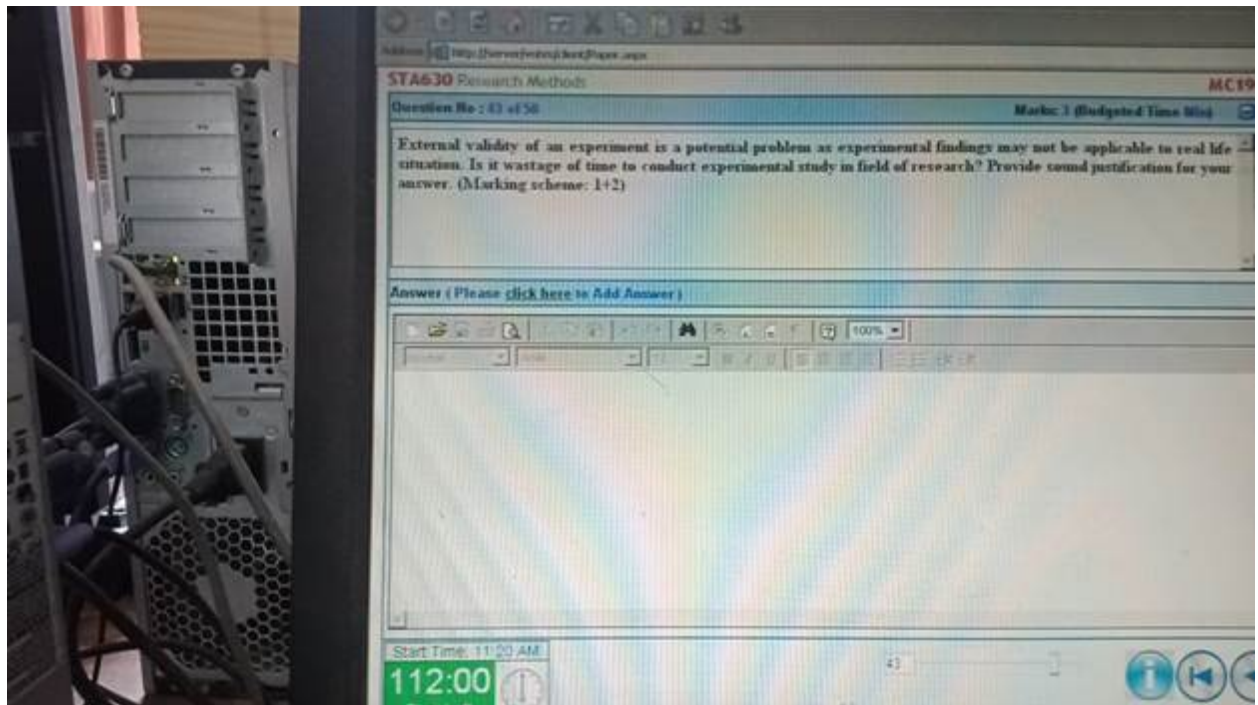
As a researcher, in which situation will you avoid using Mail Questionnaire?

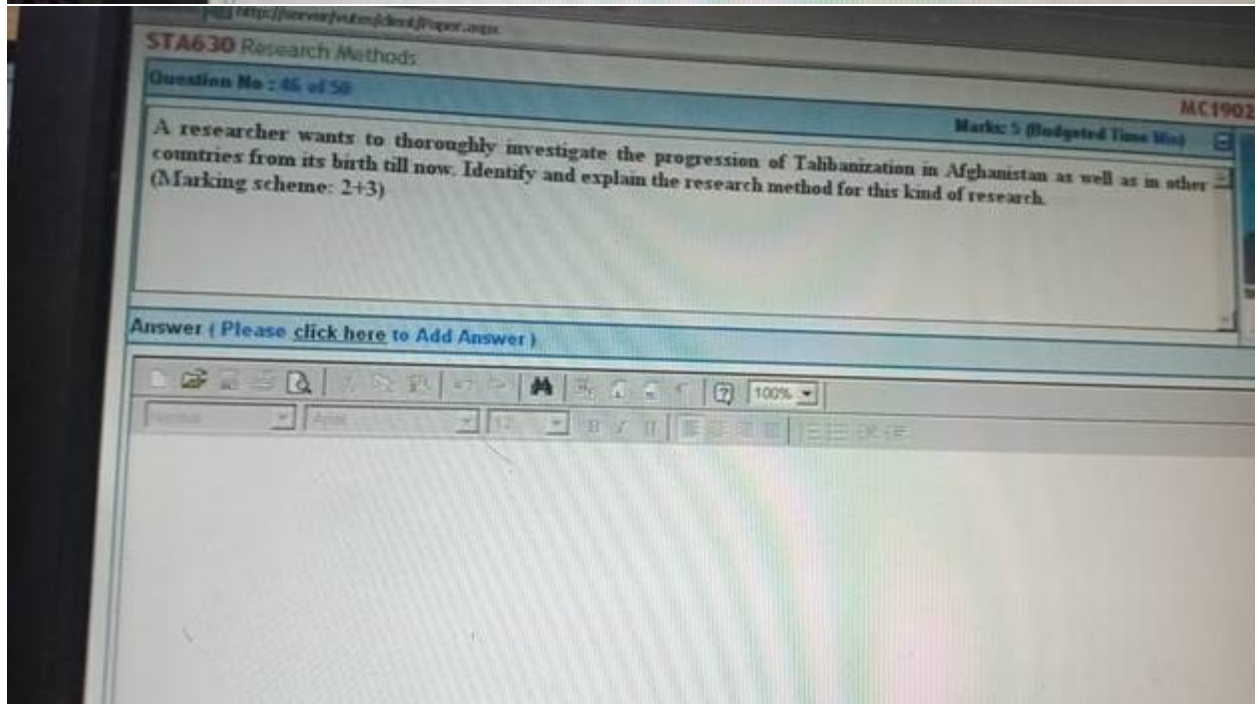
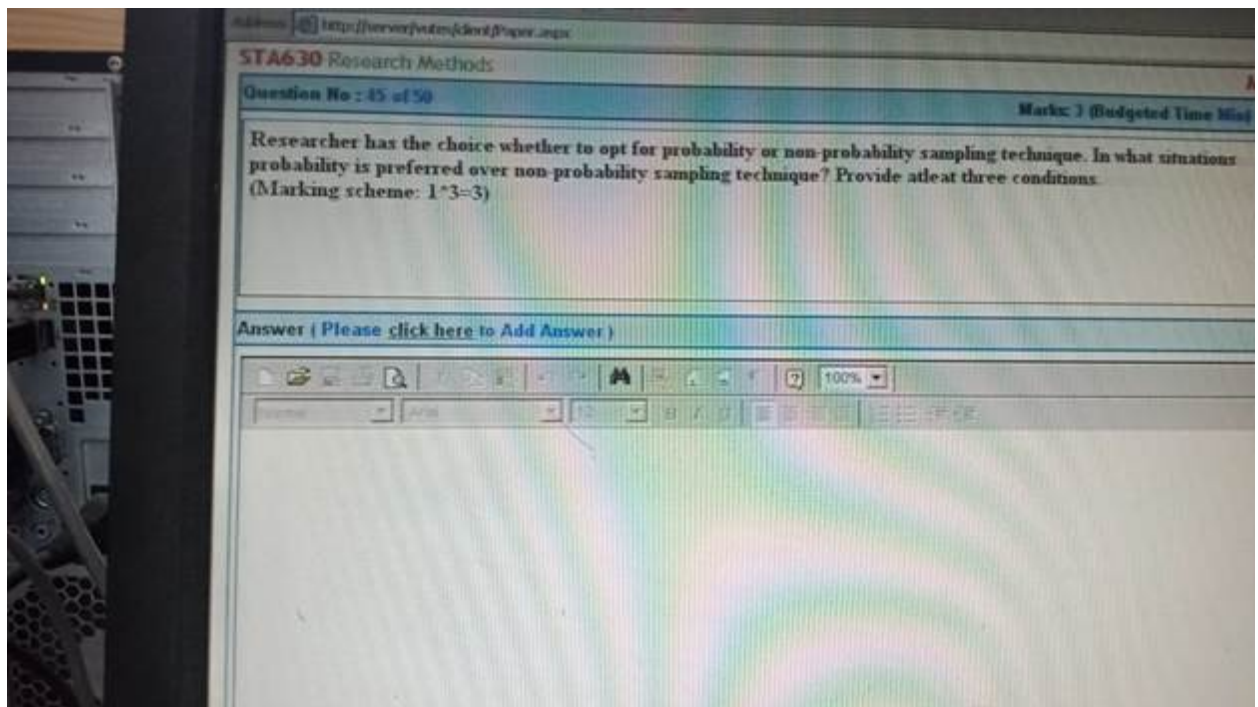
Answer ([Please click here to Add Answer](#))

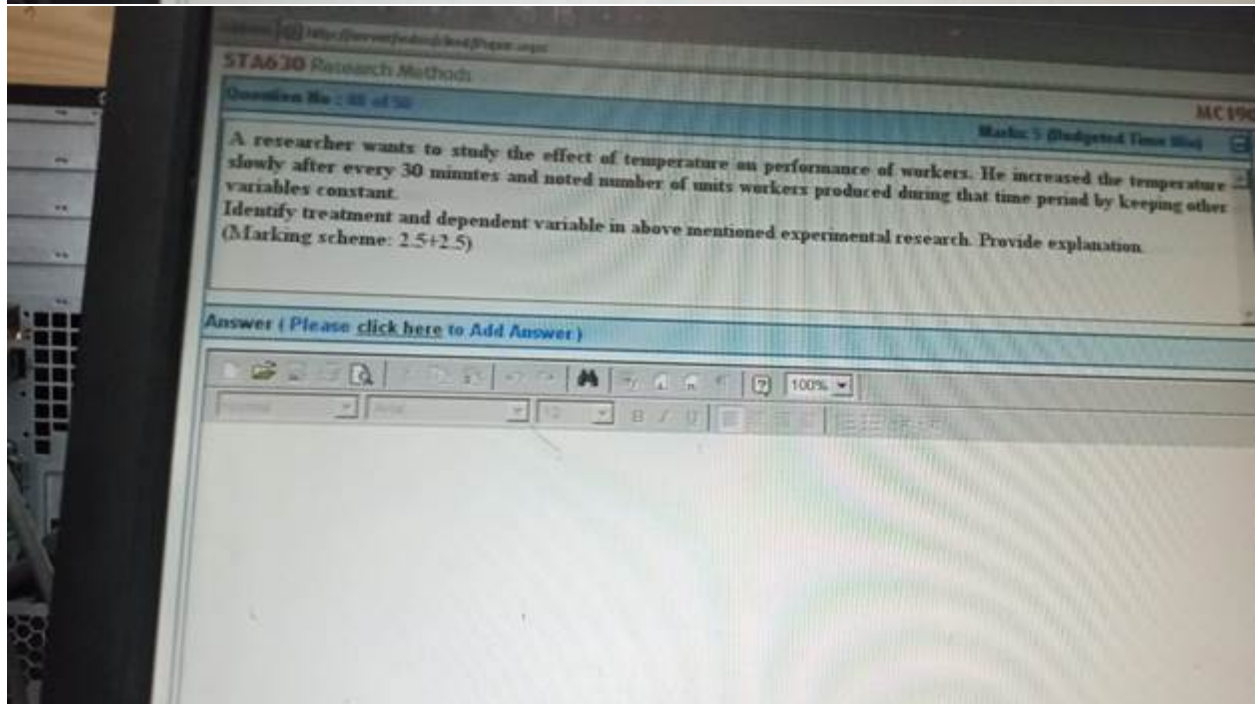
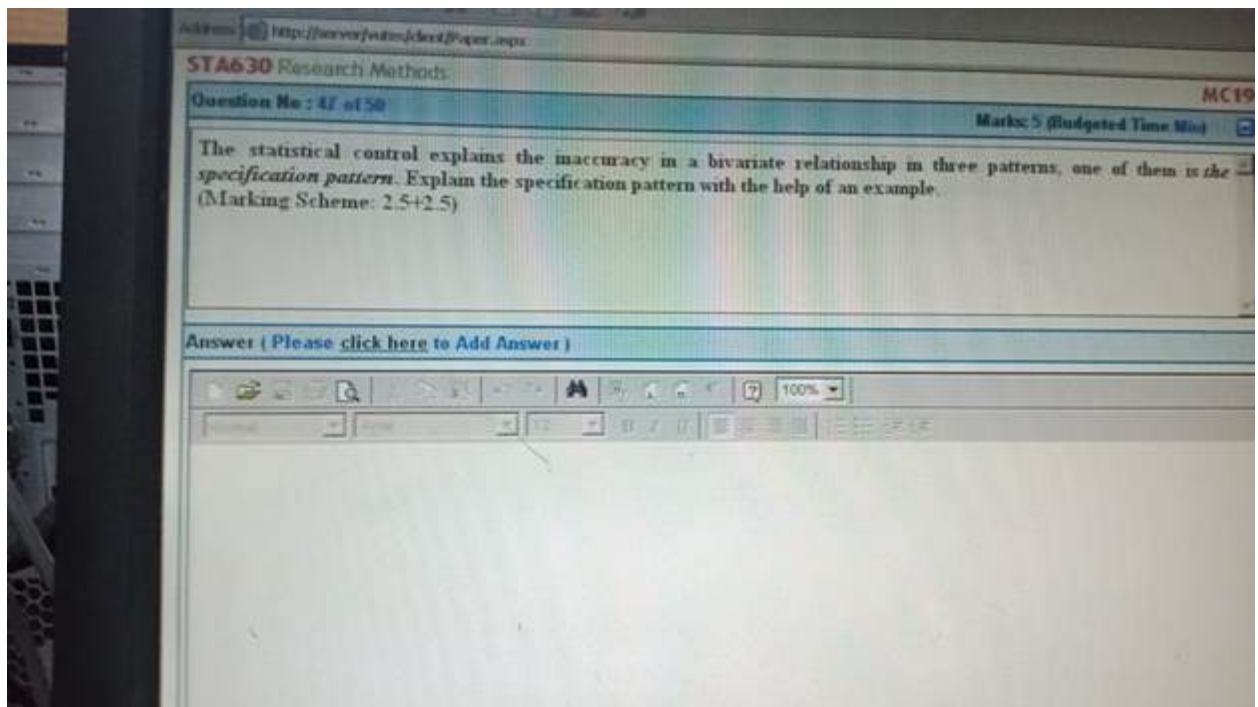
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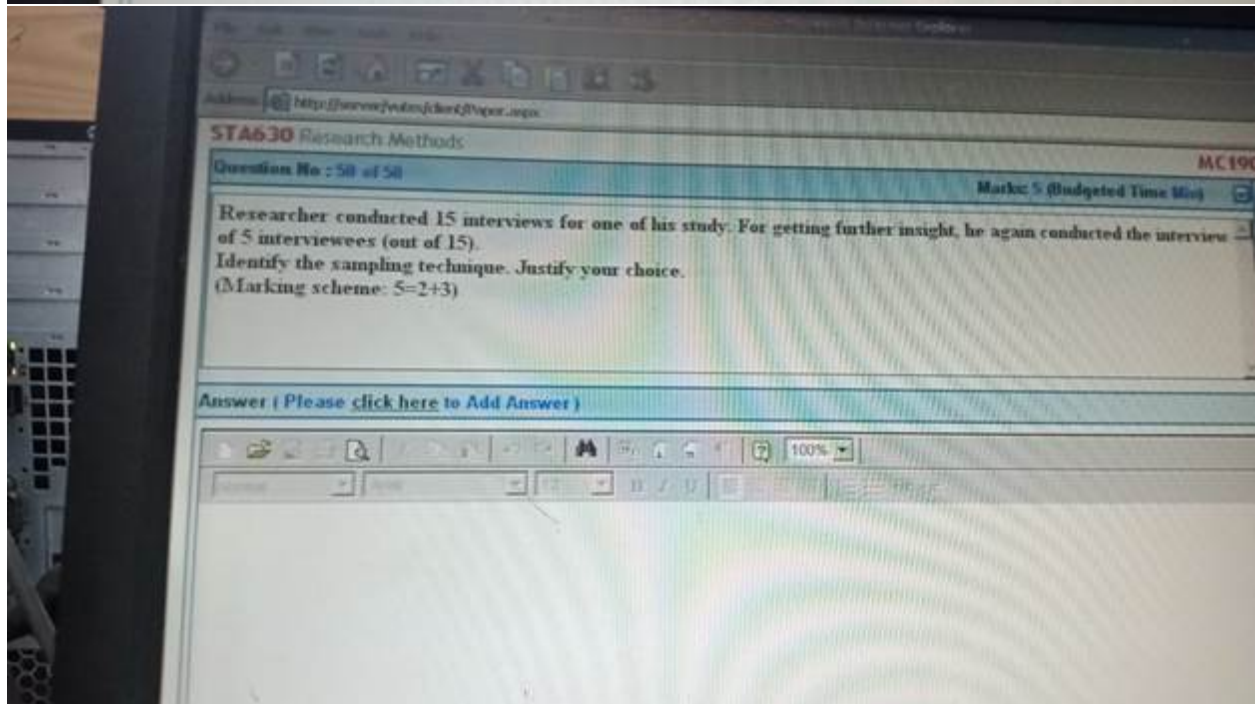
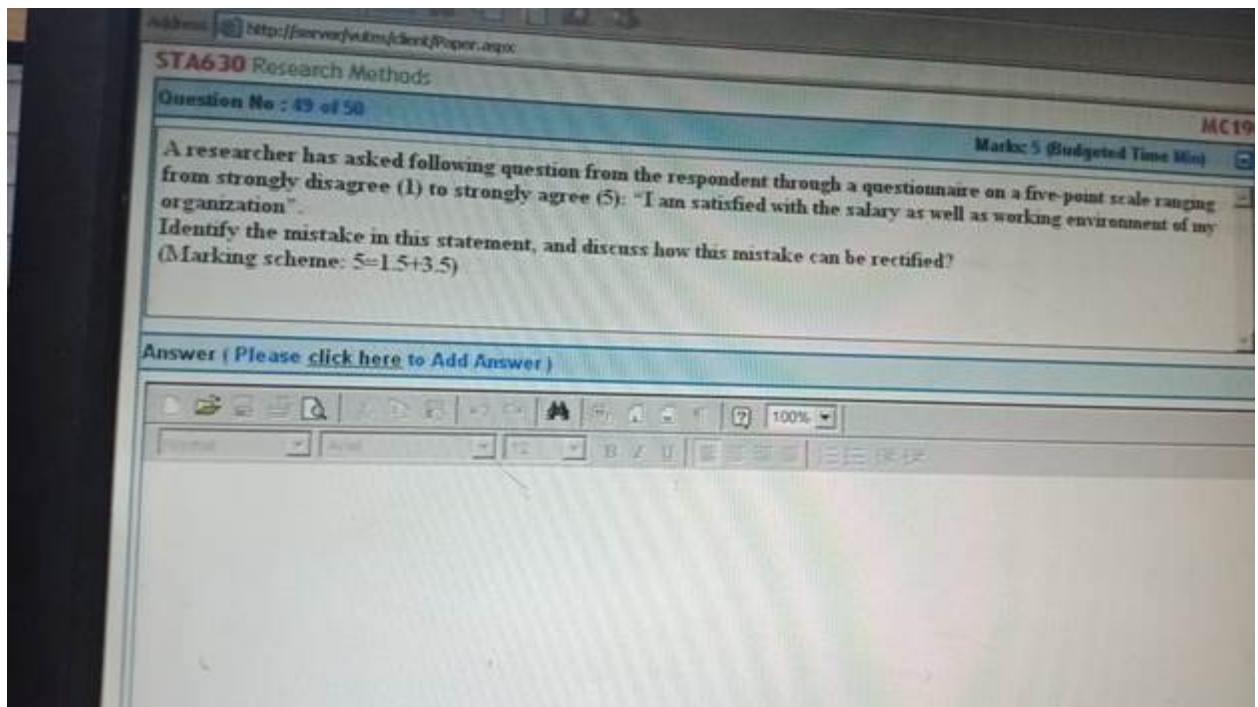
Made By: Waqar Siddhu











Address <http://server/vutes/client/login.aspx>

Question No : 33 of 39

Marks: 3 (Budgeted Time [Min](#))

In order to conduct a field research, researchers usually adopt a certain social role. Why is it recommended to adopt a social role?

(Marking scheme: 1*3)

Answer ([Please click here to Add Answer](#))

Rich text editor toolbar with icons for undo, redo, bold, italic, underline, link, unlink, list, indent, outdent, and a 100% zoom dropdown. Below the toolbar is a large text area for writing the answer.

Start Time: 11:45 AM

59:00

33



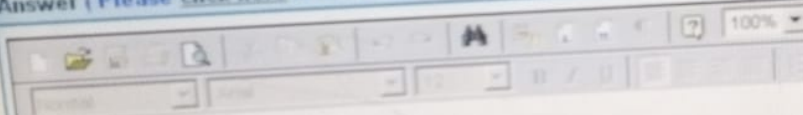
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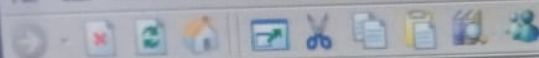
Marks: 5 (Budgeted Time Min)

Question No : 38 of 39

The ability to probe with neutral stimuli is the mark of an experienced interviewer. You are required to explain few possible tactics that can be used for probing.
(Marking scheme: 5=5*1)

Answer (Please click here to Add Answer)

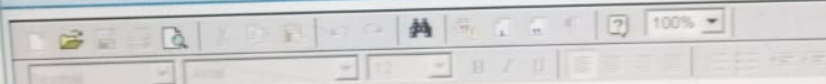




Question No : 39 of 39

A researcher wants to study the impact of organizational culture on Enterprise Resource Planning (ERP) systems. Population for this study is employees working on ERP systems in Karachi based organizations. For this purpose, he decided to contact with the respondents by using chain referrals. Identify type of sampling (Probability or Non-Probability) and sampling technique. Justify your choice. (Marking scheme: 5=1.5+1.5+2)

Answer ([Please click here to Add Answer](#))



Address <http://server/vutes/client/login.aspx>

Question No : 37 of 39

Marks: 5 (Budgeted Time Min)

A researcher wants to study the effect of temperature on performance of workers. He increased the temperature slowly after every 30 minutes and noted number of units workers produced during that time period by keeping other variables constant.

Identify treatment and dependent variable in above mentioned experimental research. Provide explanation.
(Marking scheme: 2.5+2.5)

Answer (Please [click here](#) to Add Answer)

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Address <http://server/votes/client/login.aspx>

Question No : 35 of 39

Marks: 3 (Budgeted Time Min)

One of the commonly used probability sampling techniques is stratified random sampling. In what situations, this technique is preferred over other probability sampling techniques?
(Marking scheme: 1*3=3)

Answer ([Please click here to Add Answer](#))

Rich text editor toolbar with icons for undo, redo, bold, italic, underline, link, unlink, list, indent, outdent, and a zoom dropdown set to 100%.

Address <http://server/vutes/client/login.aspx>

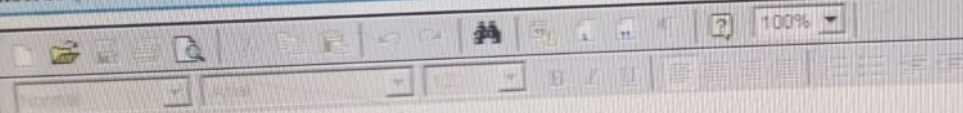
Marks: 3 (Budgeted Time [Min](#))

Question No : 34 of 39

A researcher conducted an experimental study in which he selected two groups. He provided workshop to the experimental group while control was not provided with any workshop. However, he got filled a questionnaire from subjects of two groups after workshop. What type of experimental design researcher did? Provide justification.

(Marking scheme: 1.5+1.5)

Answer ([Please click here to Add Answer](#))

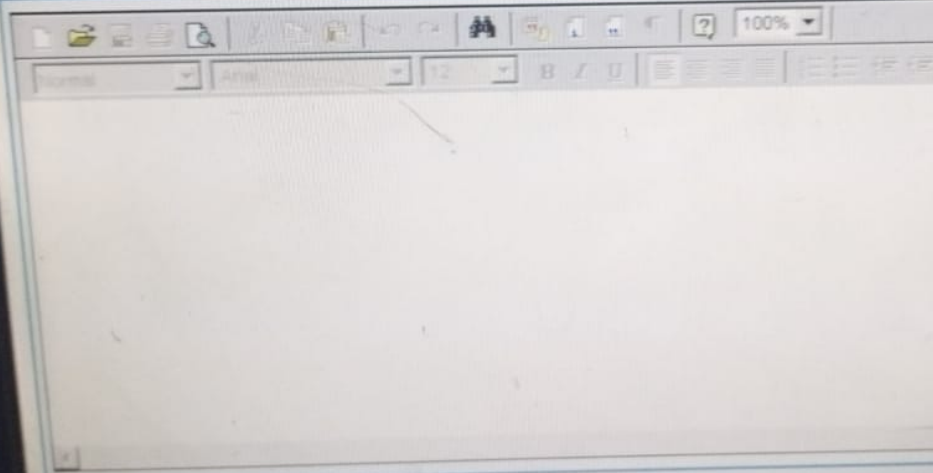


Question No : 36 of 39

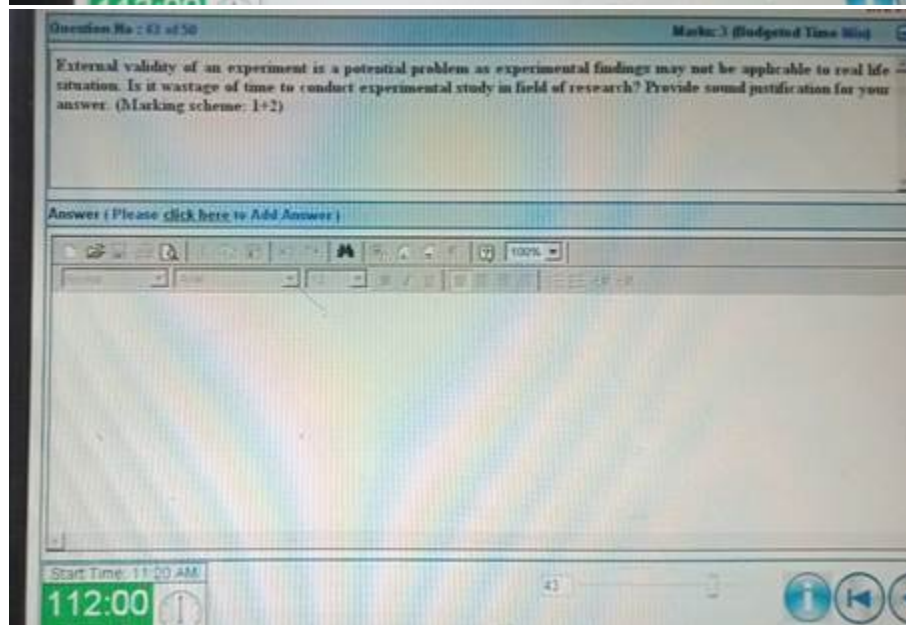
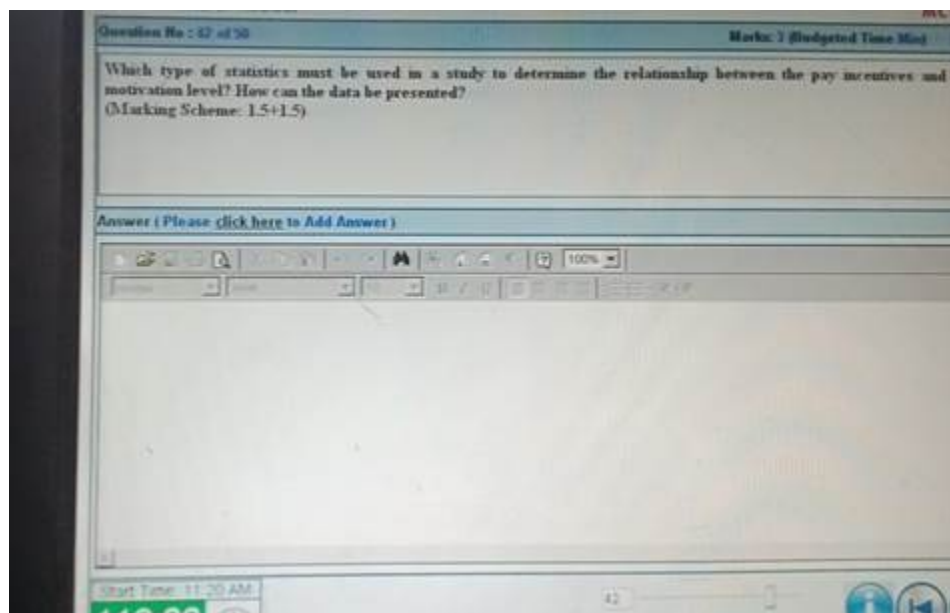
Marks: 5 (Budgeted Time [Min](#))

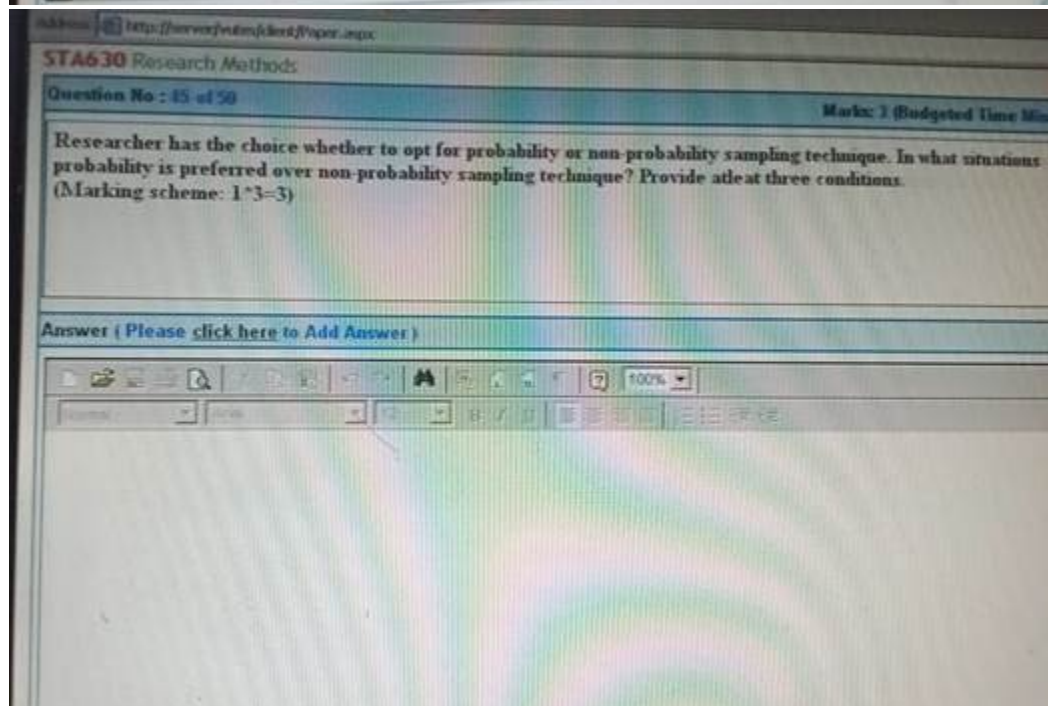
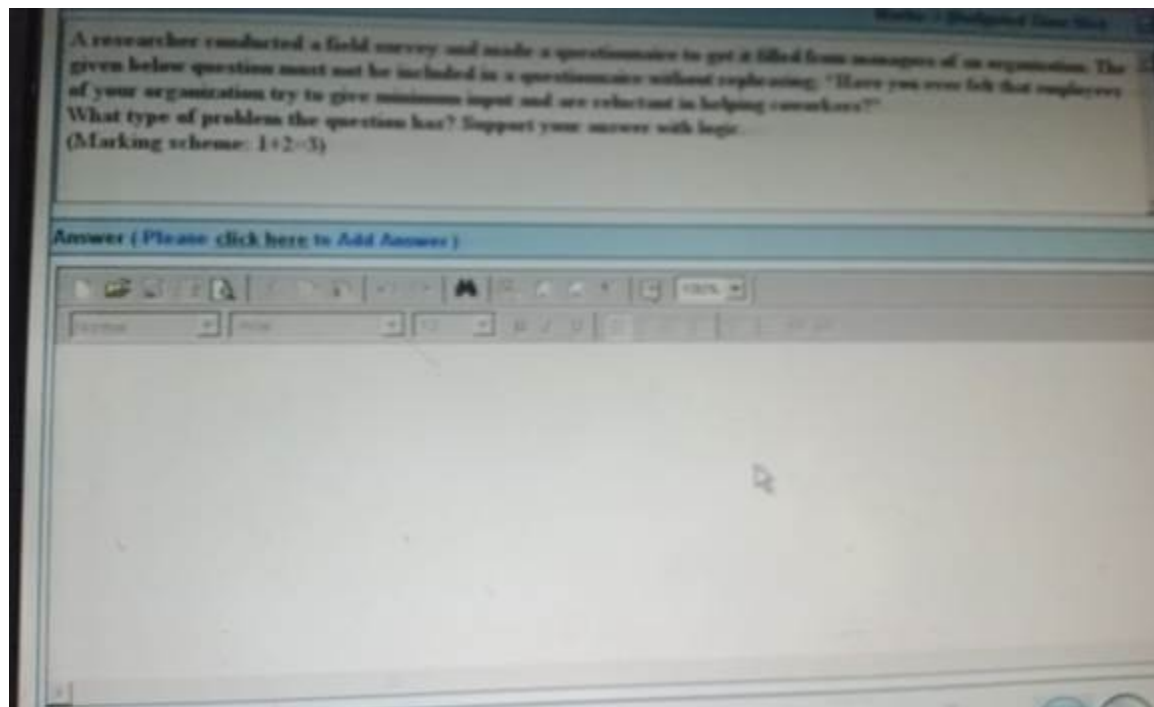
Researchers who conduct research on deviants, engaged in illegal behavior face an additional dilemma. Mention the dilemma for the field researcher?
(Marking scheme: 5 Marks)

Answer ([Please click here to Add Answer](#))



The image shows a rich text editor interface. At the top, there is a toolbar with various icons for text formatting (bold, italic, underline, strikethrough), alignment (left, center, right, justified), and editing (undo, redo, link, unlink, list, indent, outdent). Below the toolbar is a large, empty text area for writing the answer. The text area is currently blank.





ML1902

Marks: 5 (Budgeted Time: 10m)

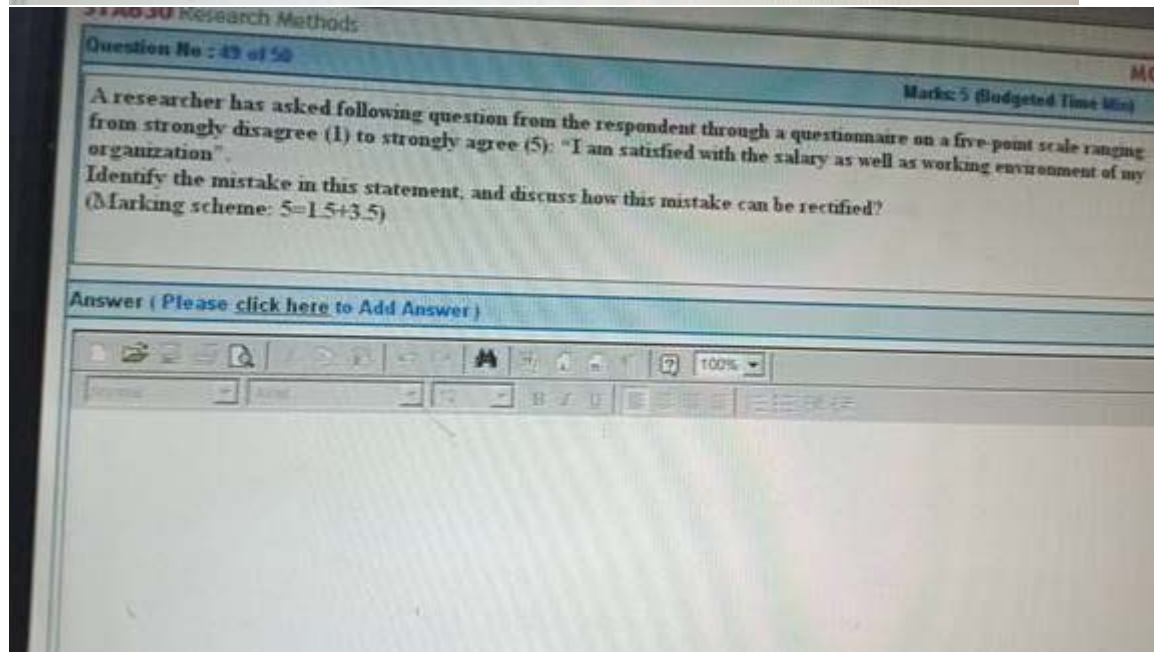
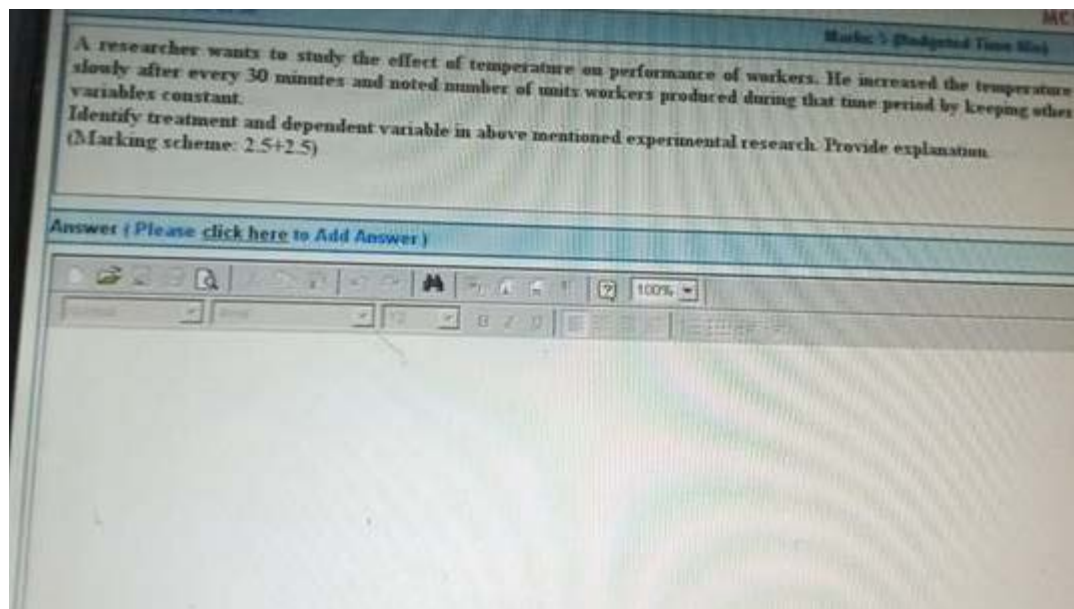
A researcher wants to thoroughly investigate the progression of Talibanization in Afghanistan as well as in other countries from its birth till now. Identify and explain the research method for this kind of research.
(Marking scheme: 2+3)

Answer (Please click here to Add Answer)

Marks: 5 (Budgeted Time: 10m)

The statistical control explains the inaccuracy in a bivariate relationship in three patterns, one of them is the *specification pattern*. Explain the specification pattern with the help of an example.
(Marking Scheme: 2.5+2.5)

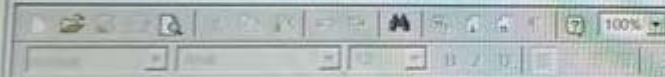
Answer (Please click here to Add Answer)



Mark: 5 (Budgeted Time: 10)

Researcher conducted 15 interviews for one of his study. For getting further insight, he again conducted the interview of 5 interviewees (out of 15).
Identify the sampling technique. Justify your choice.
(Marking scheme: 5=2+3)

Answer (Please [click here to Add Answer](#))



STA630 Research Methods

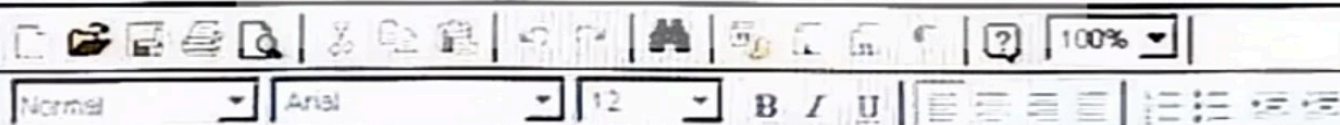
Question No : 48 of 50

Why editing is required by fieldworker during data collection process?

(Marking Scheme: 3 Marks)



Answer (Please [click here](#) to Add Answer)



STA630 Research Methods

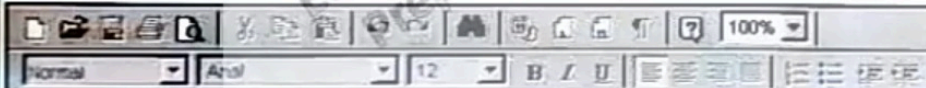
Question No: 49 of 50

Mark: 5 (Budget)

How Ex-Post Facto Experimental Design can help in establishing cause and effect relationship among independent and dependent variables?

(Marking scheme: 5 marks for explanation)

Answer (Please click here to Add Answer)



STA630 Research Methods

Question No : 50 of 50

Explain the role of pilot testing in the development of research instrument?
(Marking scheme: 5=5*1)



Answer ([Please click here to Add Answer](#))



Historical-Comparative research is one of the most sophisticated research methods in a qualitative domain. It requires a high level of expertise in successfully conducting the research as data is available in a scattered form and there are many issues of their reliability. Can you elaborate secondary sources available to the researcher for conducting historical-Comparative research?

Answer (Please click here to Add Answer)



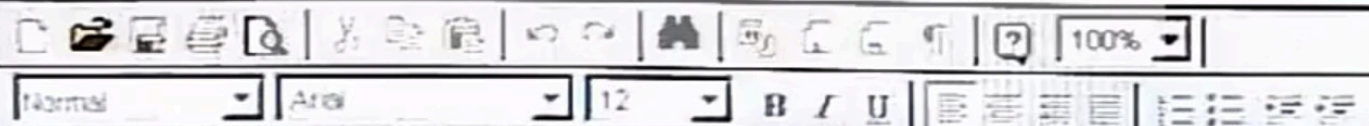
STA630 Research Methods

Question No : 46 of 50

Content analysis is "Reactive" or "Non-Reactive" research. Justify your standpoint.

(Marking Scheme: 2+3)

Answer ([Please click here to Add Answer](#))



Question No: 45 of 50

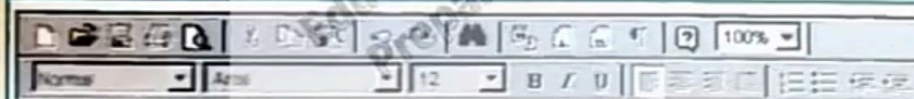
In order to obtain a good response from respondents, a researcher should adopt funnel technique in the questionnaire. What do you understand by funnel technique? Explain in your own words.
(Marking scheme: 3=3*1)

Answer (Please click here to Add Answer)



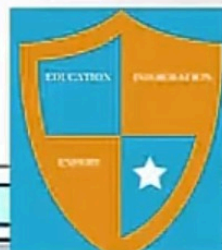
Experts generally give priority to smaller groups over larger groups in focus group discussion. What might be the reasons for this priority?
(Marking Scheme: 1*3)

Answer (Please click here to Add Answer)



Researchers transform data before data analysis. What do you understand by data transformation? Explain with the help of an example.

(Marking Scheme: 1.5+1.5)



Answer (Please [click here](#) to Add Answer.)

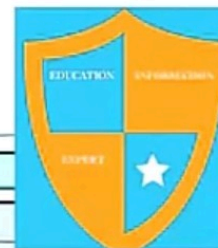


A researcher decided to conduct Solomon's Four Group Design in his experiment. What do you understand by Solomon's Four Group Design? Provide brief description.
(Marking scheme: 3 marks for explanation)

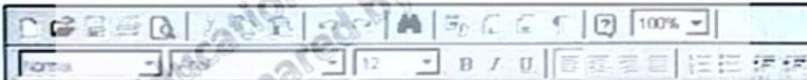
Answer ([Please click here to Add Answer](#))



Ahmad is doing content analysis research but he is confused how to select the sample in this research. Being a student of research, help Ahmad to understand sampling procedure in content analysis research with the help of an example.
(Marking Scheme: 0.5+2.5)



Answer (Please click here to Add Answer)



A researcher wants to study the impact of organizational culture on Enterprise Resource Planning (ERP) systems. Population for this study is employee working on ERP systems in Karachi based organizations. For conducting research, he decided to get responses from every available person who is working on ERP systems.

Identify the type of sampling (Probability or Non-Probability) and sampling technique. Justify your choice.
(Marking scheme: 5=1.5+1.5+2)

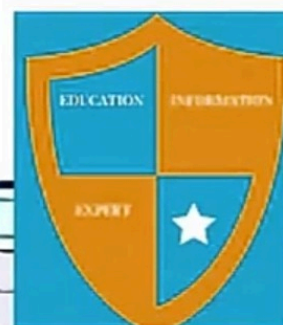
Answer: (Please click here to Add Answer)



Education Information Export
Prepared by: Muhammad Ali

Explain the role of pilot testing in the development of research instrument.
(Marking scheme: 5=5*1)

Answer (Please click here to Add Answer.)



Education Information Expert
prepared by Wubashar Ali

Identify and rectify the mistakes in the following reference of an article: (Marks: 1.5+3.5)

Grant, R. M. 1991. Porter's 'competitive advantage of nations' an assessment. Strategic management journal, 12(7), 535-548.

Answer (Please click here to Add Answer)



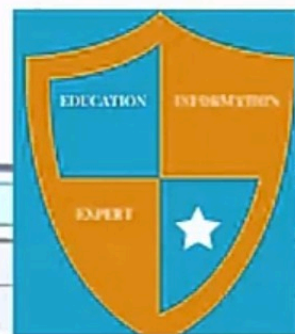
STA630 Research Methods

Question No : 47 of 50

Marks

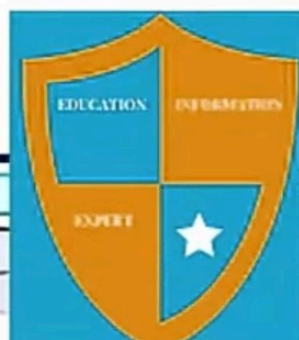
A researcher wants to analyze the theme behind pictures of a well-known artist. He visited an exhibition for collecting the data. Identify and explain the kind of research method he/she is using?
(Marking Scheme: 2+3)

Answer ([Please click here to Add Answer](#))



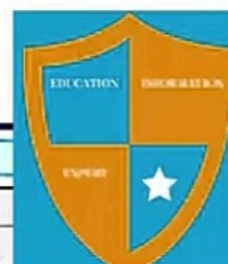
How is it possible to somehow eliminate the "Selection bias" effect while selecting participants in experiment? Give minimum three solid arguments (Mark: 5 marks for explanation)

Answer (Please click here to Add Answer)



Experimental study is mainly used in natural sciences to see the cause and effect relationship of several variables. How experimental research is useful in the field of social research? Provide minimum two justifications of your answer.
(Marking scheme: 1.5+1.5)

Answer (Please click here to Add Answer)



Education Information Expert
Prepared by Mubashar Ali

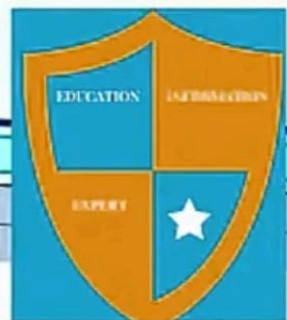
Probing in an interview helps the researcher to get the detailed answer. What are different tactics researcher can use to probe the respondents?
(Marking scheme: 1*3=3)

Answer (Please click here to Add Answer)



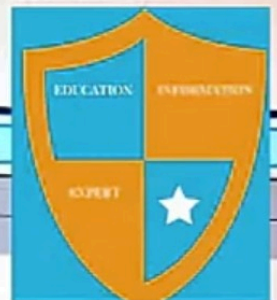
One of the most commonly used probability sampling techniques is simple random sampling but it has certain limitations. Enlist minimum three of this method.
(Marking scheme: 1*3=3)

Answer | [Please click here to Add Answer](#)



Content analysis is one of the methods of doing non-reactive research. How content analysis is non-reactive research? Provide sound justification with suitable example.
(Marking Scheme: 1+1+1)

Answer | Please click here to Add Answer



Education Information Expert
Prepared by Mubashar Ali

A researcher is doing Historical-Comparative research using different sources. How can researcher evaluate that data is authentic?
(Marking Scheme: 1*3)

Answer (Please [click here to Add Answer](#))

